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DDR5 SDRAM

From JEDEC Board Ballot JCB-24-03, formulated under the cognizance of the JC-42.3 Subcommittee on Dynamic RAMs (DDR_x), item number 1848.99O.

1 Scope

This standard defines the DDR5 SDRAM specification, including features, functionalities, AC and DC characteristics, packages, and ball/signal assignments. The purpose of this Standard is to define the minimum set of requirements for JEDEC compliant 8 Gb through 32 Gb for x4, x8, and x16 DDR5 SDRAM devices. This standard was created based on the DDR4 standards (JESD79-4) and some aspects of the DDR, DDR2, DDR3, and LPDDR4 standards (JESD79, JESD79-2, JESD79-3, and JESD209-4).

1.1 JM7 Verbal Forms and Terms

JEDEC publication JM7 provides examples and directives for the use of verbal forms (e.g., 'shall' compared with 'should' and 'may' compared with 'can').

This specification adheres to the verbal forms defined in JM7.01 July 2010 revision.

2 DDR5 SDRAM Package, Pinout Description, and Addressing

2.1 DDR5 SDRAM Row for X4, X8

The DDR5 SDRAM x4/x8 component shall have 13 electrical rows of balls. Electrical is defined as rows that contain signal ball or power/ground balls. There may be additional rows of inactive balls for mechanical support.

2.2 DDR5 SDRAM Ball Pitch

The DDR5 SDRAM component shall use a ball pitch of 0.8mm by 0.8mm.
The number of fully depopulated columns is 3.

2.3 DDR5 SDRAM Columns for X4, X8

The DDR5 SDRAM x4/x8 component shall have 6 electrical columns of balls in 2 sets of 3 columns. There shall be columns between the electrical columns where there are no balls populated. The number of these is 3. Electrical is defined as columns that contain signal ball or power/ground balls. There may be additional columns of inactive balls for mechanical support.

2.4 DDR5 SDRAM x4/8 Ballout Using MO-210

Table 1 — DDR5 SDRAM x4/8 Ballout Using MO-210

AN	1	2	3	4	5	6	7	8	9	10	11
AL		1	2	3	4	5	6	7	8	9	
A	DNU	LBDQ	VSS	VPP				ZQ ⁷	VSS	LBDQS	DNU
B		VDD	VDDQ	DQ2				DQ3	VDDQ	VDD	
C		VSS	DQ0	DQS_t				DM_n ⁴ , TDQS_t ²	DQ1	VSS	
D		VDDQ	VSS	DQS_c				TDQS_c ³	VSS	VDDQ	
E		VDD	DQ4 ¹	DQ6 ¹				DQ7 ¹	DQ5 ¹	VDD	
F		VSS	VDDQ	VSS				VSS	VDDQ	VSS	
G		CA_ODT	MIR	VDD				CK_t	VDDQ	TEN ⁵ (CS1_n ⁵)	
H		ALERT_n	VSS	CS_n				CK_c	VSS	VDD	
J		VDDQ	CA4	CA0				CA1	CA5	VDDQ	
K		VDD	CA6	CA2				CA3	CA7	VDD	
L		VDDQ	VSS	CA8				CA9	VSS	VDDQ	
M		CAI, ZQ1 ⁸	CA10	CA12				CA13	CA11	RESET_n	
N	DNU	VDD	VSS	VDD				VPP	VSS	VDD	DNU

NOTE 1

NOTE 2

NOTE 3

NOTE 4

NOTE 5

NOTE 6

NOTE 7

NOTE 8

DQ4-DQ7 are higher order DQ pins and are not connected for the x4 configuration.

TDQS_t is not valid for the x4 configuration

TDQS_c is not available for the x4 configuration

DM_n not valid for the x4 configuration

For single die package (SDP) or 3DS, this pin is TEN; CS1_n is not connected, and is used as Test Mode Enable. For dual die package (DDP), this pin is CS1_n; and TEN is disconnected, and is used as CS1_n.

The electrical parameters for DDP package are TBD and will be covered from future ballots.

For dual die package (DDP), ZQ is connected to bottom die (Rank 0).

For single die package (SDP) or 3DS, CAI is used to indicate Command Address Inversion (ZQ1 is not applicable). However, for dual die package (DDP), CAI die pads are grounded inside the package, and ZQ1 pin is connected to the top die ZQ pad (Rank 1).

MO-210-AL (x4/x8)

	1	2	3	4	5	6	7	8	9
A	○	○	○	+	+	+	○	○	○
B	○	○	○	+	+	+	○	○	○
C	○	○	○	+	+	+	○	○	○
D	○	○	○	+	+	+	○	○	○
E	○	○	○	+	+	+	○	○	○
F	○	○	○	+	+	+	○	○	○
G	○	○	○	+	+	+	○	○	○
H	○	○	○	+	+	+	○	○	○
J	○	○	○	+	+	+	○	○	○
K	○	○	○	+	+	+	○	○	○
L	○	○	○	+	+	+	○	○	○
M	○	○	○	+	+	+	○	○	○
N	○	○	○	+	+	+	○	○	○

MO-210-AN (x4/x8)
with support balls

	1	2	3	4	5	6	7	8	9	10	11
A	○	○	○	○	+	+	+	○	○	○	○
B	+	○	○	○	+	+	+	○	○	○	+
C	+	○	○	○	+	+	+	○	○	○	+
D	+	○	○	○	+	+	+	○	○	○	+
E	+	○	○	○	+	+	+	○	○	○	+
F	+	○	○	○	+	+	+	○	○	○	+
G	+	○	○	○	+	+	+	○	○	○	+
H	+	○	○	○	+	+	+	○	○	○	+
J	+	○	○	○	+	+	+	○	○	○	+
K	+	○	○	○	+	+	+	○	○	○	+
L	+	○	○	○	+	+	+	○	○	○	+
M	+	○	○	○	+	+	+	○	○	○	+
N	○	○	○	○	+	+	+	○	○	○	○

○ Populated ball
+ Ball not populated

NOTE(s):

- Additional columns and rows of inactive balls in MO-210 Terminal Pattern AN (x4/x8) with support balls are for mechanical support only, and should not be tied to either electrically high or low.
- Some of the additional support balls can be selectively populated under the supplier's discretion. Refer to supplier's datasheet.

Figure 1 — DDR5 Ball Assignments for the x4/8 Component

2.5 DDR5 SDRAM x16 Ballout using MO-210

Table 2 — DDR5 SDRAM x16 Ballout using MO-210

AU	1	2	3	4	5	6	7	8	9	10	11
AT		1	2	3	4	5	6	7	8	9	
A	DNU	LBDQ	VSS	VPP				ZQ	VSS	LBDQS	DNU
B		VDD	VDDQ	DQU2				DQU3	VDDQ	VDD	
C		VSS	DQU0	DQSU_t				DMU_n	DQU1	VSS	
D		VDDQ	VSS	DQSU_c				RFU	VSS	VDDQ	
E		VDD	DQU4	DQU6				DQU7	DQU5	VDD	
F		VDD	VDDQ	DQL2				DQL3	VDDQ	VDD	
G		VSS	DQL0	DQSL_t				DML_n	DQL1	VSS	
H		VDDQ	VSS	DQSL_c				RFU	VSS	VDDQ	
J		VDD	DQL4	DQL6				DQL7	DQL5	VDD	
K		VSS	VDDQ	VSS				VSS	VDDQ	VSS	
L		CA_ODT	MIR	VDD				CK_t	VDDQ	TEN	
M		ALERT_n	VSS	CS_n				CK_c	VSS	VDD	
N		VDDQ	CA4	CA0				CA1	CA5	VDDQ	
P		VDD	CA6	CA2				CA3	CA7	VDD	
R		VDDQ	VSS	CA8				CA9	VSS	VDDQ	
T		CAI	CA10	CA12				CA13	CA11	RESET_n	
U	DNU	VDD	VSS	VDD				VPP	VSS	VDD	DNU

MO-210-AT (x16)

	1	2	3	4	5	6	7	8	9
A	○	○	○	+	+	+	○	○	○
B	○	○	○	+	+	+	○	○	○
C	○	○	○	+	+	+	○	○	○
D	○	○	○	+	+	+	○	○	○
E	○	○	○	+	+	+	○	○	○
F	○	○	○	+	+	+	○	○	○
G	○	○	○	+	+	+	○	○	○
H	○	○	○	+	+	+	○	○	○
J	○	○	○	+	+	+	○	○	○
K	○	○	○	+	+	+	○	○	○
L	○	○	○	+	+	+	○	○	○
M	○	○	○	+	+	+	○	○	○
N	○	○	○	+	+	+	○	○	○
P	○	○	○	+	+	+	○	○	○
R	○	○	○	+	+	+	○	○	○
T	○	○	○	+	+	+	○	○	○
U	○	○	○	+	+	+	○	○	○

**MO-210-AU (x16)
with support balls**

	1	2	3	4	5	6	7	8	9	10	11
A	○	○	○	○	+	+	+	○	○	○	○
B	+	○	○	○	+	+	+	○	○	○	+
C	+	○	○	○	+	+	+	○	○	○	+
D	+	○	○	○	+	+	+	○	○	○	+
E	+	○	○	○	+	+	+	○	○	○	+
F	+	○	○	○	+	+	+	○	○	○	+
G	+	○	○	○	+	+	+	○	○	○	+
H	+	○	○	○	+	+	+	○	○	○	+
J	+	○	○	○	+	+	+	○	○	○	+
K	+	○	○	○	+	+	+	○	○	○	+
L	+	○	○	○	+	+	+	○	○	○	+
M	+	○	○	○	+	+	+	○	○	○	+
N	+	○	○	○	+	+	+	○	○	○	+
P	+	○	○	○	+	+	+	○	○	○	+
R	+	○	○	○	+	+	+	○	○	○	+
T	+	○	○	○	+	+	+	○	○	○	+
U	○	○	○	○	+	+	+	○	○	○	○

○ Populated ball
+ Ball not populated

NOTE(s):

- Additional columns and rows of inactive balls in MO-210 Terminal Pattern AU (x16) with support balls are for mechanical support only, and should not be tied to either electrically high or low.
- Some of the additional support balls can be selectively populated under the supplier's discretion. Refer to supplier's datasheet.

Figure 2 — DDR5 Ball Assignments for the x16 Component

2.6 Pinout Description

Table 3 — Pinout Description

Symbol	Type	Function
CK_t, CK_c	Input	Clock: CK_t and CK_c are differential clock inputs. All address and control input signals are sampled on the crossing of the positive edge of CK_t and negative edge of CK_c.
CS_n, (CS1_)	Input	Chip Select: All commands are masked when CS_n is registered HIGH. CS_n provides for external Rank selection on systems with multiple Ranks. CS_n is considered part of the command code. CS_n is also used to enter and exit the parts from power down modes.
DM_n, DMU_n, DML_n	Input	Input Data Mask: DM_n is an input mask signal for write data. Input data is masked when DM_n is sampled LOW coincident with that input data during a Write access. DM_n is sampled on both edges of DQS. For x8 device, the function of DM_n is enabled by MR5:OP[5]=1. DM is not supported for x4 device.
CA [13:0]	Input	Command/Address Inputs: CA signals provide the command and address inputs according to the Command Truth Table. Note: Since some commands are Multi-Cycle, the pins may not be interchanged between devices on the same bus.
RESET_n	Input	Active Low Asynchronous Reset: Reset is active when RESET_n is LOW, and inactive when RESET_n is HIGH. RESET_n must be HIGH during normal operation. RESET_n is a CMOS rail to rail signal with DC high and low at 80% and 20% of V _{DDQ} .
DQ	Input / Output	Data Input/Output: Bi-directional data bus. If CRC is enabled via Mode register then CRC code is added at the end of Data Burst.
DQS_t, DQS_c, DQSU_t, DQSU_c, DQSL_t, DQSL_c	Input / Output	Data Strobe: output with read data, input with write data. Edge-aligned with read data, centered in write data. For the x16, DQSL corresponds to the data on DQL0-DQL7; DQSU corresponds to the data on DQU0-DQU7. The data strobe DQS_t, DQSL_t and DQSU_t are paired with differential signals DQS_c, DQSL_c, and DQSU_c, respectively, to provide differential pair signaling to the system during reads and writes. DDR5 SDRAM supports differential data strobe only and does not support single-ended.
TDQS_t, TDQS_c	Output	Termination Data Strobe: TDQS_t/TDQS_c is applicable for x8 DRAMs only. When enabled via MR5:OP[4]=1, the DRAM shall enable the same termination resistance function on TDQS_t/TDQS_c that is applied to DQS_t/DQS_c. When disabled via MR5:OP[4]=0, DM_n/TDQS_t shall provide the data mask function depending on MR5:OP[5]; TDQS_c is not used. x4/x16 DRAMs must disable the TDQS function via MR5:OP[4]=0.
ALERT_n	Input/Output	Alert: If a CRC error is detected, ALERT_n goes LOW for a time interval and goes back HIGH. During Connectivity Test mode, this pin works as an input. Optional use of this signal is dependent on the system. If the ALERT_n is not used, the ALERT_n pin must be pulled to VDDQ on the board.
TEN	Input	Connectivity Test Mode Enable: Required on x4, x8 & x16 devices. HIGH in this pin shall enable Connectivity Test Mode operation along with other pins. It is a CMOS rail to rail signal with AC high and low at 80% and 20% of V _{DDQ} . Using this signal or not is dependent on System. This pin may be DRAM internally pulled low through a weak pull-down resistor to VSS.
MIR	Input	Mirror: Used to inform SDRAM device that it is being configured for Mirrored mode vs. Standard mode. With the MIR pin connected (strapped) to VDDQ, the SDRAM internally swaps even numbered CA with the next higher odd number CA. Normally the MIR pin must be tied to VSS if no CA mirror is required. Mirror pair examples: CA2 with CA3 (not CA1) CA4 with CA5 (not CA3). Note that the CA[13] function is only relevant for certain densities (including stacking) of DRAM component. In the case that CA[13] is not used, its ball location, considering whether MIR is used or not, should be connected (Strapped) to VDDQ. No active signaling requirements defined.
CAI	Input	Command & Address Inversion: With the CAI pin connected (strapped) to VDDQ, DRAM internally inverts the logic level present on all the CA signals. Normally the CAI pin must be connected to VSS if no CA inversion is required. No active signaling requirements defined.
CA_ODT	Input	ODT for Command and Address. Apply Group A settings if the pin is connected (strapped) to VSS and apply Group B settings if the pin is connected (strapped) to V _{DDQ} . No active signalling requirements defined.
LBDQ	Output	Loopback Data Output: The output of this device on the Loopback Output Select defined in MR53:OP[4:0]. When Loopback is enabled, it is in driver mode using the default RON described in the Loopback Function section. When Loopback is disabled, the pin is either terminated or HiZ based on MR36:OP[2:0].

Table 3 — Pinout Description (cont'd)

Symbol	Type	Function
LBDQS	Output	Loopback Data Strobe: This is a single ended strobe with the Rising edge-aligned with Loopback data edge, falling edge aligned with data center. When Loopback is enabled, it is in driver mode using the default RON described in the Loopback Function section. When Loopback is disabled, the pin is either terminated or HiZ based on MR36:OP[2:0].
RFU	Input/Output	Reserved for future use
NC		No Connect: No internal electrical connection is present.
VDDQ	Supply	DQ Power Supply: 1.1 V
VDD	Supply	Power Supply: 1.1 V
VSS	Supply	Ground
VPP	Supply	DRAM Activating Power Supply: 1.8 V
ZQ, (ZQ1)	Reference	Reference Pin for ZQ calibration. This ball is tied to an external 240 ohm resistor(RZQ), which is tied to V_{SS} .

2.7 DDR5 SDRAM Addressing

Table 4 — 8 Gb Addressing Table

Configuration		2 Gb x4	1 Gb x8	512 Mb x16
Bank Address	BG Address	BG0~BG2	BG0~BG2	BG0~BG1
	Bank Address in a BG	BA0	BA0	BA0
	# BG / # Banks per BG / # Banks	8 / 2 / 16	8 / 2 / 16	4 / 2 / 8
Row Address		R0~R15	R0~R15	R0~R15
Column Address		C0~C10	C0~C9	C0~C9
Page size		1KB	1KB	2KB
Chip IDs / Maximum Stack Height		CID0~3 / 16H	CID0~3 / 16H	CID0~3 / 16H

Table 5 — 16 Gb Addressing Table

Configuration		4 Gb x4	2 Gb x8	1 Gb x16
Bank Address	BG Address	BG0~BG2	BG0~BG2	BG0~BG1
	Bank Address in a BG	BA0~BA1	BA0~BA1	BA0~BA1
	# BG / # Banks per BG / # Banks	8 / 4 / 32	8 / 4 / 32	4 / 4 / 16
Row Address		R0~R15	R0~R15	R0~R15
Column Address		C0~C10	C0~C9	C0~C9
Page size		1KB	1KB	2KB
Chip IDs / Maximum Stack Height		CID0~3 / 16H	CID0~3 / 16H	CID0~3 / 16H

Table 6 — 24 Gb Addressing Table

Configuration		6 Gb x4	3 Gb x8	1.5 Gb x16
Bank Address	BG Address	BG0~BG2	BG0~BG2	BG0~BG1
	Bank Address in a BG	BA0~BA1	BA0~BA1	BA0~BA1
	# BG / # Banks per BG / # Banks	8 / 4 / 32	8 / 4 / 32	4 / 4 / 16
Row Address		R0~R16	R0~R16	R0~R16
Column Address		C0~C10	C0~C9	C0~C9
Page size		1KB	1KB	2KB
Chip IDs / Maximum Stack Height		CID0~3 / 16H	CID0~3 / 16H	CID0~3 / 16H
NOTE 1	For non-binary memory densities, a quarter of the row address space is invalid. When the MSB address bit is "HIGH", the MSB-1 address bit shall be "LOW".			

Table 7 — 32 Gb Addressing Table

Configuration		8 Gb x4	4 Gb x8	2 Gb x16
Bank Address	BG Address	BG0~BG2	BG0~BG2	BG0~BG1
	Bank Address in a BG	BA0~BA1	BA0~BA1	BA0~BA1
	# BG / # Banks per BG / # Banks	8 / 4 / 32	8 / 4 / 32	4 / 4 / 16
Row Address		R0~R16	R0~R16	R0~R16
Column Address		C0~C10	C0~C9	C0~C9
Page size		1KB	1KB	2KB
Chip IDs / Maximum Stack Height		CID0~3 / 16H	CID0~3 / 16H	CID0~3 / 16H

2.7 DDR5 SDRAM Addressing (cont'd)

Table 8 — 64 Gb Addressing Table

Configuration		16 Gb x4	8 Gb x8	4 Gb x16
Bank Address	BG Address	BG0~BG2	BG0~BG2	BG0~BG1
	Bank Address in a BG	BA0~BA1	BA0~BA1	BA0~BA1
	# BG / # Banks per BG / # Banks	8 / 4 / 32	8 / 4 / 32	4 / 4 / 16
Row Address		R0~R17	R0~R17	R0~R17
Column Address		C0~C10	C0~C9	C0~C9
Page size		1KB	1KB	2KB
Chip IDs / Maximum Stack Height		CID0~2 / 8H	CID0~2 / 8H	CID0~2 / 8H

3 Functional Description

3.1 Simplified State Diagram

This Simplified State Diagram is intended to provide an overview of the possible state transitions and the commands to control them. In particular, situations involving more than one bank, the enabling or disabling of on-die termination, and some other events are not captured in full detail.

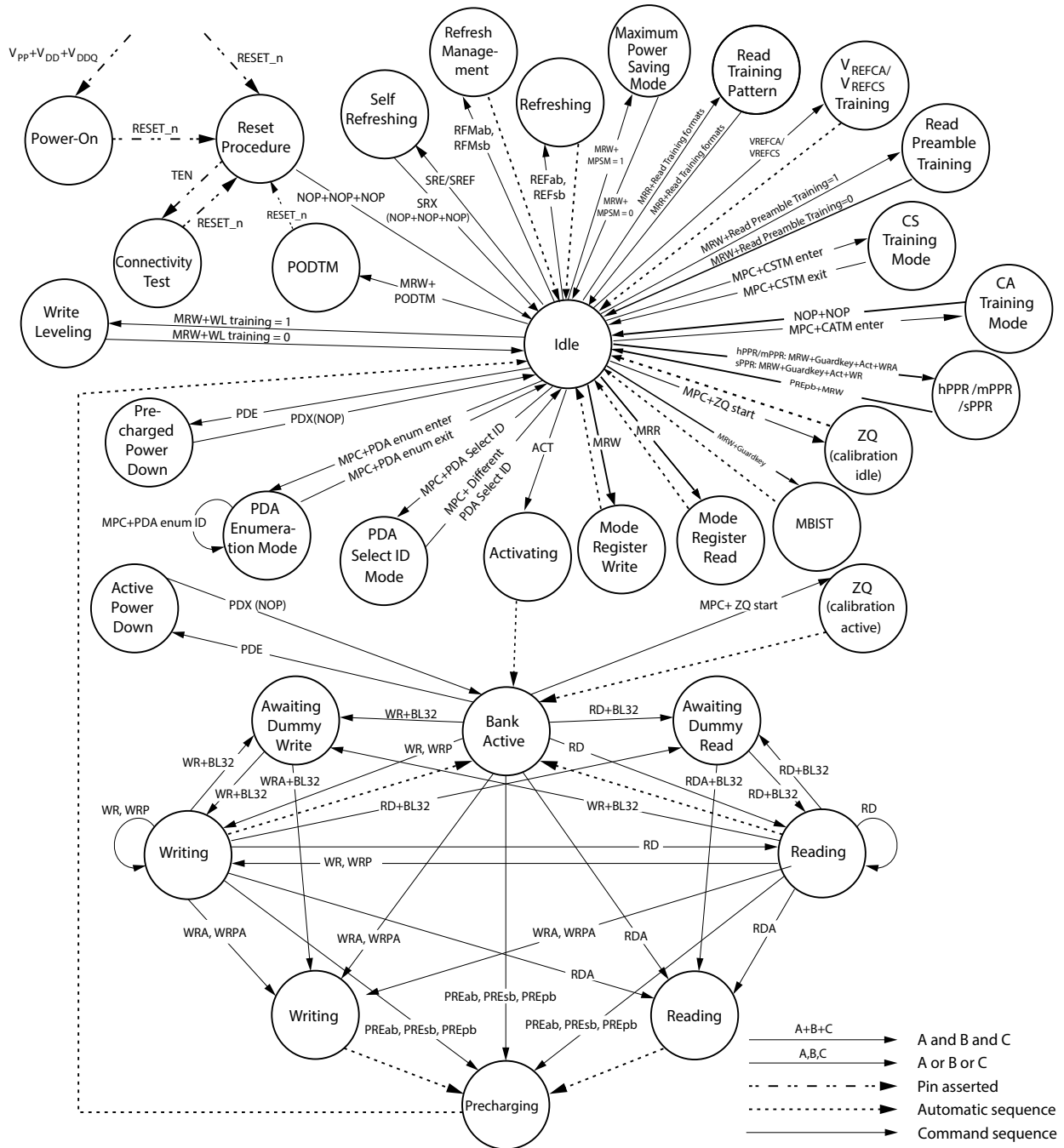


Figure 3 — Simplified State Diagram

3.2 Basic Functionality

The DDR5 SDRAM is a high-speed dynamic random-access memory. To ease transition from DDR4 to DDR5, the introductory density (8 Gb) shall be internally configured as 16-bank, 8 bank group with 2 banks for each bank group for x4/x8 and 8-bank, 4 bank group with 2 banks for each bankgroup for x16 DRAM. When the industry transitions to higher densities (≥ 16 Gb), it doubles the bank resources and internally be configured as 32-bank, 8 bank group with 4 banks for each bank group for x4/x8 and 16-bank, 4-bank group with 4 banks for each bankgroup for x16 DRAM.

The DDR5 SDRAM uses a 16n prefetch architecture to achieve high-speed operation. The 16n prefetch architecture is combined with an interface designed to transfer two data words per clock cycle at the I/O pins. A single read or write operation for the DDR5 SDRAM consists of a single 16n-bit wide, eight clock data transfer at the internal DRAM core and sixteen corresponding n-bit wide, one-half clock cycle data transfers at the I/O pins.

Read and write operation to the DDR5 SDRAM are burst oriented, start at a selected location, and continue for a burst length of sixteen or a 'chopped' burst of eight in a programmed sequence. Operation begins with the registration of an ACTIVATE Command, which is then followed by a Read or Write command. The address bits registered with the ACTIVATE Command are used to select the bank and row to be activated (i.e. in a 16Gb part, BG0-BG2 in a x4/8 and BG0-BG1 in x16 select the bankgroup; BA0-BA1 select the bank; R0-R17 select the row; refer to "DDR5 SDRAM Addressing" for specific requirements). The address bits registered with the Read or Write command are used to select the starting column location for the burst operation, determine if the auto precharge command is to be issued (CA10=L), and select BC8 on-the-fly (OTF), fixed BL16, fixed BL32 (optional), or BL32 OTF (optional) mode if enabled in the mode register.

Prior to normal operation, the DDR5 SDRAM must be powered up and initialized in a predefined manner.

The following sections provide detailed information covering device reset and initialization, register definition, command descriptions, and device operation.

3.3 RESET and Initialization Procedure

For power-up and reset initialization, in order to prevent DRAM from functioning improperly, default values for the following MR settings need to be defined.

Table 9 — MR Default Settings

Item	Mode Register	Default Setting	Description
Burst Length	MR0 OP[1:0]	00 _B	BL16
Read Latency	MR0 OP[6:2]	00010 _B	RL(CL) = 26 @3200
Write Latency	n/a	WL=RL-2 (CWL=CL-2)	Fixed based on RL (CL)
Write Recovery (tWR)	MR6 OP[3:0]	0000 _B	WR = 48nCK @3200 or 30ns
Read to Precharge Delay (tRTP)	MR6 OP[7:4]	0000 _B	tRTP=12nCK @3200 or 7.5ns
VrefDQ Value	MR10	0010 1101 _B	VREF(DQ) Range: 75% of V _{DDQ}
VrefCA Value	MR11	0010 1101 _B	VREF(CA) Range: 75% of V _{DDQ}
VrefCS Value	MR12	0010 1101 _B	VREF(CS) Range: 75% of V _{DDQ}
ECS Error Threshold Count (ETC)	MR15	011 _B	256
Post Package Repair	MR23 OP[1:0]	00 _B	hPPR and sPPR Disabled
CK ODT	MR32 OP[2:0]	CK ODT is based on strap value	Group A= RTT_OFF=000 _B Group B= 40 Ohms=111 _B
CS ODT	MR32 OP[5:3]	CS ODT is based on strap value	Group A= RTT_OFF=000 _B Group B= 40 Ohms=111 _B
CA ODT	MR33 OP[2:0]	CA ODT is based on strap value	Group A= RTT_OFF=000 _B Group B= 80 Ohms=100 _B
DQS_RTT_PARK	MR33 OP[5:3]	000 _B	RTT OFF
RTT_PARK	MR34 OP[2:0]	000 _B	RTT OFF
RTT_WR	MR34 OP[5:3]	001 _B	240 Ohm
RTT_NOM_WR	MR35 OP[2:0]	011 _B	80 Ohm
RTT_NOM_RD	MR35 OP[5:3]	011 _B	80 Ohm
RTT Loopback	MR36 OP[2:0]	000 _B	RTT OFF
RFM RAAIMT	MR58 OP[4:1]	vendor specific	vendor specific
RFM RAAMMT	MR58 OP[7:5]	vendor specific	vendor specific
RFM RAA Counter	MR59 OP[7:6]	vendor specific	vendor specific

3.3.1 Power-up Initialization Sequence

The following sequence shall be used to power up the DDR5 device. Unless specified otherwise, these steps are mandatory.

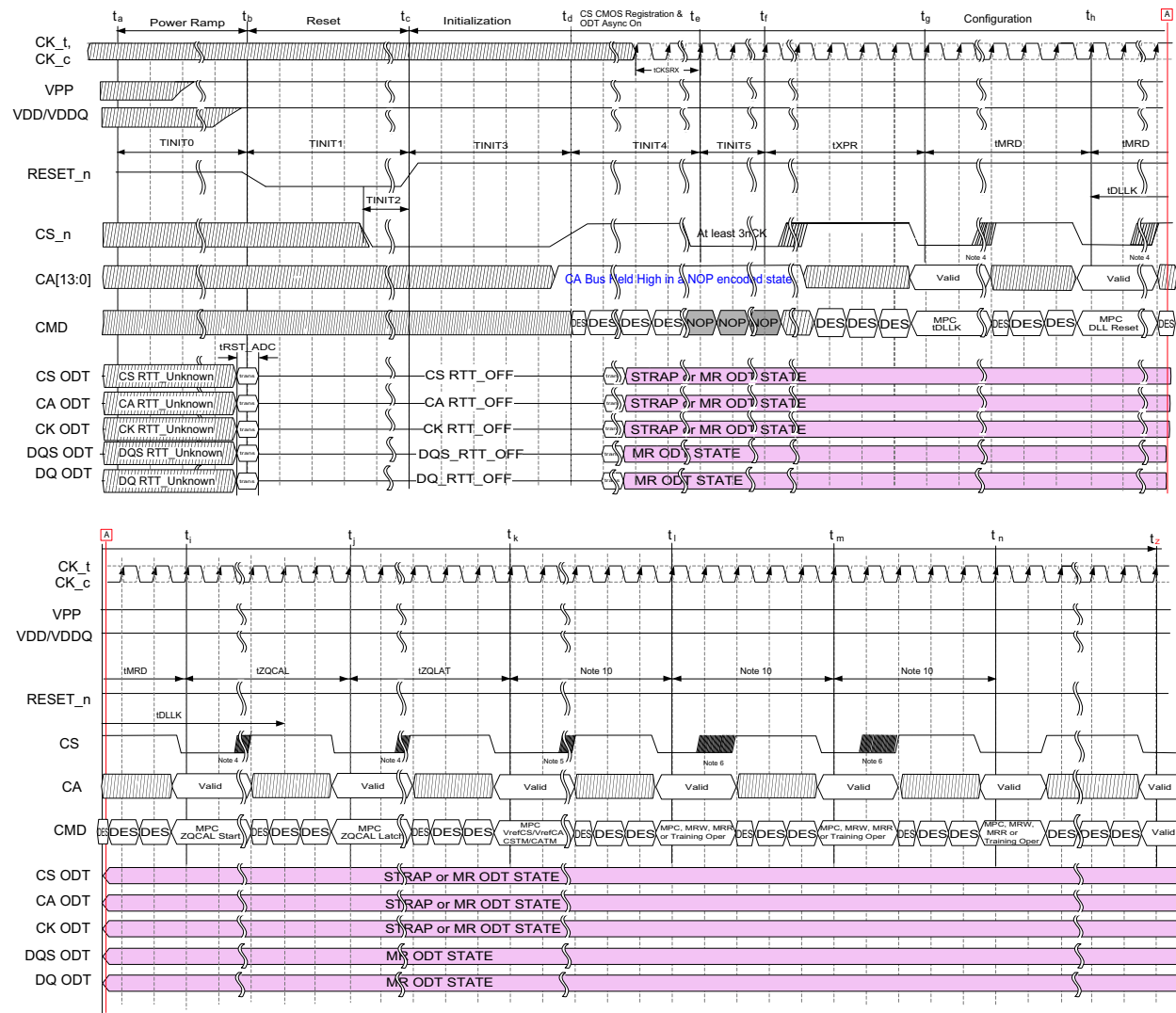
- While applying power (after Ta), RESET_n is recommended to be LOW ($\leq 0.2 \times V_{DDQ}$) and all other inputs may be undefined. The device outputs remain disabled while RESET_n is held LOW. Power supply voltage ramp requirements are provided in Table 10 VPP shall ramp at the same time or earlier than VDD.

Table 10 — Voltage Ramp Conditions

After	Application Conditions
Ta is reached	VPP shall be greater than VDD
NOTE 1 Ta is the point when any power supply first reaches 300 mV NOTE 2 Voltage ramp conditions in the table above apply between (Ta) and Power-off (controlled or uncontrolled). NOTE 3 Tb is the point at which all supply and reference voltages are within their defined ranges. NOTE 4 Power ramp duration (Tb-Ta) shall not exceed tINIT0.	

- Following the completion of the voltage ramp (Tb), RESET_n shall be maintained LOW. DQ, DQS_t, DQS_c, voltage levels shall be between VSS and VDDQ to avoid latch-up. CS_n, CK_t, CK_c and CA input levels shall be between VSS and VDDQ to avoid latch-up.
- Beginning at Tb, RESET_n shall be maintained LOW for a minimum of tINIT1 (Tb to Tc), after which RESET_n may be de-asserted to HIGH (Tc). At least tINIT2 before RESET_n de-assertion, CS_n is required to be set LOW. All other input signals are "Don't Care". The DRAM shall support the ability for RESET_n to be held indefinitely.

3.3.1 Power-up Initialization Sequence (cont'd)



NOTES:

- From time point (Ta) until (Tb), RTT_OFF may not be guaranteed due to the limitation for the DRAM to control the RTT status during the Power Ramp period.
- From time point (Td) until (Te), the command/address (CA) bus shall be held high in a NOP encoded state.
- From time point (Te) until (Tf), NOP commands shall be applied on the command/address (CA) bus.
- From time point (Tf) until (Tk), DES commands must be applied between legal commands (MRR, MRW, MPC, VREFCS & VREFCA). MRR and MRW command, while legal, may not execute properly until CS and CA training routines are completed.
- Prior to ZQcal completion at (Tk), MPC commands shall be Multi-Cycle as described in the MPC command Timings section.
- From time point (Tk) until (Tl) all MPC, VREFCS and VREFCA commands prior to CS and CA training successfully completing, shall be Multi-Cycle (MR2:OP[4]=0) as described in the MPC, VREFCS, and VREFCA timing sections.
- At time point (Tl), with successful completion of CS and CA training, an MRW command setting MR2:OP[4] = 1 is recommended and allows for MPC, VREFCS, VREFCA commands to be single-cycle, improving training duration.
- From time point (Tk) until (Tz), DES commands shall be applied between legal commands (MRR, MRW, MPC, VREFCS & VREFCA).
- Starting at Tl, MRW Commands shall be issued to all Mode Registers that require defined settings.
- Default ODT tolerances are wider prior to ZQ calibration.
- All MPC/MRW/MRR to MPC/MRW/MRR commands shall meet the timing restrictions required.
- Between (Tk to Tz), the SRX/NOP Clock-Sync feature can be enabled through an MRW command by programming MR13:OP[5]=1. When it is enabled, after CSTM/CATM prior to DCA training, host shall issue an SRE/SRX pair to ensure that the Clock-Sync is in effect during DCA training. It is also applied for CS Gear-down Initialization sequence (for example, after Power-Up Initialization, CSTM, MPC (CS Gear-down enable), Sync Pulse, CATM, MRW (SRX/NOP Clock-Sync enable), then host shall issue an SRE/SRX pair). Host (including RCD and Clock Driver) is responsible to ensure the same phases of the system clock all the time during DCA training.

Figure 4 — RESET_n and Initialization Sequence at Power-on Ramping

- After RESET_n is de-asserted (Tc), wait at least tINIT3 before driving CS_n high.

3.3.1 Power-up Initialization Sequence (cont'd)

5. After setting CS_n high (Td), wait a minimum of tINIT4 to allow the DRAM CMOS based receiver to register the exit and allow the CS_n, CK, CA, DQ and DQS ODT to go to the defined strap or MRS state (Te). Clock (CK_t, CK_c) is required to be started and stabilized for tCKSRX before exit of tINIT4 (Te). Upon the completion of Te, all ODT states (CA, CS_n, CK, DQ and DQS ODT) should be valid and the DRAM's CS_n receiver should no longer be in its CMOS based mode. ODT termination states will be uncalibrated until completion of ZQCal at (Tj)
6. Upon (Te), NOP commands shall be issued for a minimum of tINIT5 to conclude exit of initialization process and start tXPR timer at (Tf). The system shall wait at least tXPR before issuing any legal configuration commands (Tg). During early configuration steps (Tg to Tk), only MRR, MRW, MPC, VREFCS and VREFCA commands are legal. MRR and MRW command, while legal, may not execute properly until CS and CA training routines are completed.
7. Between (Tg to Tj), the following initial configuration modes shall be completed prior to other training modes:
 - MPC for setting MR13 (tCCD_L*/tDLLK) shall be issued before the MPC command to reset the DLL.
 - MPC to execute DLL RESET shall be issued before ZQCal Start
 - MPC to execute ZQCal Start followed by ZQCal Latch shall be issued before any other training modes such as VrefCS, VrefCA, CS and CA Training.
8. Between (Tk to Tz), any number of legal configuration commands are allowed. Many training based commands are optional and may be done at the system architect's discretion and may vary depending on the systems, though proper setting of certain registers, such as those related to Write Leveling Training, is required. The host shall follow Multi-Cycle MPC timing requirements in cases where the alignment between CS_n, CA, and CK is unknown, (ie., prior to successfully completing VREFCS, VREFCA, CS, and CA training commands or host provided successful training solutions). MRW and MRR commands can be used once the alignment between CS_n, CA and CK are known. Single-cycle CS_n assertion are allowed after setting MR2:OP[4]=1.
9. Between (Tk to Tz), the SRX/NOP Clock-Sync feature can be enabled through an MRW command by programming MR13:OP[5]=1. When it is enabled, after CSTM/CATM prior to DCA training, host shall issue an SRE/SRX pair to ensure that the Clock-Sync is in effect during DCA training. It is also applied for CS Geardown Init sequence (for example, after PowerUp Initialization, CSTM, MPC (CS Geardown enable), Sync Pulse, CATM, MRW (SRX/NOP Clock-Sync enable), then host shall issue an SRE/SRX pair). Host (including RCD and Clock Driver) is responsible to ensure the same phases of the system clock all the time during DCA training to make the DCA training effective with the consistent phases of system and DRAM internal clocks.
10. After (Tz), and the completion of any training or calibration timing parameters (i.e. tZQLAT is satisfied), the DDR5 device is ready for normal operation and is able to accept any valid command. Any additional mode registers that have not previously been set up for normal operation should be written at this time. If host use writeback suppression mode, it should be set after the initial write process to prevent aliasing to 2-bit errors.
11. After all mode registers have been programmed for normal operation, optional MBIST mode can be entered by writing MR23:OP[4] to HIGH, followed by subsequent MR24 PPR guard keys, then DRAM will drive ALERT_n to LOW for a maximum of tSELFTEST time. DRAM will drive ALERT_n to HIGH to indicate that this operation is completed. After ALERT_n is driven high, the DRAM is immediately ready to receive valid commands. The MBIST/mPPR transparency status shall subsequently be checked in MR22:OP[2:0] in order to determine whether mPPR should be performed. Please refer to Ch. 4.30 MBIST/mPPR for more detailed operation procedures.

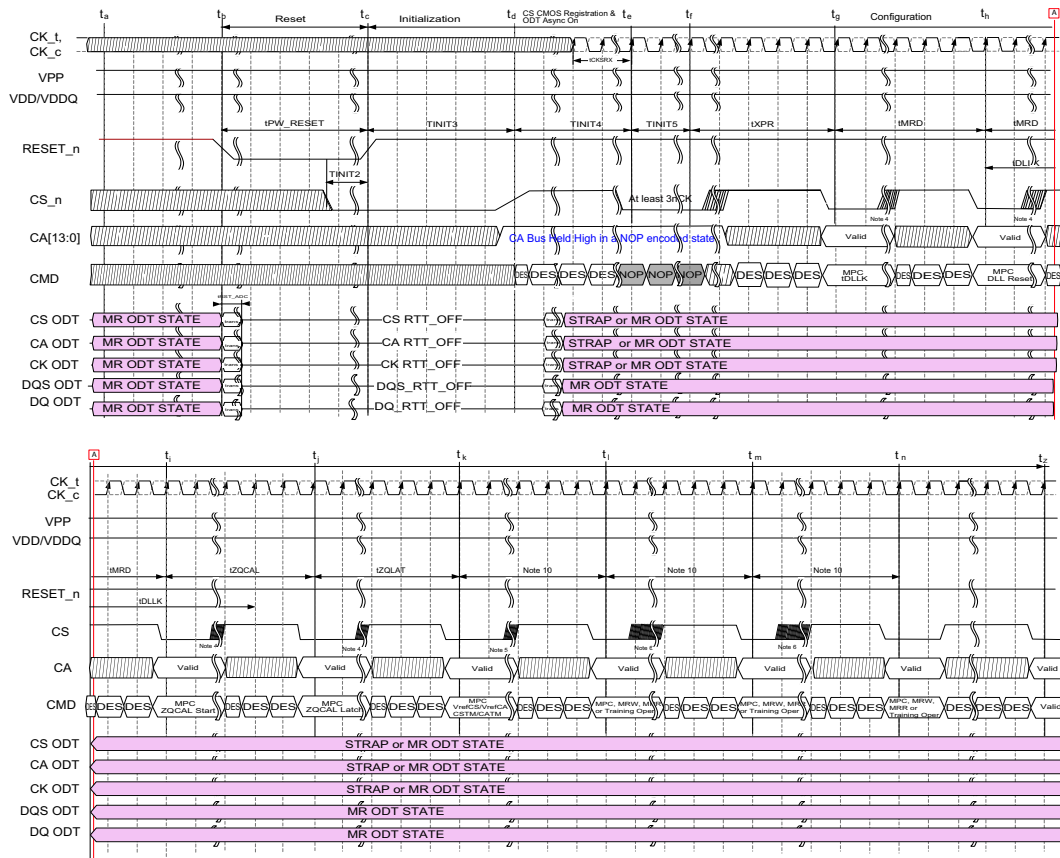
Table 11 — Initialization Timing Parameters

Parameter	Symbol	Value		Units	Note(s)
		MIN	MAX		
Maximum voltage-ramp time	tINIT0	-	20	ms	
Minimum RESET_n LOW time after completion of voltage ramp	tINIT1	200		μs	
Minimum CS_n LOW time before RESET_n HIGH	tINIT2	10	-	ns	
Minimum CS_n LOW time after RESET_n HIGH	tINIT3	4	-	ms	
Minimum time for DRAM to register EXIT on CS_n with CMOS.	tINIT4	2	-	μs	
Minimum cycles required after CS_n HIGH	tINIT5	3	-	nCK	1
Minimum time from Exit Reset to first valid Configuration Command	tXPR	tXS	-	ns	
Minimum stable clock time	tCKSRX	SEE Self Refresh Timing Table			
NOTE 1 Min number of NOP commands issued after CS_n High (tINIT4).					

3.3.2 Reset Initialization with Stable Power

The following sequence is required for RESET at no power interruption initialization as shown in Figure 5.

1. Assert RESET_n below $0.2 \times VDDQ$ anytime when reset is needed. RESET_n needs to be maintained for a minimum of tPW_RESET. CS_n shall be pulled LOW at least tINIT2 before de-asserting RESET_n.
2. Repeat steps 4 to 11 in Section 3.3.1



NOTES:

1. From time point (Td) until (Te), the command/address (CA) bus shall be held high in a NOP encoded state.
2. From time point (Te) until (Tf), NOP commands shall be applied on the command/address (CA) bus.
3. From time point (Tf) until (Tk), DES commands shall be applied between legal commands (MRR, MRW, MPC, VREFCS & VREFCA). MRR and MRW command while legal, may not execute properly until CS and CA training routines are completed.
4. Prior to ZQcal completion at (Tk), MPC commands shall be Multi-Cycle as described in the MPC command Timings section.
5. From time point (Tk) until (Tl) all MPC, VREFCS and VREFCA commands prior to CS and CA training successfully completing, shall be Multi-Cycle (MR2:OP[4]=0) as described in the MPC, VREFCS, and VREFCA timing sections.
6. At time point (Tl), with successful completion of CS and CA training, an MRW command setting MR2:OP[4] = 1 is recommended and allows for MPC, VREFCS, VREFCA commands to be single-cycle, improving training duration.
7. From time point (Tk) until (Tz), DES commands shall be applied between legal commands (MRR, MRW, MPC, VREFCS & VREFCA).
8. Starting at Tl, MRW Commands shall be issued to all Mode Registers that require defined settings.
9. Default ODT tolerances are wider prior to ZQ calibration.
10. All MPC/MRW/MRR to MPC/MRW/MRR commands shall meet the timing restrictions required.
11. MRR is legal after DLL_reset.
12. Between (Tk to Tz), the SRX/NOP Clock-Sync feature can be enabled through an MRW command by programming MR13:OP[5]=1. When it is enabled, after CSTM/CATM prior to DCA training, host shall issue an SRE/SRX pair to ensure that the Clock-Sync is in effect during DCA training. It is also applied for CS Geardown Initialization sequence (for example, after Power-Up Initialization, CSTM, MPC (CS Geardown enable), Sync Pulse, CATM, MRW (SRX/NOP Clock-Sync enable), then host shall issue an SRE/SRX pair). Host (including RCD and Clock Driver) is responsible to ensure the same phases of the system clock all the time during DCA training.

Figure 5 — Reset Procedure at Stable Power

3.3.2 Reset Initialization with Stable Power (cont'd)

Table 12 — Reset Timing Parameters

Parameter	Symbol	Value		Units	Note(s)
		Min	Max		
Minimum RESET_n low time for Reset Initialization with stable power	tPW_RESET	1	-	μS	
Maximum time after RESET_n assertion to ODT off	tRST_ADC	-	50	nS	

3.3.3 Input Voltage Power-up and Power-Down Sequence

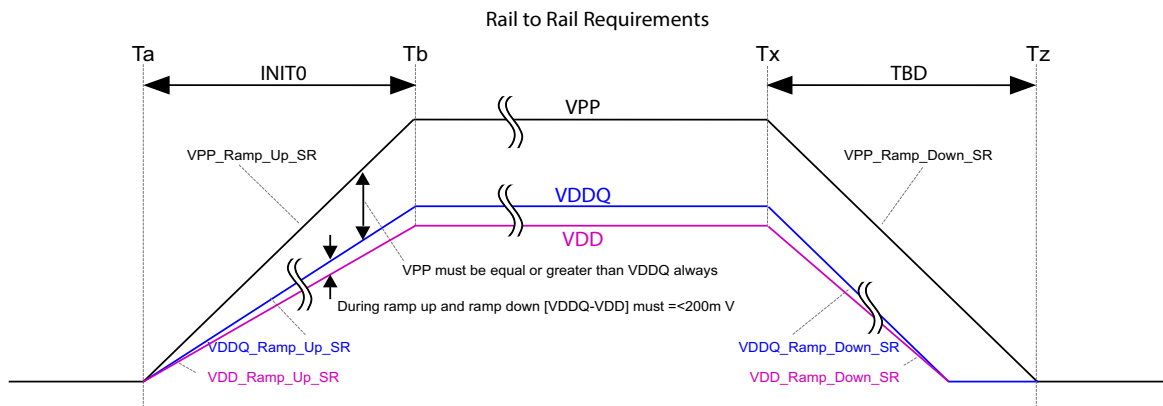


Table 13 — Input Voltage Slew Rates and Power Ramp Down Time

Description	Symbol	Min	Max	Units	Notes
VPP Rail	VPP_Ramp_Up_SR	0.2	5	V/ms	1,2,3,4,5
	VPP_Ramp_Down_SR	0.1	4.5	V/ms	
VDD Rail	VDD_Ramp_Up_SR	0.1	4.5	V/ms	
	VDD_Ramp_Down_SR	0.1	4.5	V/ms	
VDDQ Rail	VDDQ_Ramp_Up_SR	0.1	4.5	V/ms	
	VDDQ_Ramp_Down_SR	0.1	4.5	V/ms	
Power Ramp Down Time	tPRD	-	19.08	ms	
NOTE 1 Both VDD and VPP supply measurements made between 10% and 90% nominal voltage					
NOTE 2 1Mhz bandwidth limited measurement					
NOTE 3 VPP shall be equal to or greater than VDD/VDDQ at all times.					
NOTE 4 During ramp up and ramp down [VDDQ-VDD] must be equal or less than 200 mV.					
NOTE 5 After tINIT0, all supplies shall be within their specified tolerance, as defined in the DC Operating Tables.					

3.4 Mode Register Definition

3.4.1 Mode Register Read (MRR)

The Mode Register Read (MRR) command is used to read configuration and status data from the DDR5-SDRAM registers. The MRR command is initiated with CS_n and CA[13:0] in the proper state as defined by the Command Truth Table. The mode register address operands (MA[7:0]) allow the user to select one of 256 registers. The mode register contents are available on the second 8 UI's of the burst and are repeated across all DQ's after the CL following the MRR command. To avoid a potentially worst-case pattern, every odd DQ bit (represented with !) shall have its contents inverted. Data in the burst (BL0-7) shall be either "0" or "1", with "1" indicating that the content of the later UI's (BL8-15) are inverted.

MRR commands require all banks to be idle on the DRAM. For 3DS, MRR commands require all banks to be idle on all logical ranks. The 3DS base die (CID[3:0]=0x0) is the targeted logical rank for most Mode Register Reads, however the CID setting of MR14:OP[3:0] designates the targeted logical rank for MRR commands to MR14-MR20, MR22 and MR54-MR57, and still requires all banks to be idle not only on the targeted logical rank but also on all other logical ranks.

DQS is toggled for the duration of the MRR burst. The MRR has a command burst length 16 regardless of the MR0 setting, the training mode or the mode register address. MRR termination control and ODT timings are the same as for the Read command. The MRR operation must not be interrupted. Non-Target ODT encoding is available for MRR, just like a normal Read. MRR NT ODT termination control and ODT timings are the same as for the Read NT command.

In the case that CRC is enabled, MRR's output will come with BL18 (BL16 plus 2 CRC-bit), but the host has the option to consider the 17th and 18th bits "don't care" for MRR handling. Regardless on if the host uses the 17th and 18th bits, while CRC is enabled, the strobe needs to toggle for BL18.

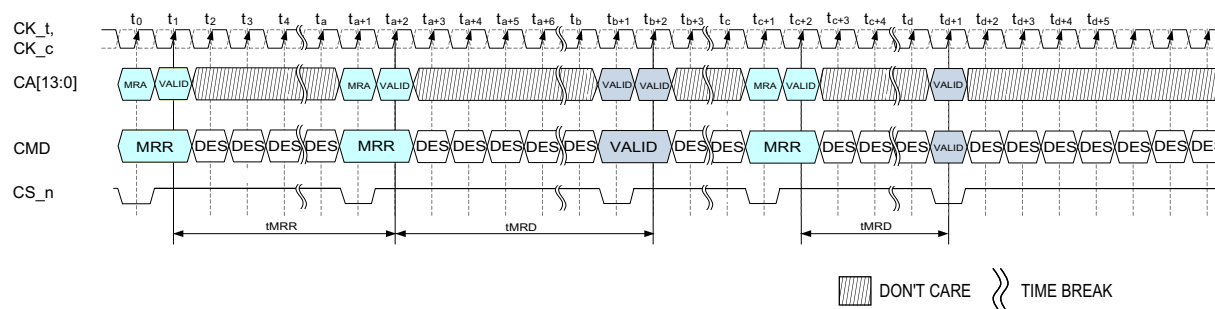


Figure 7 — Mode Register Read Timing

Table 14 — DQ Output Mapping for x4 Device

BL	0-7	8	9	10	11	12	13	14	15
DQ0	0	OP0	OP1	OP2	OP3	OP4	OP5	OP6	OP7
DQ1	1	!OP0	!OP1	!OP2	!OP3	!OP4	!OP5	!OP6	!OP7
DQ2	0	OP0	OP1	OP2	OP3	OP4	OP5	OP6	OP7
DQ3	1	!OP0	!OP1	!OP2	!OP3	!OP4	!OP5	!OP6	!OP7

NOTE 1 The read pre-amble and post-amble of MRR are same as normal read.

3.4.1 Mode Register Read (MRR) (cont'd)

Table 15 — DQ Output Mapping for x8 Device

BL	0-7	8	9	10	11	12	13	14	15
DQ0	0	OP0	OP1	OP2	OP3	OP4	OP5	OP6	OP7
DQ1	1	!OP0	!OP1	!OP2	!OP3	!OP4	!OP5	!OP6	!OP7
DQ2	0	OP0	OP1	OP2	OP3	OP4	OP5	OP6	OP7
DQ3	1	!OP0	!OP1	!OP2	!OP3	!OP4	!OP5	!OP6	!OP7
DQ4	0	OP0	OP1	OP2	OP3	OP4	OP5	OP6	OP7
DQ5	1	!OP0	!OP1	!OP2	!OP3	!OP4	!OP5	!OP6	!OP7
DQ6	0	OP0	OP1	OP2	OP3	OP4	OP5	OP6	OP7
DQ7	1	!OP0	!OP1	!OP2	!OP3	!OP4	!OP5	!OP6	!OP7

NOTE 1 The read pre-amble and post-amble of MRR are same as normal read.

Table 16 — DQ Output Mapping for x16 Device (OSC Count - MR46 and MR47 only)

BL	0-7	8	9	10	11	12	13	14	15
DQ0	0	OP0	OP1	OP2	OP3	OP4	OP5	OP6	OP7
DQ1	1	!OP0	!OP1	!OP2	!OP3	!OP4	!OP5	!OP6	!OP7
DQ2	0	OP0	OP1	OP2	OP3	OP4	OP5	OP6	OP7
DQ3	1	!OP0	!OP1	!OP2	!OP3	!OP4	!OP5	!OP6	!OP7
DQ4	0	OP0	OP1	OP2	OP3	OP4	OP5	OP6	OP7
DQ5	1	!OP0	!OP1	!OP2	!OP3	!OP4	!OP5	!OP6	!OP7
DQ6	0	OP0	OP1	OP2	OP3	OP4	OP5	OP6	OP7
DQ7	1	!OP0	!OP1	!OP2	!OP3	!OP4	!OP5	!OP6	!OP7
DQ8	0	OP0	OP1	OP2	OP3	OP4	OP5	OP6	OP7
DQ9	1	!OP0	!OP1	!OP2	!OP3	!OP4	!OP5	!OP6	!OP7
DQ10	0	OP0	OP1	OP2	OP3	OP4	OP5	OP6	OP7
DQ11	1	!OP0	!OP1	!OP2	!OP3	!OP4	!OP5	!OP6	!OP7
DQ12	0	OP0	OP1	OP2	OP3	OP4	OP5	OP6	OP7
DQ13	1	!OP0	!OP1	!OP2	!OP3	!OP4	!OP5	!OP6	!OP7
DQ14	0	OP0	OP1	OP2	OP3	OP4	OP5	OP6	OP7
DQ15	1	!OP0	!OP1	!OP2	!OP3	!OP4	!OP5	!OP6	!OP7

NOTE 1 The read pre-amble and post-amble of MRR are same as normal read.

NOTE 2 Output map excludes per bit DFE, DCA and VrefDQ mode registers (MR103 through MR255)

Table 17 — DQ Output Mapping for x16 Device (DFE Registers Excluded)

NOTE 1	The read pre-amble and post-amble of MRR are same as normal read.
NOTE 2	Output of mode register data is only duplicated and inverted across the first 8 bits of a x16 device.
NOTE 3	Output map excludes per bit DFE, DCA and VrefDQ mode registers (MR103 through MR255)

Table 18 — DQ Output Mapping for x16 Device (DFE Lower Byte - DQ[0:7], DML)

NOTE 1	The read pre-amble and post-amble of MRR are same as normal read.
NOTE 2	Output of mode register data is only duplicated and inverted across the first 8 bits of a x16 device when reading from a DFE register associated with a lower byte DQ or DML.
NOTE 3	Output map is ONLY for per bit DFE, DCA and VrefDQ mode registers (MR103 through MR255)

3.4.1 Mode Register Read (MRR) (cont'd)

Table 19 — DQ Output Mapping for x16 Device (DFE Upper Byte - DQ[15:8], DMU)

BL	0-7	8	9	10	11	12	13	14	15
DQ0	Don't Care								
DQ1									
DQ2									
DQ3									
DQ4									
DQ5									
DQ6									
DQ7									
DQ8	0	OP0	OP1	OP2	OP3	OP4	OP5	OP6	OP7
DQ9	1	!OP0	!OP1	!OP2	!OP3	!OP4	!OP5	!OP6	!OP7
DQ10	0	OP0	OP1	OP2	OP3	OP4	OP5	OP6	OP7
DQ11	1	!OP0	!OP1	!OP2	!OP3	!OP4	!OP5	!OP6	!OP7
DQ12	0	OP0	OP1	OP2	OP3	OP4	OP5	OP6	OP7
DQ13	1	!OP0	!OP1	!OP2	!OP3	!OP4	!OP5	!OP6	!OP7
DQ14	0	OP0	OP1	OP2	OP3	OP4	OP5	OP6	OP7
DQ15	1	!OP0	!OP1	!OP2	!OP3	!OP4	!OP5	!OP6	!OP7

NOTE 1 The read pre-amble and post-amble of MRR are same as normal read.

NOTE 2 Output of mode register data is only duplicated and inverted across the last 8 bits of a x16 device when reading from a DFE register associated with an upper byte DQ or DMU.

NOTE 3 Output map is ONLY for per bit DFE, DCA and VrefDQ mode registers (MR103 through MR255)

3.4.2 Mode Register WRITE (MRW)

The Mode Register Write (MRW) command is used to write configuration data to the mode registers. The MRW command is initiated with CS_n and CA[13:0] in the proper state as defined by the Command Truth Table. The mode register address and the data written to the mode registers is contained in CA[13:0] according to the Command Truth Table. The MRW command period is defined by tMRW. Mode Register Writes to read-only registers have no impact on the functionality of the device.

MRW commands require all banks to be idle on the DRAM. For 3DS, MRW commands are broadcast across all logical ranks, requiring all banks to be idle on all logical ranks. For 3DS, MBIST Enable (MR23:OP[4]=1) is only enabled on the target logical rank designated by CID[3:0] and programmed by MRW via MR14:OP[3:0], but still requires all banks to be idle on all logical ranks.

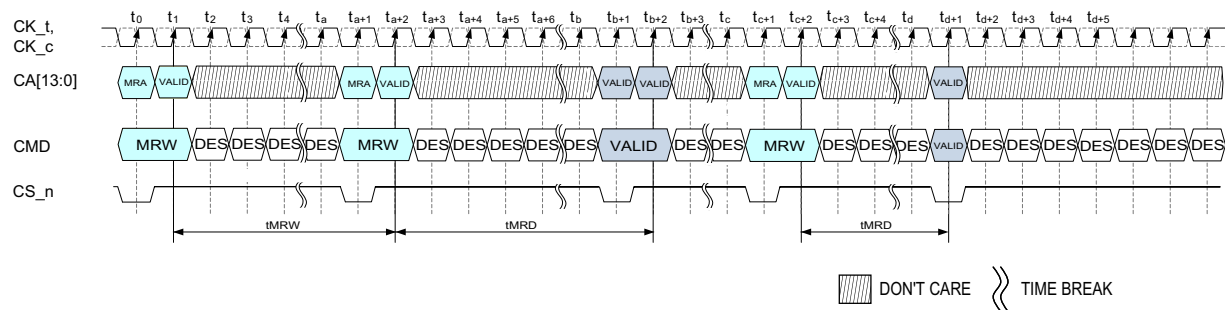


Figure 8 — Mode Register Write Timing

3.4.3 DFE Mode Register Write Update Timing

This Mode Register update timing parameter applies for MR112 (MA[7:0]=70H) thru MR248 (MA[7:0]=F8H) - Mode Registers for DFE including DFE Gain Bias, DFE Tap-1, DFE Tap-2, DFE Tap-3, DFE Tap-4 mode registers

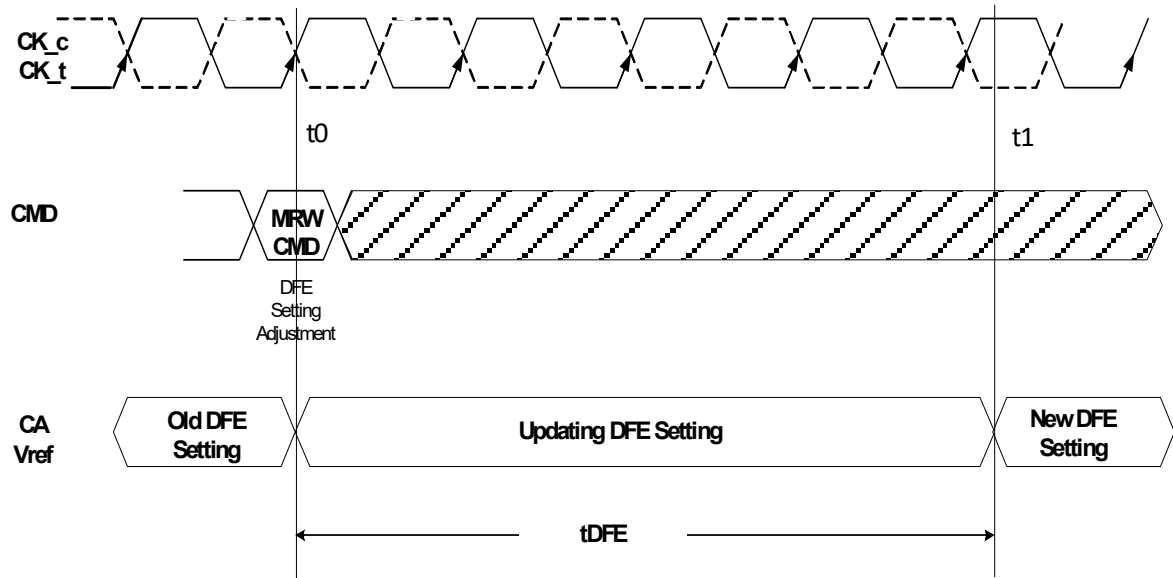


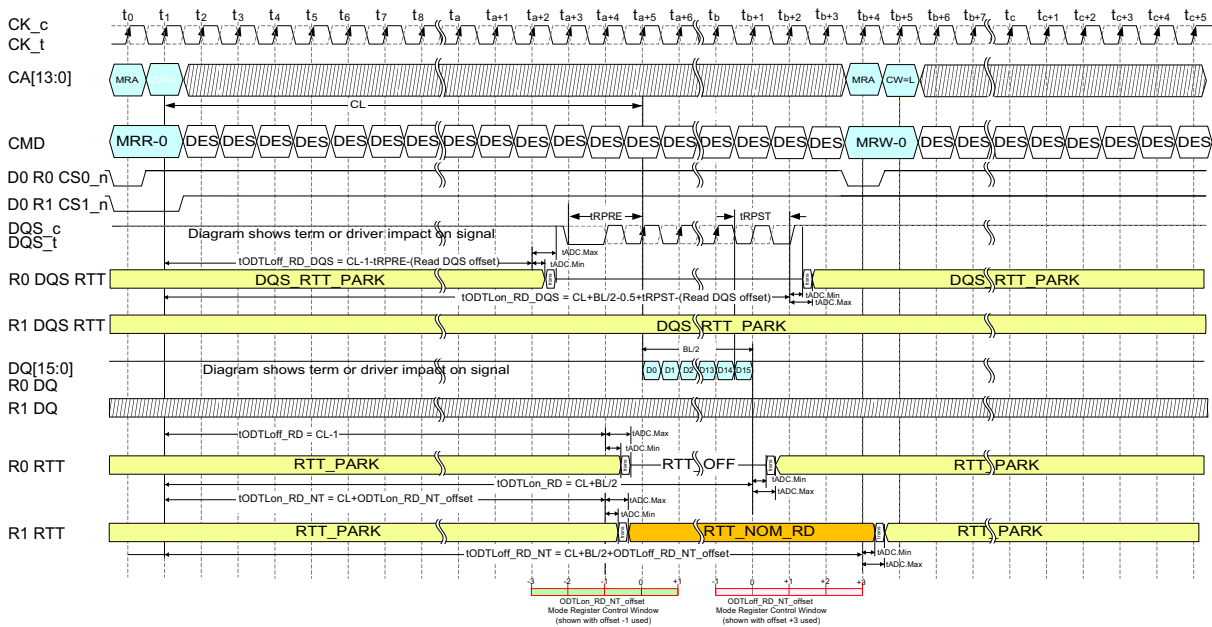
Figure 9 — DFE Update Setting

3.4.4 Mode Register Truth Tables and Timing Constraints

Table 20 — Mode Register Read/Write AC Timing

Parameter	Symbol	Min/Max	Value	Unit	Note
Mode Register Read command period	tMRR	Min	max(14ns, 16nCK)	nCK	1
Mode Register Read Pattern to Mode Register Read Pattern Command spacing	tMRR_p	Min	8	nCK	
Mode Register Write command period	tMRW	Min	max(5ns, 8nCK)	nCK	1
Mode Register Set command delay	tMRD	Min	max(14ns, 16nCK)	nCK	
DFE Mode Register Write Update Delay Time	tDFE	Min	80	ns	2

NOTE 1 MRR and MRW commands require all banks idle.
 NOTE 2 This parameter applies only to MRW's to DFE registers and is defined as the settling time before a new DFE setting is active.



NOTES:

- Example details ODTLoff_RD_Offset configured for -1, ODTLon_RD_Offset configured for 0, ODTLon_RD_NT_offset configured for -1, ODTLoff_RD_NT_offset configured for +3. Read DQS Offset programmed at 0.
- Timing constraints for 1-cycle MPC/VrefCS/VrefCA commands are affected by ODTLoff_RD_NT_Offset the same as MRW in the diagram.
- The tODTLoff_RD_NT minimum timing constraint begins when the MRR command is registered in the second cycle of the command, and ends on the first cycle of the MRW command. The MRW command will not be registered until the second cycle of the command, however the minimum timing spec shall be maintained as illustrated.

Figure 10 — Example MRR to MRW Timing Diagram for Same Physical Rank

3.4.4 Mode Register Truth Tables and Timing Constraints (cont'd)**Table 21 — Truth Table for Mode Register Read (MRR) and Mode Register Write (MRW)**

Current State	Command	Intermediate State	Next State
SDRAM		SDRAM	SDRAM
All Banks Idle	MRR	Mode Register Reading (All Banks Idle)	All Banks Idle
	MRW	Mode Register Writing (All Banks Idle)	All Banks Idle
NOTE 1 For 3DS, both MRR and MRW commands require all banks to be idle on all logical ranks			

Table 22 — MRR/MRW Timing Constraints: DQ ODT is Disable

From Command	To Command	Minimum Delay between “From Command” and “To Command”	Unit	Note
MRR	MRR	tMRR/tMRR_p	-	2,4
	MRW	CL+BL/2+max[1,ODTLoff_RD_NT_Offset]	tCK	2,3
	MPC	CL+BL/2+max[1,ODTLoff_RD_NT_Offset]	tCK	2,3
	VrefCA/VrefCS	CL+BL/2+max[1,ODTLoff_RD_NT_Offset]	tCK	2,3
	Any other valid command	tMRD	-	1,2
MRW	MRW	tMRW	-	
	Any other valid command	tMRD	-	
WRA	MRR	CWL + BL/2 + tWR + tRP	-	
	MRW	CWL + BL/2 + tWR + tRP	-	
RDA	MRR	tRTP + tRP		
	MRW	tRTP + tRP		
PRE	MRR	tRP	-	
	MRW	tRP	-	
REF	MRR	tRFC	-	
	MRW	tRFC	-	

NOTE 1 All data should be completed before entry into self refresh or power down.

NOTE 2 MRR can refer to both Target ODT MRR and Non-Target ODT MRR

NOTE 3 Minimum delay is 1 clock after the burst or until the ODT offset requirements for DQS/DQ are met.

NOTE 4 During Read Training, MRR to MRR command spacing can be tMRR_p.

3.4.4 Mode Register Truth Tables and Timing Constraints (cont'd)

Table 23 — MRR/MRW Timing Constraints: DQ ODT is Enable

From Command	To Command	Minimum Delay between “From Command” and “To Command”	Unit	Note
MRR	MRR	Same as ODT Disable Case	-	
	MRW		-	
	MPC		-	
	VrefCA/VrefCS		-	
	Any other valid command		-	
			-	
MRW	MRW	Same as ODT Disable Case	-	
	Any other valid command		-	
			-	
RDA	MRR	Same as ODT Disable Case	-	
	MRW		-	
WRA	MRR	Same as ODT Disable Case	-	
	MRW		-	
PRE	MRR	Same as ODT Disable Case	-	
	MRW		-	
REF	MRR	Same as ODT Disable Case	-	
	MRW		-	

3.5 Mode Registers

With DDR5, the utilization and programming method shall change from the traditional addressing scheme found in DDR3 and DDR4, and shall move to the method used by LPDDR, where the Mode Register Addresses (MRA) and Payload placed in Op Codes (OP) are all packeted in the command bus encoding method. Please refer to the Command Truth Table 30 for Mode Register Read (MRR) and Mode Register Write (MRW) command protocol.

For DDR5, the SDRAM shall support up to 8 MRA's, each with a byte-wide payload. Allowing for up to 256 byte-wide registers.

3.5.1 Mode Register Assignment and Definition in DDR5 SDRAM

Table 24 shows the mode registers for DDR5 SDRAM. Each bit in a register byte (MR#) is denoted as "R" if it can be read but not written, "W" if it can be written but reads shall always produce a ZERO for those specific bits, and "R/W" if it can be read and written. Additionally, a DRAM read-only bit combined with a Host write-only bit is denoted as a "SR/W" bit. This bit allows the DRAM to return a defined status during a read of that bit (SR = Status Read), independent of what the Host may have written to the bit.

A defined register byte (MR#), is any MR# that has at least one of the bits defined.

When the entire MR# is marked RFU, then it is considered undefined and all the bits from the DRAM shall be don't care for reads or writes. These undefined mode registers (completely empty bytes, not individual bits of an MR) may not be supported in the DRAM. When a defined register byte (MR#) contains an "RFU" bit, the host must write a ZERO for those specific bits and the DRAM does not guarantee any operation of those specific RFU bits. When the host issues an MRR to a defined register (MR#) that contains RFU bits in it, those specific bits shall always produce a ZERO.

For cases in which a mode register is specific to a particular device configuration (x16, x8, x4) and/or density (32Gb, 16Gb, 8Gb), the following rules shall be applied:

- When the DRAM is configured as a x4/x8, an entire MR# used only for a x16 shall be considered RFU. These bits are don't care for reads and writes, and they may be unsupported.
- When a bit field within a register is used by a different configuration or density than a given DRAM, the host may write/read programmed values to these fields, but DRAM operation will not be affected.

A Mode Register Read command is used to read a mode register. A Mode Register Write command is used to write a mode register.

Table 24 — Mode Register Assignment in DDR5 SDRAM

MR#	OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
0	RFU	CAS Latency (RL)					Burst Length	
1	PDA Select ID				PDA Enumerate ID			
2	Internal Write Timing	Reserved	Device 15 MPSM	CS Assertion Duration (MPC)	Max Power Saving Mode (MPSM)	2N Mode	Write Leveling Training	Read Preamble Training
3	Write Leveling Internal Cycle Alignment - Upper Byte				Write Leveling Internal Cycle Alignment - Lower Byte			
4	TUF	RFU	Wide Range (Optional)	Refresh tRFC Mode	Refresh Interval Rate Indicator	Minimum Refresh Rate		
5	Pull-Down Output Driver Impedance		DM Enable	TDQS Enable	PODTM Support	Pull-up Output Driver Impedance		Data Output Disable
6	tRTP				Write Recovery Time			
7	RFU						(Optional) Write Leveling Internal +0.5tCK Alignment Offset - Upper Byte	(Optional) Write Leveling Internal +0.5tCK Alignment Offset - Lower Byte

Table 24 — Mode Register Assignment in DDR5 SDRAM (cont'd)

8	Write Postamble Settings	Read Postamble Settings	RFU	Write Preamble Settings		Read Preamble Settings		
9	TM	RFU				x4 Writes	ECS Write-back	
10	VrefDQ Calibration Value							
11	VrefCA Calibration Value							
12	VrefCS Calibration Value							
13	RFU				tCCD_L / tDLLK			
14	ECS Mode	Reset ECS Counter	Row Mode/ Code Word Mode	RFU	CID3	CID2	CID1	CID0
15	x4 Writes	ECS Writeback	RFU		Automatic ECS in Self Refresh	ECS Error Threshold Count (ETC)		
16	Transparency - Row Address with Max Errors 1 - See MR for encoding details							
17	Transparency - Row Address with Max Errors 2 - See MR for encoding details							
18	Transparency - Row Address with Max Errors 3 - See MR for encoding details							
19	PASR	RFU	Transparency - Max Row Error Count - See MR for encoding details					
20	Transparency - Error Count (EC) - See MR for encoding details							
21	RFU					Rx DQS CTLE Control (Optional)		
22	Rx CS_n CTLE Control (Optional)		Rx CA CTLE Control (Optional)		Rx CTLE Support	MBIST/mPPR Transparency (Optional)		
23	RFU		RFU	MBIST (Optional)	mPPR (Optional)	sPPR		hPPR
24	PPR Guard Key							
25	RFU				Continuous Burst Mode	LFSR1 Pattern Option	LFSR0 Pattern Option	Read Training Pattern Format
26	Read Training Pattern Data0 / LFSR0 Seed							
27	Read Training Pattern Data1 / LFSR1 Seed							
28	Read Training Pattern Invert DQL7:0 (DQ7:0)							
29	Read Training Pattern Invert DQU7:0 (DQ15:8)							
30	LFSR Assignment DQL7/ DQU7	LFSR Assignment DQL6/ DQU6	LFSR Assignment DQL5/ DQU5	LFSR Assignment DQL4/ DQU4	LFSR Assignment DQL3/ DQU3	LFSR Assignment DQL2/ DQU2	LFSR Assignment DQL1/ DQU1	LFSR Assignment DQL0/ DQU0
31	Read Training Pattern Address							
32	RFU	CA_ODT Strap Value	CS ODT			CK ODT		
33	RFU		DQS_RTT_PARK			CA ODT		
34	RFU		RTT_WR			RTT_PARK		
35	RFU		RTT_NOM_RD			RTT_NOM_WR		
36	RFU					RTT_Loopback		
37	RFU		ODTLoFF_WR_offset			ODTLoN_WR_offset		
38	RFU		ODTLoFF_WR_NT_offset			ODTLoN_WR_NT_offset		
39	RFU		ODTLoFF_RD_NT_offset			ODTLoN_RD_NT_offset		
40	RFU					Read DQS offset timing		
41	RFU							
42	DCA Training Assist Mode II (Optional)				DCA Training Assist Mode I		DCA Types Supported	

LIGHT GREY - All Light Grey text is defined as something that should be considered TBD. The content may be accurate or the same as previous technologies but has not yet been reviewed or determined to be the working assumption.

Table 24 — Mode Register Assignment in DDR5 SDRAM (cont'd)

43	Sign Bit for OP[6:4]	DCA for IBCLK in 4-phase clocks			Sign Bit for OP[2:0]	DCA for single/two-phase clock(s) or QCLK in 4-phase clocks		
44	RFU				Sign Bit for QBCLK in 4-phase clocks	DCA for QBCLK in 4-phase clocks		
45	DQS Interval Timer Run Time							
46	DQS Oscillator Count - LSB							
47	DQS Oscillator Count - MSB							
48	Write Pattern Mode							
49	RFU							
50	RFU	RFU	Write CRC auto-disable status	Write CRC auto-disable enable	Write CRC error status	Write CRC enable upper nibble	Write CRC enable lower nibble	Read CRC enable
51	RFU	Write CRC Auto-Disable Threshold - See MR for encoding details						
52	RFU	Write CRC Auto-Disable Window - See MR for encoding details						
53	Loopback Output Mode	Loopback Select Phase		Loopback Output Select				
54	hPPR Resource BG1 Bank 3	hPPR Resource BG1 Bank 2	hPPR Resource BG1 Bank 1	hPPR Resource BG1 Bank 0	hPPR Resource BG0 Bank 3	hPPR Resource BG0 Bank 2	hPPR Resource BG0 Bank 1	hPPR Resource BG0 Bank 0
55	hPPR Resource BG3 Bank 3	hPPR Resource BG3 Bank 2	hPPR Resource BG3 Bank 1	hPPR Resource BG3 Bank 0	hPPR Resource BG2 Bank 3	hPPR Resource BG2 Bank 2	hPPR Resource BG2 Bank 1	hPPR Resource BG2 Bank 0
56	hPPR Resource BG5 Bank 3	hPPR Resource BG5 Bank 2	hPPR Resource BG5 Bank 1	hPPR Resource BG5 Bank 0	hPPR Resource BG4 Bank 3	hPPR Resource BG4 Bank 2	hPPR Resource BG4 Bank 1	hPPR Resource BG4 Bank 0
57	hPPR Resource BG7 Bank 3	hPPR Resource BG7 Bank 2	hPPR Resource BG7 Bank 1	hPPR Resource BG7 Bank 0	hPPR Resource BG6 Bank 3	hPPR Resource BG6 Bank 2	hPPR Resource BG6 Bank 1	hPPR Resource BG6 Bank 0
58	RAAMMT[2:0]			RAAIMT[3:0]				RFM Required
59	RFM RAA Counter		ARFM		BRC Support Level	Bounded Refresh Configuration		DRFM Enable
60	PASR Segment Mask							
61	RSVD			Package Output Driver Test Mode				
62	Vendor Specified							
63	DRAM Scratch Pad							
64	Reserved							
65	Serial Number 1							
66	Serial Number 2							
67	Serial Number 3							
68	Serial Number 4							
69	Serial Number 5							
70	ALERT_n Verification	ALERT_n Verification Support (Optional)	Alert Back-Off Flag	RFU	Activation Counter Initialization Complete	Activation Counter Initialization	Per Row Activation Counting and Alert Back-Off Enable/Disable	Per Row Activation Counting and Alert Back-Off Support (Optional)
71	RFU	PRAC Testing Initialization	PRAC Testing Enable/Disable	PRAC Testing Support (Optional)	Adaptive Per Row Activation Counting		Min RFMab Commands during Recovery Period (ABO_RFM) and min ACT commands during Alert Back-Off Delay (ABO_Delay)	

Table 24 — Mode Register Assignment in DDR5 SDRAM (cont'd)

72	RFU					Counter Address Report Flag	Counter Error Flag
73	8 bits of the row address						
74	8 bits of the row address						
75	RFU	3 bits of the bank group		2 bits of the bank address		2 bits of the row address	
76~102	RFU						
103	DQSL_t IBCLK Sign	RFU	DQSL_t DCA for IBCLK	DQSL_t QCLK Sign	RFU	DQSL_t DCA for QCLK	
104	RFU			DQSL_t QBCLK Sign	RFU	DQSL_t DCA for QBCLK	
105	DQSL_c IBCLK Sign	RFU	DQSL_c DCA for IBCLK	DQSL_c QCLK Sign	RFU	DQSL_c DCA for QCLK	
106	RFU			DQSL_c QBCLK Sign	RFU	DQSL_c DCA for QBCLK	
107	DQSU_t IBCLK Sign	RFU	DQSU_t DCA for IBCLK	DQSU_t QCLK Sign	RFU	DQSU_t DCA for QCLK	
108	RFU			DQSU_t QBCLK Sign	RFU	DQSU_t DCA for QBCLK	
109	DQSU_c IBCLK Sign	RFU	DQSU_c DCA for IBCLK	DQSU_c QCLK Sign	RFU	DQSU_c DCA for QCLK	
110	RFU			DQSU_c QBCLK Sign	RFU	DQSU_c DCA for QBCLK	
111	RFU		Global DFE Tap-4 Enable	Global DFE Tap-3 Enable	Global DFE Tap-2 Enable	Global DFE Tap-1 Enable	Global DFE gain Enable
112	RFU			DML DFE Gain Bias - See MR for encoding details			
113	DML DFE Tap-1 Bias - See MR for encoding details						
114	DML DFE Tap-2 Bias - See MR for encoding details						
115	DML DFE Tap-3 Bias - See MR for encoding details						
116	DML DFE Tap-4 Bias - See MR for encoding details						
117	RFU						
118	DML VREFDQ sign	DML VREFDQ Offset		RFU			
119	RFU						
120	RFU			DMU DFE Gain Bias - See MR for encoding details			
121	DMU DFE Tap-1 Bias - See MR for encoding details						
122	DMU DFE Tap-2 Bias - See MR for encoding details						
123	DMU DFE Tap-3 Bias - See MR for encoding details						
124	DMU DFE Tap-4 Bias - See MR for encoding details						
125	RFU						
126	DMU VREFDQ sign	DMU VREFDQ Offset		RFU			
127	RFU						
128	RFU			DQL0 DFE Gain Bias - See MR for encoding details			
129	DQL0 DFE Tap-1 Bias - See MR for encoding details						
130	DQL0 DFE Tap-2 Bias - See MR for encoding details						
131	DQL0 DFE Tap-3 Bias - See MR for encoding details						
132	DQL0 DFE Tap-4 Bias - See MR for encoding details						
133	DQL0 IBCLK Sign	RFU	DQL0 DCA for IBCLK	DQL0 QCLK Sign	RFU	DQL0 DCA for QCLK	
134	DQL0 VREFDQ Sign	DQL0 VREFDQ Offset		DQL0 QBCLK Sign	RFU	DQL0 DCA for QBCLK	
135	RFU						

Table 24 — Mode Register Assignment in DDR5 SDRAM (cont'd)

136	RFU			DQL1 DFE Gain Bias - See MR for encoding details		
137	DQL1 DFE Tap-1 Bias - See MR for encoding details					
138	DQL1 DFE Tap-2 Bias - See MR for encoding details					
139	DQL1 DFE Tap-3 Bias - See MR for encoding details					
140	DQL1 DFE Tap-4 Bias - See MR for encoding details					
141	DQL1 IBCLK Sign	RFU	DQL1 DCA for IBCLK	DQL1 QCLK Sign	RFU	DQL1 DCA for QCLK
142	DQL1 VREFDQ Sign	DQL1 VREFDQ Offset		DQL1 QBCLK Sign	RFU	DQL1 DCA for QBCLK
143	RFU					
144	RFU			DQL2 DFE Gain Bias - See MR for encoding details		
145	DQL2 DFE Tap-1 Bias - See MR for encoding details					
146	DQL2 DFE Tap-2 Bias - See MR for encoding details					
147	DQL2 DFE Tap-3 Bias - See MR for encoding details					
148	DQL2 DFE Tap-4 Bias - See MR for encoding details					
149	DQL2 IBCLK Sign	RFU	DQL2 DCA for IBCLK	DQL2 QCLK Sign	RFU	DQL2 DCA for QCLK
150	DQL2 VREFDQ Sign	DQL2 VREFDQ Offset		DQL2 QBCLK Sign	RFU	DQL2 DCA for QBCLK
155	RFU					
152	RFU			DQL3 DFE Gain Bias - See MR for encoding details		
153	DQL3 DFE Tap-1 Bias - See MR for encoding details					
154	DQL3 DFE Tap-2 Bias - See MR for encoding details					
155	DQL3 DFE Tap-3 Bias - See MR for encoding details					
156	DQL3 DFE Tap-4 Bias - See MR for encoding details					
157	DQL3 IBCLK Sign	RFU	DQL3 DCA for IBCLK	DQL3 QCLK Sign	RFU	DQL3 DCA for QCLK
158	DQL3 VREFDQ Sign	DQL3 VREFDQ Offset		DQL3 QBCLK Sign	RFU	DQL3 DCA for QBCLK
159	RFU					
160	RFU			DQL4 DFE Gain Bias - See MR for encoding details		
161	DQL4 DFE Tap-1 Bias - See MR for encoding details					
162	DQL4 DFE Tap-2 Bias - See MR for encoding details					
163	DQL4 DFE Tap-3 Bias - See MR for encoding details					
164	DQL4 DFE Tap-4 Bias - See MR for encoding details					
165	DQL4 IBCLK Sign	RFU	DQL4 DCA for IBCLK	DQL4 QCLK Sign	RFU	DQL4 DCA for QCLK
166	DQL4 VREFDQ Sign	DQL4 VREFDQ Offset		DQL4 QBCLK Sign	RFU	DQL4 DCA for QBCLK
167	RFU					
168	RFU			DQL5 DFE Gain Bias - See MR for encoding details		
169	DQL5 DFE Tap-1 Bias - See MR for encoding details					
170	DQL5 DFE Tap-2 Bias - See MR for encoding details					
171	DQL5 DFE Tap-3 Bias - See MR for encoding details					
172	DQL5 DFE Tap-4 Bias - See MR for encoding details					
173	DQL5 IBCLK Sign	RFU	DQL5 DCA for IBCLK	DQL5 QCLK Sign	RFU	DQL5 DCA for QCLK
174	DQL5 VREFDQ Sign	DQL5 VREFDQ Offset		DQL5 QBCLK Sign	RFU	DQL5 DCA for QBCLK
175	RFU					
176	RFU			DQL6 DFE Gain Bias - See MR for encoding details		
177	DQL6 DFE Tap-1 Bias - See MR for encoding details					
178	DQL6 DFE Tap-2 Bias - See MR for encoding details					

Table 24 — Mode Register Assignment in DDR5 SDRAM (cont'd)

179	DQL6 DFE Tap-3 Bias - See MR for encoding details					
180	DQL6 DFE Tap-4 Bias - See MR for encoding details					
181	DQL6 IBCLK Sign	RFU	DQL6 DCA for IBCLK	DQL6 QCLK Sign	RFU	DQL6 DCA for QCLK
182	DQL6 VREFDQ Sign	DQL6 VREFDQ Offset		DQL6 QBCLK Sign	RFU	DQL6 DCA for QBCLK
183	RFU					
184	RFU			DQL7 DFE Gain Bias - See MR for encoding details		
185	DQL7 DFE Tap-1 Bias - See MR for encoding details					
186	DQL7 DFE Tap-2 Bias - See MR for encoding details					
187	DQL7 DFE Tap-3 Bias - See MR for encoding details					
188	DQL7 DFE Tap-4 Bias - See MR for encoding details					
189	DQL7 IBCLK Sign	RFU	DQL6 DCA for IBCLK	DQL7 QCLK Sign	RFU	DQL7 DCA for QCLK
190	DQL7 VREFDQ Sign	DQL7 VREFDQ Offset		DQL7 QBCLK Sign	RFU	DQL7 DCA for QBCLK
191	RFU					
192	RFU			DQU0 DFE Gain Bias - See MR for encoding details		
193	DQU0 DFE Tap-1 Bias - See MR for encoding details					
194	DQU0 DFE Tap-2 Bias - See MR for encoding details					
195	DQU0 DFE Tap-3 Bias - See MR for encoding details					
196	DQU0 DFE Tap-4 Bias - See MR for encoding details					
197	DQU0 IBCLK Sign	RFU	DQU0 DCA for IBCLK	DQU0 QCLK Sign	RFU	DQU0 DCA for QCLK
198	DQU0 VREFDQ Sign	DQU0 VREFDQ Offset		DQU0 QBCLK Sign	RFU	DQU0 DCA for QBCLK
199	RFU					
200	RFU			DQU1 DFE Gain Bias - See MR for encoding details		
201	DQU1 DFE Tap-1 Bias - See MR for encoding details					
202	DQU1 DFE Tap-2 Bias - See MR for encoding details					
203	DQU1 DFE Tap-3 Bias - See MR for encoding details					
204	DQU1 DFE Tap-4 Bias - See MR for encoding details					
205	DQU1 IBCLK Sign	RFU	DQU1 DCA for IBCLK	DQU1 QCLK Sign	RFU	DQU1 DCA for QCLK
206	DQU1 VREFDQ Sign	DQU1 VREFDQ Offset		DQU1 QBCLK Sign	RFU	DQU1 DCA for QBCLK
207	RFU					
208	RFU			DQU2 DFE Gain Bias - See MR for encoding details		
209	DQU2 DFE Tap-1 Bias - See MR for encoding details					
210	DQU2 DFE Tap-2 Bias - See MR for encoding details					
211	DQU2 DFE Tap-3 Bias - See MR for encoding details					
212	DQU2 DFE Tap-4 Bias - See MR for encoding details					
213	DQU2 IBCLK Sign	RFU	DQU2 DCA for IBCLK	DQU2 QCLK Sign	RFU	DQU2 DCA for QCLK
214	DQU2 VREFDQ Sign	DQU2 VREFDQ Offset		DQU2 QBCLK Sign	RFU	DQU2 DCA for QBCLK
215	RFU					
216	RFU			DQU3 DFE Gain Bias - See MR for encoding details		
217	DQU3 DFE Tap-1 Bias - See MR for encoding details					
218	DQU3 DFE Tap-2 Bias - See MR for encoding details					
219	DQU3 DFE Tap-3 Bias - See MR for encoding details					
220	DQU3 DFE Tap-4 Bias - See MR for encoding details					
221	DQU3 IBCLK Sign	RFU	DQU3 DCA for IBCLK	DQU3 QCLK Sign	RFU	DQU3 DCA for QCLK

Table 24 — Mode Register Assignment in DDR5 SDRAM (cont'd)

222	DQU3 VREFDQ Sign	DQU3 VREFDQ Offset		DQU3 QBCLK Sign	RFU	DQU3 DCA for QBCLK
223	RFU					
224	RFU			DQU4 DFE Gain Bias - See MR for encoding details		
225	DQU4 DFE Tap-1 Bias - See MR for encoding details					
226	DQU4 DFE Tap-2 Bias - See MR for encoding details					
227	DQU4 DFE Tap-3 Bias - See MR for encoding details					
228	DQU4 DFE Tap-4 Bias - See MR for encoding details					
229	DQU4 IBCLK Sign	RFU	DQU4 DCA for IBCLK	DQU4 QCLK Sign	RFU	DQU4 DCA for QCLK
230	DQU4 VREFDQ Sign	DQU4 VREFDQ Offset		DQU4 QBCLK Sign	RFU	DQU4 DCA for QBCLK
231	RFU					
232	RFU			DQU5 DFE Gain Bias - See MR for encoding details		
233	DQU5 DFE Tap-1 Bias - See MR for encoding details					
234	DQU5 DFE Tap-2 Bias - See MR for encoding details					
235	DQU5 DFE Tap-3 Bias - See MR for encoding details					
236	DQU5 DFE Tap-4 Bias - See MR for encoding details					
237	DQU5 IBCLK Sign	RFU	DQU5 DCA for IBCLK	DQU5 QCLK Sign	RFU	DQU5 DCA for QCLK
238	DQU5 VREFDQ Sign	DQU5 VREFDQ Offset		DQU5 QBCLK Sign	RFU	DQU5 DCA for QBCLK
239	RFU					
240	RFU			DQU6 DFE Gain Bias - See MR for encoding details		
241	DQU6 DFE Tap-1 Bias - See MR for encoding details					
242	DQU6 DFE Tap-2 Bias - See MR for encoding details					
243	DQU6 DFE Tap-3 Bias - See MR for encoding details					
244	DQU6 DFE Tap-4 Bias - See MR for encoding details					
245	DQU6 IBCLK Sign	RFU	DQU6 DCA for IBCLK	DQU6 QCLK Sign	RFU	DQU6 DCA for QCLK
246	DQU6 VREFDQ Sign	DQU6 VREFDQ Offset		DQU6 QBCLK Sign	RFU	DQU6 DCA for QBCLK
247	RFU					
248	RFU			DQU7 DFE Gain Bias - See MR for encoding details		
249	DQU7 DFE Tap-1 Bias - See MR for encoding details					
250	DQU7 DFE Tap-2 Bias - See MR for encoding details					
251	DQU7 DFE Tap-3 Bias - See MR for encoding details					
252	DQU7 DFE Tap-4 Bias - See MR for encoding details					
253	DQU7 IBCLK Sign	RFU	DQU7 DCA for IBCLK	DQU7 QCLK Sign	RFU	DQU7 DCA for QCLK
254	DQU7 VREFDQ Sign	DQU7 VREFDQ Offset		DQU7 QBCLK Sign	RFU	DQU7 DCA for QBCLK
255	RFU					

3.5.2 MR0 (MA[7:0]=00H) Burst Length and CAS Latency**MR0 Register Information**

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
CAS Latency (CL)						Burst Length	

Function	Register Type	Operand	Data	Notes
Burst Length	R/W	OP[1:0]	00B: BL16 01B: BC8 OTF 10B: BL32 (Optional) 11B: BL32 OTF (Optional)	
CAS Latency (CL)	R/W	OP[7:2]	000000B: 22 000001B: 24 000010B: 26 000011B: 28 ... 010011B: 60 010100B: 62 010101B: 64 010110B: 66 010111B: 68 011000B: 70 011001B: 72 ... 011101B: 80 011110B: 82 011111B: 84 100000B: 86 100001B: 88 100010B: 90 All other encodings reserved.	1, 2

NOTE 1 Range covers both Monolithic DDR5 and 3DS-DDR5 devices up to 8800

NOTE 2 CWL=CL-2, also known as WL=RL-2.

3.5.3 MR1 (MA [7:0] = 01_H) - PDA Mode Details**MR1 Register Information**

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
PDA Select ID				PDA Enumerate ID			

Function	Register Type	Operand	Data	Notes
PDA Enumerate ID	R	OP[3:0]	This is a Read Only MR field, which is only programmed through an MPC command with the PDA Enumerate ID opcode. xxxx_B Encoding is set with MPC command with the PDA Enumerate ID opcode. This can only be set when PDA Enumerate Programming Mode is enabled and the associated DRAM's DQ0 is asserted LOW. The PDA Enumerate ID opcode includes 4 bits for this encoding. Default setting is 1111_B	
PDA Select ID	R	OP[7:4]	This is a Read Only MR field, which is only programmed through an MPC command with the PDA Select ID opcode. xxxx_B Encoding is set with MPC command with the PDA Select ID opcode. The PDA Select ID opcode includes 4 bits for this encoding. 1111_B = all DRAMs execute MRW, MPC, and VrefCA commands For all other encodings, DRAMs execute MRW, MPC, and VrefCA commands only if PDA Select ID[3:0] = PDA Enumerate ID[3:0], with some exceptions for specific MPC commands that execute regardless of PDA Select ID. Default setting is 1111_B	

3.5.4 MR2 (MA [7:0] = 02_H) - Functional Modes**MR2 Register Information**

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
Internal Write Timing	Reserved	Device 15 MPSM	CS Assertion Duration (MPC)	Max Power Saving Mode (MPSM)	2N Mode	Write Leveling Training	Read Preamble Training

Function	Register Type	Operand	Data	Notes
Read Preamble Training	R/W	OP[0]	0B: Normal Mode (Default) 1B: Read Preamble Training	
Write Leveling	R/W	OP[1]	0B: Normal Mode (Default) 1B: Write Leveling	1, 2, 3
2N Mode	R	OP[2]	0B: 2N Mode (Default) 1B: 1N Mode	4
Max Power Saving Mode	R/W	OP[3]	0B: Disable (Default) 1B: Enable	
CS Assertion Duration (MPC)	R/W	OP[4]	0B: Only Multiple cycles of CS assertion supported for MPC, VrefCA and VrefCS commands (Default) 1B: Only a single cycle of CS assertion supported for MPC, VrefCA and VrefCS commands	
Device 15 Maximum Power Savings Mode	R/W	OP[5]	0B: Disable (Default) 1B: Enable	
Reserved	Reserved	OP[6]	Reserved	
Internal Write Timing	R/W	OP[7]	0B: Disable 1B: Enable	5

NOTE 1 To enter WL Training Mode the MR field must be programmed to 1. WL Training Mode is used when Internal Write Timing = 0 (External WL Training) and when Internal Write Timing = 1 (Internal WL Training).

NOTE 2 To exit WL Training Mode the MR field must be programmed to 0.

NOTE 3 MRR's are not supported during Write Leveling.

NOTE 4 This mode register is programmed via an explicit MPC command only.

NOTE 5 This is set during WL Training, after the host DQS has been aligned to the ideal External WL timings. The Internal Write Timing is enabled and the WL Internal Timing Alignment is set to ensure the internal Write Enable aligns within tDQS2CK of the external WL Trained location. When Internal Write Timing is Disabled, the WL Internal Cycle Alignment setting does not change the behavior of the write timings

3.5.5 MR3 (MA[7:0]=03_H) - DQS Training**MR3 Register Information**

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
Write Leveling Internal Cycle Alignment - Upper Byte				Write Leveling Internal Cycle Alignment - Lower Byte			

Function	Register Type	Operand	Data	Notes
Write Leveling Internal Cycle Alignment - Lower Byte	R/W	OP[3:0]	0000 _B : 0 tCK (Default) 0001 _B : -1 tCK 0010 _B : -2 tCK 0011 _B : -3 tCK 0100 _B : -4 tCK 0101 _B : -5 tCK 0110 _B : -6 tCK (Optional OPcode: 0111 _B through 1111 _B) 0111 _B : -7 tCK 1000 _B : -8 tCK ... 1110 _B : -14 tCK 1111 _B : -15 tCK	1, 2, 3, 5
Write Leveling Internal Cycle Alignment - Upper Byte	R/W	OP[7:4]	0000 _B : 0 tCK (Default) 0001 _B : -1 tCK 0010 _B : -2 tCK 0011 _B : -3 tCK 0100 _B : -4 tCK 0101 _B : -5 tCK 0110 _B : -6 tCK (Optional OPcode: 0111 _B through 1111 _B) 0111 _B : -7 tCK 1000 _B : -8 tCK ... 1110 _B : -14 tCK 1111 _B : -15 tCK	1, 2, 4, 5

- NOTE 1 This is set during WL Training, after the host DQS has been aligned to the ideal External WL timings. The Internal Write Timing is enabled and the WL Internal Timing Alignment is set to ensure the internal Write Enable aligns within t_{DQS2CK} of the external WL Trained location. When Internal Write Timing is Disabled, the WL Internal Cycle Alignment setting does not change the behavior of the write timings.
- NOTE 2 The DRAM implementation may optionally have the same behavior when the Internal Write Timing is enabled vs disabled. This would mean that the CK and DQS timing paths remain matched internally. The WL Internal Cycle Alignment setting must still support pulling the Internal WL Pulse earlier so that the same WL Training Flow will produce the correct result.
- NOTE 3 Lower Byte WL Internal Cycle Alignment is intended for x4, x8, and x16 configurations.
- NOTE 4 Upper Byte WL Internal Cycle Alignment is intended for x16 configuration only. Although training of the Lower and Upper Bytes is independent, contact the DRAM vendor regarding recommendations for setting the WICA values to the same offset.
- NOTE 5 Optional OPcode may be needed for certain speed bins.

3.5.6 MR4 (MA[7:0]=04_H) - Refresh Settings**MR4 Register Information**

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
TUF	RFU	Wide Range (Optional)	Refresh tRFC Mode	Refresh Interval Rate Indicator	Minimum Refresh Rate		

Refresh Settings MR4 Register Information Table

Function	Register Type	Operand	Data	Notes
Minimum Refresh Rate	R	OP[2:0]	If Wide Range is not supported (OP[5]=0): 000_B: RFU 001_B: tREFI x1 (1x Refresh Rate), <80 °C nominal 010_B: tREFI x1 (1x Refresh Rate), 80-85 °C nominal 011_B: tREFI /2 (2x Refresh Rate), 85-90 °C nominal 100_B: tREFI /2 (2x Refresh Rate), 90-95 °C nominal 101_B: tREFI /2 (2x Refresh Rate), >95 °C nominal 110_B: RFU 111_B: RFU If Wide Range is supported (OP[5]=1): 000_B: tREFI x1 (1x Refresh Rate), <75 °C nominal 001_B: tREFI x1 (1x Refresh Rate), 75-80 °C nominal 010_B: tREFI x1 (1x Refresh Rate), 80-85 °C nominal 011_B: tREFI /2 (2x Refresh Rate), 85-90 °C nominal 100_B: tREFI /2 (2x Refresh Rate), 90-95 °C nominal 101_B: tREFI /2 (2x Refresh Rate), 95-100 °C nominal 110_B: tREFI /2 (2x Refresh Rate), >100 °C nominal 111_B: RFU	1, 2, 3, 4, 5, 6, 7, 8
Refresh Interval Rate Indicator	SR/W	OP[3]	DRAM Status Read (SR): 0_B: Not implemented (Default) 1_B: Implemented Host Write (W): 0_B: Disabled (Default) 1_B: Enabled	
Refresh tRFC Mode	R/W	OP[4]	0_B: Normal Refresh Mode (tRFC1) 1_B: Fine Granularity Refresh Mode (tRFC2)	
Wide Range (Optional)	R	OP[5]	0_B: Wide range is not supported 1_B: Wide range is supported (Optional)	
RFU	RFU	OP[6]	RFU	
TUF (Temperature Update Flag)	R	OP[7]	0_B: No change in OP[2:0] since last MR4 read (default) 1_B: Change in OP[2:0] since last MR4 read	

Refresh Settings MR4 Register Information Table (cont'd)

NOTE 1	The minimum required refresh rate for each OP[2:0] setting applies to tREFI1 and tREFI2. Each OP[2:0] setting specifies a nominal temperature range. The ranges defined by OP[2:0] are determined by temperature thresholds used by the system for proper operation.
NOTE 2	When OP[5]=0, the four temperature thresholds are nominally at 80 °C, 85 °C, 90 °C, and 95 °C. The <80 °C threshold has no minimum value specified and the >95°C threshold has no maximum temperature value specified. When OP[5]=1, the six temperature thresholds are nominally at 75 °C, 80 °C, 85 °C, 90 °C, 95 °C, and 100 °C. The <75 °C threshold has no minimum value specified and the >100 °C threshold has no maximum temperature value specified
NOTE 3	DRAM vendors must report all of the possible settings over the operating temperature range of the device. Each vendor guarantees that their device will work at any temperature within the range when the system refresh interval follows there guidelines: ● Threshold $\leq 85\text{ }^{\circ}\text{C}$: tREFI x1 (1x Refresh Rate) or faster may be used ● Threshold $> 85\text{ }^{\circ}\text{C}$: tREFI /2 (2x Refresh Rate) or faster is required ● Data integrity at thresholds $>95^{\circ}\text{C}$ is not assured regardless of refresh rate
NOTE 4	The 2x Refresh Rate must be provided by the system before the DRAM Tj has gone up by more than 2 °C (Temperature Margin) since the first report out of OP[2:0]=011B. This condition is reset when OP[2:0] is equal to 010B.
NOTE 5	The device may not operate properly when OP[2:0]=101B, if the DRAM Tj has gone up by more than 2 °C (Temperature Margin) since the first report out of OP[2:0]=101B. This condition is reset when OP[2:0] is equal to 100B. OP[2:0]=101B must be a temporary condition of the DRAM, to be addressed by immediately reducing the Tj of the DRAM by throttling its power, and/or the power of nearby devices.
NOTE 6	OP[7] = 0 at power-up. OP[2:0] bits are valid after initialization sequence (Te).
NOTE 7	See the section on "Temperature Sensor" for information on the recommended frequency of reading MR4
NOTE 8	Support for wide range temperature sensor ranges does not indicate that the DDR5 DRAM device may operate properly at temperature ranges above 95 °C – side effects may include loss of data integrity

3.5.7 MR5 (MA[7:0]=05_H) - IO Settings**MR5 Register Information**

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
Pull-Down Output Driver Impedance		DM Enable	TDQS Enable	PODTM Support	Pull-up Output Driver Impedance		Data Output Disable

Function	Register Type	Operand	Data	Notes
Data Output Disable	W	OP[0]	0 _B : Normal Operation (Default) 1 _B : Outputs Disabled	
Pull-up Output Driver Impedance	R/W	OP[2:1]	00 _B : RZQ/7 (34) 01 _B : RZQ/6 (40) 10 _B : RZQ/5 (48) 11 _B : RFU	
Package Output Driver Test Mode Supported	R	OP[3]	0 _B : Function Not Supported 1 _B : Function Supported	
TDQS Enable	R/W	OP[4]	0 _B : Disable (Default) 1 _B : Enable	
DM Enable	R/W	OP[5]	0 _B : Disable (Default) 1 _B : Enable	
Pull-Down Output Driver Impedance	R/W	OP[7:6]	00 _B : RZQ/7 (34) 01 _B : RZQ/6 (40) 10 _B : RZQ/5 (48) 11 _B : RFU	

3.5.8 MR6 (MA[7:0]=06_H) - Write Recovery Time and tRTP**MR6 Register Information**

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
tRTP				Write Recovery Time			

Function	Register Type	Operand	Data	Notes
Write Recovery Time Legacy / PRAC	R/W	OP[3:0]	0000 _B : 48nCK (Legacy) / 16nCK (PRAC) 0001 _B : 54nCK (Legacy) / 18nCK (PRAC) 0010 _B : 60nCK (Legacy) / 20nCK (PRAC) 0011 _B : 66nCK (Legacy) / 22nCK (PRAC) 0100 _B : 72nCK (Legacy) / 24nCK (PRAC) 0101 _B : 78nCK (Legacy) / 26nCK (PRAC) 0110 _B : 84nCK (Legacy) / 28nCK (PRAC) 0111 _B : 90nCK (Legacy) / 30nCK (PRAC) 1000 _B : 96nCK (Legacy) / 32nCK (PRAC) 1001 _B : 102nCK (Legacy) / 34nCK (PRAC) 1010 _B : 108nCK (Legacy) / 36nCK (PRAC) 1011 _B : 114nCK (Legacy) / 38nCK (PRAC) 1100 _B : 120nCK (Legacy) / 40nCK (PRAC) 1101 _B : 126nCK (Legacy) / 42nCK (PRAC) 1110 _B : 132nCK (Legacy) / 44nCK (PRAC) 1111 _B : RFU	1
tRTP Legacy / PRAC	R/W	OP[7:4]	0000 _B : 12nCK (Legacy) / 8nCK (PRAC) 0001 _B : 14nCK (Legacy) / 9nCK (PRAC) 0010 _B : 15nCK (Legacy) / 10nCK (PRAC) 0011 _B : 17nCK (Legacy) / 11nCK (PRAC) 0100 _B : 18nCK (Legacy) / 12nCK (PRAC) 0101 _B : 20nCK (Legacy) / 13nCK (PRAC) 0110 _B : 21nCK (Legacy) / 14nCK (PRAC) 0111 _B : 23nCK (Legacy) / 15nCK (PRAC) 1000 _B : 24nCK (Legacy) / 16nCK (PRAC) 1001 _B : 26nCK (Legacy) / 17nCK (PRAC) 1010 _B : 27nCK (Legacy) / 18nCK (PRAC) 1011 _B : 29nCK (Legacy) / 19nCK (PRAC) 1100 _B : 30nCK (Legacy) / 20nCK (PRAC) 1101 _B : 32nCK (Legacy) / 21nCK (PRAC) 1110 _B : 33nCK (Legacy) / 22nCK (PRAC) 1111 _B : RFU	2
NOTE 1 tWR,min is defined in the "Timing Parameters" tables (Table 330 - Table 332). Host must operate with MR settings resulting in tCK * MR6:OP[3:0] >= tWR,min. NOTE 2 tRTP,min is defined in the "Timing Parameters" tables (Table 328 - Table 330). Host must operate with MR settings resulting in tCK * MR6:OP[7:4] >= tRTP,min. NOTE 3 All nCK conversions require rounding algorithm consideration.				

3.5.9 MR7 (MA[7:0]=07_H) - Write Leveling Internal +0.5tCK Alignment Offset**MR7 Register Information**

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
RFU						(Optional) Write Leveling Internal +0.5tCK Alignment Offset - Upper Byte	(Optional) Write Leveling Internal +0.5tCK Alignment Offset - Lower Byte

Function	Register Type	Operand	Data	Notes
(Optional) Write Leveling Internal +0.5tCK Alignment Offset - Lower Byte	R/W	OP[0]	0B: Disabled (Default) 1B: 0.5tCK	1, 2
(Optional) Write Leveling Internal +0.5tCK Alignment Offset - Upper Byte	R/W	OP[1]	0B: Disabled (Default) 1B: 0.5tCK	1, 3
RFU	RFU	OP[7:2]	RFU	
NOTE 1 The WICA 0.5 tCK offset is a positive adjustment to the target WICA value. (Ex. MR3:OP[3:0] = -3 tCK (0011B) and MR7:OP[0] = 1, WICA + WICAhalfCycle = -3 tCK + 0.5 tCK = -2.5 tCK) NOTE 2 Lower Byte WL Internal Cycle Alignment is intended for x4, x8, and x16 configurations. NOTE 3 Upper Byte WL Internal Cycle Alignment is intended for x16 configuration only. Although training of the Lower and Upper Bytes is independent, contact the DRAM vendor regarding recommendations for setting the WICA values to the same offset. NOTE 4 When operating at 1980-2100 MT/s (CL=22), the host shall not apply the WICA 0.5tCK offset.				

3.5.10 MR8 (MA[7:0]=08_H) - Preamble / Postamble**MR8 Register Information**

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
Write Postamble Settings	Read Postamble Settings	RFU	Write Preamble Settings		Read Preamble Settings		

Function	Register Type	Operand	Data	Notes
Read Preamble Settings	R/W	OP[2:0]	000B: 1 tCK - 10 Pattern 001B: 2 tCK - 0010 Pattern 010B: 2 tCK - 1110 Pattern (DDR4 Style) 011B: 3 tCK - 000010 Pattern 100B: 4 tCK - 00001010 Pattern 101B: Reserved 110B: Reserved 111B: Reserved	1
Write Preamble Settings	R/W	OP[4:3]	00B: Reserved 01B: 2 tCK - 0010 Pattern (Default) 10B: 3 tCK - 000010 Pattern 11B: 4 tCK - 00001010 Pattern	
RFU	RFU	OP[5]	RFU	
Read Postamble Settings	R/W	OP[6]	0B: 0.5 tCK - 0 Pattern 1B: 1.5 tCK - 010 Pattern	
Write Postamble Settings	R/W	OP[7]	0B: 0.5 tCK - 0 Pattern 1B: 1.5 tCK - 000 Pattern	
NOTE 1 Please refer to the Preamble Specification for details on the Read Preamble modes and patterns.				

3.5.11 MR9 (MA[7:0]=09H) - Writeback Suppression and TM**MR9 Register Information**

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
TM	RFU					x4 Write	ECS Write-back

Function	Register Type	Operand	Data	Notes
ECS Writeback	R/W	OP[0]	0B: Do not suppress writeback of Data and ECC Check Bits (Default) 1B: Suppress writeback of Data and ECC Check Bits (Optional)	1
x4 Writes	R/W	OP[1]	0B: Do not suppress writeback of Data during RMW (Default) 1B: Suppress writeback of Data during RMW (Optional)	1
RFU	RFU	OP[6:2]	RFU	
TM	W	OP[7]	0B: Normal (Default) 1B: Test Mode	

NOTE 1 DDR5 SPD Byte 14 Bits[2:1] indicates if feature is supported and will also indicate whether to use MR9 or MR15 for enabling the modes.

3.5.12 MR10 (MA[7:0]=0A_H) - VrefDQ Calibration Value**MR10 Register Information**

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
VrefDQ Calibration Value							

Function	Register Type	Operand	Data	Notes
VrefDQ Cal Value	R/W	OP[7:0]	0000:0000B: --Thru-- 1111:1111B: See Table 25	

Table 25 — VrefDQ Setting Range

Function	Operand						Notes
VrefDQ Cal Value for MR10	OP	0000 0000 _B : 97.5%	0001 1011 _B : 84.0%	0011 0110 _B : 70.5%	0101 0001 _B : 57.0%	0110 1100 _B : 43.5%	
		0000 0001 _B : 97.0%	0001 1100 _B : 83.5%	0011 0111 _B : 70.0%	0101 0010 _B : 56.5%	0110 1101 _B : 43.0%	
		0000 0010 _B : 96.5%	0001 1101 _B : 83.0%	0011 1000 _B : 69.5%	0101 0011 _B : 56.0%	0110 1110 _B : 42.5%	
		0000 0011 _B : 96.0%	0001 1110 _B : 82.5%	0011 1001 _B : 69.0%	0101 0100 _B : 55.5%	0110 1111 _B : 42.0%	
		0000 0100 _B : 95.5%	0001 1111 _B : 82.0%	0011 1010 _B : 68.5%	0101 0101 _B : 55.0%	0111 0000 _B : 41.5%	
		0000 0101 _B : 95.0%	0010 0000 _B : 81.5%	0011 1011 _B : 68.0%	0101 0110 _B : 54.5%	0111 0001 _B : 41.0%	
		0000 0110 _B : 94.5%	0010 0001 _B : 81.0%	0011 1100 _B : 67.5%	0101 0111 _B : 54.0%	0111 0010 _B : 40.5%	
		0000 0111 _B : 94.0%	0010 0010 _B : 80.5%	0011 1101 _B : 67.0%	0101 1000 _B : 53.5%	0111 0011 _B : 40.0%	
		0000 1000 _B : 93.5%	0010 0011 _B : 80.0%	0011 1110 _B : 66.5%	0101 1001 _B : 53.0%	0111 0100 _B : 39.5%	
		0000 1001 _B : 93.0%	0010 0100 _B : 79.5%	0011 1111 _B : 66.0%	0101 1010 _B : 52.5%	0111 0101 _B : 39.0%	
		0000 1010 _B : 92.5%	0010 0101 _B : 79.0%	0100 0000 _B : 65.5%	0101 1011 _B : 52.0%	0111 0110 _B : 38.5%	
		0000 1011 _B : 92.0%	0010 0110 _B : 78.5%	0100 0001 _B : 65.0%	0101 1100 _B : 51.5%	0111 0111 _B : 38.0%	
		0000 1100 _B : 91.5%	0010 0111 _B : 78.0%	0100 0010 _B : 64.5%	0101 1101 _B : 51.0%	0111 1000 _B : 37.5%	
		0000 1101 _B : 91.0%	0010 1000 _B : 77.5%	0100 0011 _B : 64.0%	0101 1110 _B : 50.5%	0111 1001 _B : 37.0%	
		0000 1110 _B : 90.5%	0010 1001 _B : 77.0%	0100 0100 _B : 63.5%	0101 1111 _B : 50.0%	0111 1010 _B : 36.5%	
		0000 1111 _B : 90.0%	0010 1010 _B : 76.5%	0100 0101 _B : 63.0%	0110 0000 _B : 49.5%	0111 1011 _B : 36.0%	
		0001 0000 _B : 89.5%	0010 1011 _B : 76.0%	0100 0110 _B : 62.5%	0110 0001 _B : 49.0%	0111 1100 _B : 35.5%	
		0001 0001 _B : 89.0%	0010 1100 _B : 75.5%	0100 0111 _B : 62.0%	0110 0010 _B : 48.5%	0111 1101 _B : 35.0%	
		0001 0010 _B : 88.5%	0010 1101 _B : 75.0%	0100 1000 _B : 61.5%	0110 0011 _B : 48.0%	All Others: Reserved	
		0001 0011 _B : 88.0%	0010 1110 _B : 74.5%	0100 1001 _B : 61.0%	0110 0100 _B : 47.5%		
		0001 0100 _B : 87.5%	0010 1111 _B : 74.0%	0100 1010 _B : 60.5%	0110 0101 _B : 47.0%		
		0001 0101 _B : 87.0%	0011 0000 _B : 73.5%	0100 1011 _B : 60.0%	0110 0110 _B : 46.5%		
		0001 0110 _B : 86.5%	0011 0001 _B : 73.0%	0100 1100 _B : 59.5%	0110 0111 _B : 46.0%		
		0001 0111 _B : 86.0%	0011 0010 _B : 72.5%	0100 1101 _B : 59.0%	0110 1000 _B : 45.5%		
		0001 1000 _B : 85.5%	0011 0011 _B : 72.0%	0100 1110 _B : 58.5%	0110 1001 _B : 45.0%		
		0001 1001 _B : 85.0%	0011 0100 _B : 71.5%	0100 1111 _B : 58.0%	0110 1010 _B : 44.5%		
		0001 1010 _B : 84.5%	0011 0101 _B : 71.0%	0101 0000 _B : 57.5%	0110 1011 _B : 44.0%		

MR11 Register Information

Function	Register Type	Operand	Data	Notes
VrefCA Cal Value	R	OP[6:0]	000:0000B: --Thru-- 111:1111B: See Table 26	1, 2
	R	OP[7]	V: Valid	
NOTE 1 Since VREF CA Calibration setting has an explicit command (VrefCA COMMAND), it can only be programmed via that command and its mode register is therefore read only.				
NOTE 2 Since the state of CA12 is used to differentiate the VrefCA vs VrefCS command, the MR11/12 OP[7] value is defined as valid.				

Function	Operand						Notes
VrefCA Cal Value for MR11	OP	V000 0000 _B : 97.5%	V001 1011 _B : 84.0%	V011 0110 _B : 70.5%	V101 0001 _B : 57.0%	V110 1100 _B : 43.5%	
		V000 0001 _B : 97.0%	V001 1100 _B : 83.5%	V011 0111 _B : 70.0%	V101 0010 _B : 56.5%	V110 1101 _B : 43.0%	
		V000 0010 _B : 96.5%	V001 1101 _B : 83.0%	V011 1000 _B : 69.5%	V101 0011 _B : 56.0%	V110 1110 _B : 42.5%	
		V000 0011 _B : 96.0%	V001 1110 _B : 82.5%	V011 1001 _B : 69.0%	V101 0100 _B : 55.5%	V110 1111 _B : 42.0%	
		V000 0100 _B : 95.5%	V001 1111 _B : 82.0%	V011 1010 _B : 68.5%	V101 0101 _B : 55.0%	V111 0000 _B : 41.5%	
		V000 0101 _B : 95.0%	V010 0000 _B : 81.5%	V011 1011 _B : 68.0%	V101 0110 _B : 54.5%	V111 0001 _B : 41.0%	
		V000 0110 _B : 94.5%	V010 0001 _B : 81.0%	V011 1100 _B : 67.5%	V101 0111 _B : 54.0%	V111 0010 _B : 40.5%	
		V000 0111 _B : 94.0%	V010 0010 _B : 80.5%	V011 1101 _B : 67.0%	V101 1000 _B : 53.5%	V111 0011 _B : 40.0%	
		V000 1000 _B : 93.5%	V010 0011 _B : 80.0%	V011 1110 _B : 66.5%	V101 1001 _B : 53.0%	V111 0100 _B : 39.5%	
		V000 1001 _B : 93.0%	V010 0100 _B : 79.5%	V011 1111 _B : 66.0%	V101 1010 _B : 52.5%	V111 0101 _B : 39.0%	
		V000 1010 _B : 92.5%	V010 0101 _B : 79.0%	V100 0000 _B : 65.5%	V101 1011 _B : 52.0%	V111 0110 _B : 38.5%	
		V000 1011 _B : 92.0%	V010 0110 _B : 78.5%	V100 0001 _B : 65.0%	V101 1100 _B : 51.5%	V111 0111 _B : 38.0%	
		V000 1100 _B : 91.5%	V010 0111 _B : 78.0%	V100 0010 _B : 64.5%	V101 1101 _B : 51.0%	V111 1000 _B : 37.5%	
		V000 1101 _B : 91.0%	V010 1000 _B : 77.5%	V100 0011 _B : 64.0%	V101 1110 _B : 50.5%	V111 1001 _B : 37.0%	
		V000 1110 _B : 90.5%	V010 1001 _B : 77.0%	V100 0100 _B : 63.5%	V101 1111 _B : 50.0%	V111 1010 _B : 36.5%	
		V000 1111 _B : 90.0%	V010 1010 _B : 76.5%	V100 0101 _B : 63.0%	V110 0000 _B : 49.5%	V111 1011 _B : 36.0%	
		V001 0000 _B : 89.5%	V010 1011 _B : 76.0%	V100 0110 _B : 62.5%	V110 0001 _B : 49.0%	V111 1100 _B : 35.5%	
		V001 0001 _B : 89.0%	V010 1100 _B : 75.5%	V100 0111 _B : 62.0%	V110 0010 _B : 48.5%	V111 1101 _B : 35.0%	
		V001 0010 _B : 88.5%	V010 1101 _B : 75.0%	V100 1000 _B : 61.5%	V110 0011 _B : 48.0%	All Others: Reserved	
		V001 0011 _B : 88.0%	V010 1110 _B : 74.5%	V100 1001 _B : 61.0%	V110 0100 _B : 47.5%		
		V001 0100 _B : 87.5%	V010 1111 _B : 74.0%	V100 1010 _B : 60.5%	V110 0101 _B : 47.0%		
		V001 0101 _B : 87.0%	V011 0000 _B : 73.5%	V100 1011 _B : 60.0%	V110 0110 _B : 46.5%		
		V001 0110 _B : 86.5%	V011 0001 _B : 73.0%	V100 1100 _B : 59.5%	V110 0111 _B : 46.0%		
		V001 0111 _B : 86.0%	V011 0010 _B : 72.5%	V100 1101 _B : 59.0%	V110 1000 _B : 45.5%		
		V001 1000 _B : 85.5%	V011 0011 _B : 72.0%	V100 1110 _B : 58.5%	V110 1001 _B : 45.0%		
		V001 1001 _B : 85.0%	V011 0100 _B : 71.5%	V100 1111 _B : 58.0%	V110 1010 _B : 44.5%		
		V001 1010 _B : 84.5%	V011 0101 _B : 71.0%	V101 0000 _B : 57.5%	V110 1011 _B : 44.0%		

NOTE "V" = Valid.

MR12 Register Information

Function	Register Type	Operand	Data	Notes
VrefCS Cal Value	R	OP[6:0]	000:0000B: --Thru-- 111:1111B: See Table 27	1, 2
	R	OP[7]	V: Valid	
NOTE 1	Since VREF CS Calibration setting has an explicit command (VrefCS COMMAND), it can only be programmed via that command and its mode register is therefore read only.			
NOTE 2	Since the state of CA12 is used to differentiate the VrefCA vs VrefCS command, the MR11/12 OP[7] value is defined as valid.			

Table 27 — VrefCS Setting Range

Function	Operand						Notes
VrefCS Cal Value for MR12	OP	V000 0000B: 97.5%	V001 1011B: 84.0%	V011 0110B: 70.5%	V101 0001B: 57.0%	V110 1100B: 43.5%	
		V000 0001B: 97.0%	V001 1100B: 83.5%	V011 0111B: 70.0%	V101 0010B: 56.5%	V110 1101B: 43.0%	
		V000 0010B: 96.5%	V001 1101B: 83.0%	V011 1000B: 69.5%	V101 0011B: 56.0%	V110 1110B: 42.5%	
		V000 0011B: 96.0%	V001 1110B: 82.5%	V011 1001B: 69.0%	V101 0100B: 55.5%	V110 1111B: 42.0%	
		V000 0100B: 95.5%	V001 1111B: 82.0%	V011 1010B: 68.5%	V101 0101B: 55.0%	V111 0000B: 41.5%	
		V000 0101B: 95.0%	V010 0000B: 81.5%	V011 1011B: 68.0%	V101 0110B: 54.5%	V111 0001B: 41.0%	
		V000 0110B: 94.5%	V010 0001B: 81.0%	V011 1100B: 67.5%	V101 0111B: 54.0%	V111 0010B: 40.5%	
		V000 0111B: 94.0%	V010 0010B: 80.5%	V011 1101B: 67.0%	V101 1000B: 53.5%	V111 0011B: 40.0%	
		V000 1000B: 93.5%	V010 0011B: 80.0%	V011 1110B: 66.5%	V101 1001B: 53.0%	V111 0100B: 39.5%	
		V000 1001B: 93.0%	V010 0100B: 79.5%	V011 1111B: 66.0%	V101 1010B: 52.5%	V111 0101B: 39.0%	
		V000 1010B: 92.5%	V010 0101B: 79.0%	V100 0000B: 65.5%	V101 1011B: 52.0%	V111 0110B: 38.5%	
		V000 1011B: 92.0%	V010 0110B: 78.5%	V100 0001B: 65.0%	V101 1100B: 51.5%	V111 0111B: 38.0%	
		V000 1100B: 91.5%	V010 0111B: 78.0%	V100 0010B: 64.5%	V101 1101B: 51.0%	V111 1000B: 37.5%	
		V000 1101B: 91.0%	V010 1000B: 77.5%	V100 0011B: 64.0%	V101 1110B: 50.5%	V111 1001B: 37.0%	
		V000 1110B: 90.5%	V010 1001B: 77.0%	V100 0100B: 63.5%	V101 1111B: 50.0%	V111 1010B: 36.5%	
		V000 1111B: 90.0%	V010 1010B: 76.5%	V100 0101B: 63.0%	V110 0000B: 49.5%	V111 1011B: 36.0%	
		V001 0000B: 89.5%	V010 1011B: 76.0%	V100 0110B: 62.5%	V110 0001B: 49.0%	V111 1100B: 35.5%	
		V001 0001B: 89.0%	V010 1100B: 75.5%	V100 0111B: 62.0%	V110 0010B: 48.5%	V111 1101B: 35.0%	
		V001 0010B: 88.5%	V010 1101B: 75.0%	V100 1000B: 61.5%	V110 0011B: 48.0%	All Others: Reserved	
		V001 0011B: 88.0%	V010 1110B: 74.5%	V100 1001B: 61.0%	V110 0100B: 47.5%		
		V001 0100B: 87.5%	V010 1111B: 74.0%	V100 1010B: 60.5%	V110 0101B: 47.0%		
		V001 0101B: 87.0%	V011 0000B: 73.5%	V100 1011B: 60.0%	V110 0110B: 46.5%		
		V001 0110B: 86.5%	V011 0001B: 73.0%	V100 1100B: 59.5%	V110 0111B: 46.0%		
		V001 0111B: 86.0%	V011 0010B: 72.5%	V100 1101B: 59.0%	V110 1000B: 45.5%		
		V001 1000B: 85.5%	V011 0011B: 72.0%	V100 1110B: 58.5%	V110 1001B: 45.0%		
		V001 1001B: 85.0%	V011 0100B: 71.5%	V100 1111B: 58.0%	V110 1010B: 44.5%		
V001 1010B: 84.5%	V011 0101B: 71.0%	V101 0000B: 57.5%	V110 1011B: 44.0%				
NOTE “V” = Valid.							

3.5.15 MR13 (MA [7:0] = 0DH) - SRX/NOP Clock-Sync / CS Geardown / tCCD_L / tCCD_L_WR / tCCD_L_WR2 / tDLLK

MR13 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
RFU		SRX/NOP Clock-Sync Support and Enable/Disable at next SRX (Optional)	CS Gear-down Enable/Disable at next SREF	tCCD_L / tCCD_L_WR / tCCD_L_WR2 / tDLLK			

Function	Register Type	Operand	Data	Notes
tCCD_L / tCCD_L_WR / tCCD_L_WR2 / tDLLK	R	OP[3:0]	0000B: --Thru-- 1111B: See Table Below	
CS Geardown Enable/Disable at next SREF	W/R	OP[4]	0B: CS Geardown is disabled at next SREF (Default). 1B: CS Geardown is enabled at next SREF.	
SRX/NOP Clock-Sync Support and Enable/Disable at next SRX (Optional)	SR/W	OP[5]	DRAM Status Read (SR): 0B: SRX/NOP Clock-Sync not supported (Default). 1B: SRX/NOP Clock-Sync supported. Host Write (W): 0B: Disable SRX/NOP Clock-Sync (Default). 1B: Enable SRX/NOP Clock-Sync.	1
RFU	RFU	OP[7:6]	RFU	

NOTE 1 Host shall program MR13:OP[5]=1 via MRW command to update to enable/disable the SRX/NOP Clock-Sync feature at next SRX, if needed. During the next Self-Refresh, DRAM will automatically apply the update to the function.

Table 28 — tCCD_L/tCCD_L_WR/tCCD_L_WR2/tDLLK Encoding Details

Function	OP[3:0]	tCCD_L.min (nCK)	tCCD_L_WR2.min (nCK)	tCCD_L_WR.min (nCK)	tDLLK.min (nCK)	Details	Notes
tCCD_L / tCCD_L_WR / tCCD_L_WR2 / tDLLK	0000	8	16	32	1024	1980 MT/s ≤ Data Rate ≤ 2100 MT/s & 2933 MT/s ≤ Data Rate ≤ 3200 MT/s	1, 2, 3
	0001	9	18	36	1024	3200 MT/s < Data Rate ≤ 3600 MT/s	
	0010	10	20	40	1280	3600 MT/s < Data Rate ≤ 4000 MT/s	
	0011	11	22	44	1280	4000 MT/s < Data Rate ≤ 4400 MT/s	
	0100	12	24	48	1536	4400 MT/s < Data Rate ≤ 4800 MT/s	
	0101	13	26	52	1536	4800 MT/s < Data Rate ≤ 5200 MT/s	
	0110	14	28	56	1792	5200 MT/s < Data Rate ≤ 5600 MT/s	
	0111	15	30	60	1792	5600 MT/s < Data Rate ≤ 6000 MT/s	
	1000	16	32	64	2048	6000 MT/s < Data Rate ≤ 6400 MT/s	
	1001	17	34	68	2048	6400 MT/s < Data Rate ≤ 6800 MT/s	
	1010	18	36	72	2304	6800 MT/s < Data Rate ≤ 7200 MT/s	
	1011	19	38	76	2304	7200 MT/s < Data Rate ≤ 7600 MT/s	
	1100	20	40	80	2560	7600 MT/s < Data Rate ≤ 8000 MT/s	
	1101	21	42	84	2560	8000 MT/s < Data Rate ≤ 8400 MT/s	
1110	22	44	88	2816	8400 MT/s < Data Rate ≤ 8800 MT/s		
	1111	Reserved					
NOTE 1	tCCD_L/tCCD_L_WR/tCCD_L_WR2/tDLLK are programmed according to the value defined in the AC parametric table per operating frequency.						
NOTE 2	The register type is "R" (read only) since MR13 is set by the "Configure tDLLK/tCCD_L/tCCD_L_WR/tCCD_L_WR2" MPC command.						
NOTE 3	Data rate ranges align with Speed Bin Table definitions.						

3.5.16 MR14 (MA[7:0]=0EH) - Transparency ECC Configuration**MR14 Register Information**

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
ECS Mode	Reset ECS Counter	Row Mode/ Code Word Mode	RFU	CID3	CID2	CID1	CID0

Function	Register Type	Operand	Data	Notes
ECS Error Register Index/ MBIST Rank Select	R/W	OP[3:0]	CID[3:0]	1, 2, 3, 4
RFU	RFU	OP[4]	RFU	
Code Word/Row Count	R/W	OP[5]	0B: ECS counts Rows with errors 1B: ECS counts Code words with errors	1
ECS Reset Counter	W	OP[6]	0B: Normal (Default) 1B: Reset ECC Counter	1,4
ECS Mode	R/W	OP[7]	0B: Manual ECS Mode Disabled (Default) 1B: Manual ECS Mode Enabled	1

NOTE 1 MR14:OP[3:0] must be setup by MRW to indicate which slice in the 3DS-DDR5 stack is referenced by the MRR for MR14MR20 ECS transparency data, MR22 MBIST transparency data, and MR54-MR57 hPPR resource availability. On 3DS devices that support optional MBIST/mPPR, prior to MBIST initialization via MR23:OP[4] followed by guard keys, MR14:OP[3:0] must be programmed by MRW according to the logical rank that is desired to perform MBIST.

NOTE 2 CID[3:0] encoding is based on the stack height of the device and varies depending on the number of dies in the stack.

NOTE 3 For Monolithic DDR5, CID[3:0] should be set to 0.

NOTE 4 ECS stands for Error Check Scrub operation.

3.5.17 MR15 (MA[7:0]=0F_H) - Transparency ECC Threshold per Gb of Memory Cells and Automatic ECS in Self Refresh

MR15 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
x4 Writes	ECS Writeback	RFU		Automatic ECS in Self Refresh	ECS Error Threshold Count (ETC)		

Function	Register Type	Operand	Data	Notes
ECS Error Threshold Count (ETC)	R/W	OP[2:0]	000B: 4 001B: 16 010B: 64 011B: 256 (Default) 100B: 1024 101B: 4096 110B: RFU 111B: RFU	
Automatic ECS in Self Refresh	W	OP[3]	0B: Automatic ECS disabled in Self-Refresh in Manual ECS mode (default) 1B: Automatic ECS enabled in Self-Refresh in Manual ECS mode	
RFU	RFU	OP[5:4]	RFU	
ECS Writeback	R/W	OP[6]	0B: Do not suppress writeback of Data and ECC Check Bits (Default) 1B: Suppress writeback of Data and ECC Check Bits (Optional)	4
x4 Writes	R/W	OP[7]	0B: Do not suppress writeback of Data during RMW (Default) 1B: Suppress writeback of Data during RMW (Optional)	4
NOTE 1 MR14:OP[3:0] applies to CID[3:0] for 3DS-DDR5 and must be setup to indicate which slice in the 3DS-DDR5 stack is referenced in the MR14 through MR20 transparency data.				
NOTE 2 DDR5 performs Automatic ECS operation while in Self-Refresh mode either by enabling MR15:OP[3]=1 _B (Automatic ECS in Self-Refresh enable) or disabling MR14:OP[7]=0 _B (Automatic ECS mode enable).				
NOTE 3 If the Automatic ECS in Self-Refresh is enabled, transparency mode-registers updated cannot be controlled by the number of Manual ECS operation MPC command since the ECS counter is increased by both manual ECS command and the Automatic ECS Operation in Self-Refresh mode.				
NOTE 4 DDR5 SPD Byte 14 Bits[2:1] indicates if feature is supported and will also indicate whether to use MR9 or MR15 for enabling the modes.				

3.5.18 MR16 (MA [7:0] = 10_H) - Row Address with Max Errors 1

MR16 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
R7	R6	R5	R4	R3	R2	R1	R0

Function	Register Type	Operand	Data	Notes
Max Row Error Address R[7:0]	R	OP[7:0]	Contains 8 bits of the row address with the highest error count	1
NOTE 1 MR14:OP[3:0] applies to CID[3:0] for 3DS-DDR5 and must be setup to indicate which slice in the 3DS-DDR5 stack is referenced in the MR14 through MR20 transparency data				

3.5.19 MR17 (MA [7:0] = 11_H) - Row Address with Max Errors 2

MR17 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
R15	R14	R13	R12	R11	R10	R9	R8

Function	Register Type	Operand	Data	Notes
Max Row Error Address R[15:8]	R	OP[7:0]	Contains 8 bits of the row address with the highest error count	1
NOTE 1 MR14:OP[3:0] applies to CID[3:0] for 3DS-DDR5 and must be setup to indicate which slice in the 3DS-DDR5 stack is referenced in the MR14 through MR20 transparency data				

3.5.20 MR18 (MA [7:0] = 12_H) - Row Address with Max Errors 3

MR18 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
RFU	BG2	BG1	BG0	BA1	BA0	R17	R16

Function	Register Type	Operand	Data	Notes
Max Row Error Address BG[2:0],BA[1,0], R[17,16]	R	OP[6:0]	Contains 7 bits of the row address with the highest error count	1, 2, 3, 4, 5
RFU	RFU	OP[7]	RFU	
NOTE 1 MR14:OP[3:0] applies to CID[3:0] for 3DS-DDR5 and must be setup to indicate which slice in the 3DS-DDR5 stack is referenced in the MR14 through MR20 transparency data.				
NOTE 2 BG2 is don't care for x16.				
NOTE 3 BA1 is don't care for 8 Gb.				
NOTE 4 R16 is don't care for 8 Gb and 16 Gb.				
NOTE 5 R17 is don't care for 8 Gb, 16 Gb, 24 Gb, and 32 Gb.				

3.5.21 MR19 (MA [7:0] = 13_H) - Max Row Error Count**MR19 Register Information**

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
PASR	RFU	REC5	REC4	REC3	REC2	REC1	REC0

Function	Register Type	Operand	Data	Notes
Max Row Error Count REC[5:0]	R	OP[5:0]	Contains number of errors within the row with the most errors	1
RFU	RFU	OP[6]	RFU	
PASR	R	OP[7]	0 = PASR not supported 1 = PASR supported	2

NOTE 1 MR14:OP[3:0] applies to CID[3:0] for 3DS-DDR5 and must be setup to indicate which slice in the 3DS-DDR5 stack is referenced in the MR14 through MR20 transparency data.

NOTE 2 Support of PASR has been deprecated starting with spec working revision 1.90 of JESD79-5C-v1.30.

3.5.22 MR20 (MA [7:0] = 14_H) - Error Count (EC)**MR20 Register Information**

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
EC7	EC6	EC5	EC4	EC3	EC2	EC1	EC0

Function	Register Type	Operand	Data	Notes
Error Count EC[7:0]	R	OP[7:0]	Contains the error count range data	1

NOTE 1 MR14:OP[3:0] applies to CID[3:0] for 3DS-DDR5 and must be setup to indicate which slice in the 3DS-DDR5 stack is referenced in the MR14 through MR20 transparency data.

3.5.23 MR21 (MA [7:0] = 15_H) - Rx CTLE Control Setting (DQS)

MR21 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
RFU	RFU	RFU	RFU	RFU	Rx DQS CTLE Control (Optional)		

Function	Register Type	Operand	Data	Notes
Rx DQS CTLE Control (Optional)	W	OP[2:0]	000B: Vendor Optimized Setting (Default) 001B: Vendor defined 010B: Vendor defined 011B: Vendor defined 100B: Vendor defined 101B: Vendor defined 110B: Vendor defined 111B: Vendor defined	1, 2, 3, 4, 5
RFU	RFU	OP[7:3]	RFU	

NOTE 1 Rx CTLE is an optional feature on DDR5. It may be needed for DRAMs that operate at >=6000Mbps. MR22:OP[3] indicates host whether Rx CTLE is supported or not.

NOTE 2 Refer to the vendor data sheets for more information regarding Rx CTLE Control Settings.

NOTE 3 Since CTLE circuits can not be typically bypassed, a disable option is not provided. Instead, a vendor optimized setting is given. It should be noted that the settings are not specifically linear in relationship to the vendor optimized setting, so the host may opt to instead walk through all the provided options and use the setting that works best in their environment.

NOTE 4 The host can step through all possible combinations of MR bit allocation and choose the settings that is best optimized for the system based on the performance metric of interest.

NOTE 5 MR21:OP[2:0] controls both upper and lower DQS (U/LDQS) for X16 DRAMs.

3.5.24 MR22 (MA [7:0] = 16_H) - MBIST/mPPR Transparency, Rx CTLE Control Setting (Support Indicator, CA, and CS_n)

MR22 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
Rx CS _n CTLE Control (Optional)		Rx CA CTLE Control (Optional)		Rx CTLE Support	MBIST/mPPR Transparency (Optional)		

Function	Register Type	Operand	Data	Notes
MBIST/mPPR Transparency (Optional)	R	OP[2:0]	000B: MBIST hasn't run since INIT OR no fails remain after most recent run (Default)	1, 2
			001B: Fails remain	2
			010B: Unrepairable fails remain	2
			011B: MBIST should be run again	2
			100B-111B: Reserved	2
Rx CTLE Support	R	OP[3]	0B: Rx CTLE not supported 1B: Rx CTLE supported	3
Rx CA CTLE Control	R/W	OP[5:4]	00B: Vendor Optimized Setting (Default) 01B: Vendor defined 10B: Vendor defined 11B: Vendor defined	3, 4, 5, 6
Rx CS CTLE Control	R/W	OP[7:6]	00B: Vendor Optimized Setting (Default) 01B: Vendor defined 10B: Vendor defined 11B: Vendor defined	3, 4, 5, 6

- NOTE 1 The host should track whether MBIST has run since INIT. If MBIST is run and finds no fails, this transparency state will remain set to 000_B.
- NOTE 2 MR14:OP[3:0] must be programmed by MRW to indicate which slice in the 3DS-DDR5 stack is referenced by the MRR for reading MR22 MBIST transparency data.
- NOTE 3 Rx CTLE is an optional feature on DDR5. It may be needed for DRAMs that operate at >=6000Mbps. MR22:OP[3] indicates host whether Rx CTLE is supported or not.
- NOTE 4 Refer to the vendor data sheets for more information regarding Rx CTLE Control Settings.
- NOTE 5 Since CTLE circuits can not be typically bypassed, a disable option is not provided. Instead, a vendor optimized setting is given. It should be noted that the settings are not specifically linear in relationship to the vendor optimized setting, so the host may opt to instead walk through all the provided options and use the setting that works best in their environment.
- NOTE 6 The host can step through all possible combinations of MR bit allocation and choose the settings that is best optimized for the system based on the performance metric of interest.

3.5.25 MR23 (MA [7:0] = 17_H) - MBIST/PPR Settings

MR23 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
RFU		RFU	MBIST (Optional)	mPPR (Optional)	sPPR		hPPR

Function	Register Type	Operand	Data	Notes
hPPR	R/W	OP[0]	0B: Disable 1B: Enable	1
sPPR	R/W	OP[1]	0B: Disable 1B: Enable See OP[2] for definition with sPPR Undo/Lock Implemented	1
	SR/W	OP[2]	DRAM Status Read (SR): 0B: Not Implemented (Default) 1B: sPPR Undo/Lock Implemented Host Write (W) for OP[2:1] 00B: Disabled (Normal Operation) 01B: sPPR Enabled 10B: sPPR Undo Enabled 11B: sPPR Lock Enabled	
mPPR	W	OP[3]	0B: Disable 1B: Enable (Optional)	1
MBIST	SR/W	OP[4]	DRAM Status Read (SR): 0B: No MBIST/mPPR Support 1B: Supports MBIST/mPPR (Optional) Host Write (W): 0B: MBIST Disabled 1B: MBIST Enable	1,2
RFU	RFU	OP[7:5]	RFU	

NOTE 1 Only one of these opcode bits may be programmed by the host to 1 at any given time. If any one of these opcode bits are enabled, the remaining bits must be programmed to 0.

NOTE 2 DRAM will automatically write to 0 when MBIST completes. Therefore, the host is not required to program to zero before performing MBIST again.

NOTE 3 For 3DS-DDR5 devices, MBIST Enable (MR23:OP[4]=1) is only enabled on the target logical rank designated by CID[3:0] and programmed by MRW via MR14:OP[3:0].

3.5.26 MR24 (MA [7:0] = 18_H) - PPR Guard Key

MR24 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
PPR Guard Key							

Function	Register Type	Operand	Data	Notes
PPR Guard Key	W	OP[7:0]	See PPR Section for Sequence	

3.5.27 MR25 (MA[7:0]=19_H) - Read Training Mode Settings**MR25 Register Information**

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
RFU				Continuous Burst Mode	LFSR1 Pattern Option	LFSR0 Pattern Option	Read Training Pattern Format

Function	Register Type	Operand	Data	Notes
Read Training Pattern Format	R/W	OP[0]	0B: Serial 1B: LFSR	
LFSR0 Pattern Option	R/W	OP[1]	0B: LFSR 1B: Clock	
LFSR1 Pattern Option	R/W	OP[2]	0B: LFSR 1B: Clock	
Continuous Burst Mode	R/W	OP[3]	0B: MRR command based (Default) 1B: Continuous Burst Output	
RFU	RFU	OP[7:4]	RFU	

3.5.28 MR26 (MA[7:0]=1A_H) - Read Pattern Data0 / LFSR0

MR26 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
Read Training Pattern Data0 / LFSR0 Seed							

Function	Register Type	Operand	Data	Notes
Read Pattern / LFSR Seed UI 0	R/W	OP[0]	UI<7:0> data for serial mode, LFSR0 seed for LFSR mode	1
Read Pattern / LFSR Seed UI 1	R/W	OP[1]		
Read Pattern / LFSR Seed UI 2	R/W	OP[2]		
Read Pattern / LFSR Seed UI 3	R/W	OP[3]		
Read Pattern / LFSR Seed UI 4	R/W	OP[4]		
Read Pattern / LFSR Seed UI 5	R/W	OP[5]		
Read Pattern / LFSR Seed UI 6	R/W	OP[6]		
Read Pattern / LFSR Seed UI 7	R/W	OP[7]		
NOTE 1 The default value for the Read Training Pattern Data0/LFSR0 register setting is: 0x5A.				

3.5.29 MR27 (MA[7:0]=1B_H) - Read Pattern Data1 / LFSR1

MR27 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
Read Training Pattern Data1 / LFSR1 Seed							

Function	Register Type	Operand	Data	Notes
Read Pattern / LFSR Seed UI 8	R/W	OP[0]	UI<15:8> data for serial mode, LFSR1 seed for LFSR mode	1
Read Pattern / LFSR Seed UI 9	R/W	OP[1]		
Read Pattern / LFSR Seed UI 10	R/W	OP[2]		
Read Pattern / LFSR Seed UI 11	R/W	OP[3]		
Read Pattern / LFSR Seed UI 12	R/W	OP[4]		
Read Pattern / LFSR Seed UI 13	R/W	OP[5]		
Read Pattern / LFSR Seed UI 14	R/W	OP[6]		
Read Pattern / LFSR Seed UI 15	R/W	OP[7]		
NOTE 1 The default value for the Read Training Pattern Data1/LFSR1 register setting is: 0x3C.				

3.5.30 MR28 (MA[7:0]=1C_H) - Read Pattern Invert DQL7:0 (DQ7:0)**MR28 Register Information**

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
Read Training Pattern Invert DQL7:0 (DQ7:0)							

Function	Register Type	Operand	Data	Notes
DQ Invert (Lower DQ Bits)	R/W	OP[0]	DQL0 (DQ0) 0B: Normal 1B: Invert	1
	R/W	OP[1]	DQL1 (DQ1) 0B: Normal 1B: Invert	
	R/W	OP[2]	DQL2 (DQ2) 0B: Normal 1B: Invert	
	R/W	OP[3]	DQL3 (DQ3) 0B: Normal 1B: Invert	
	R/W	OP[4]	DQL4 (DQ4) 0B: Normal 1B: Invert	
	R/W	OP[5]	DQL5 (DQ5) 0B: Normal 1B: Invert	
	R/W	OP[6]	DQL6 (DQ6) 0B: Normal 1B: Invert	
	R/W	OP[7]	DQL7 (DQ7) 0B: Normal 1B: Invert	
NOTE 1 The default value for the Read Training Pattern Invert DQL7:0 (DQ7:0) register setting is: 0x00.				

3.5.31 MR29 (MA[7:0]= D_H) - Read Pattern Invert DQU7:0 (DQ15:8)

MR29 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
Read Training Pattern Invert DQU7:0 (DQ15:8)							

Function	Register Type	Operand	Data	Notes
DQ Invert (Upper DQ Bits)	R/W	OP[0]	DQU0 (DQ8) 0B: Normal 1B: Invert	1
	R/W	OP[1]	DQU1 (DQ9) 0B: Normal 1B: Invert	
	R/W	OP[2]	DQU2 (DQ10) 0B: Normal 1B: Invert	
	R/W	OP[3]	DQU3 (DQ11) 0B: Normal 1B: Invert	
	R/W	OP[4]	DQU4 (DQ12) 0B: Normal 1B: Invert	
	R/W	OP[5]	DQU5 (DQ13) 0B: Normal 1B: Invert	
	R/W	OP[6]	DQU6 (DQ14) 0B: Normal 1B: Invert	
	R/W	OP[7]	DQU7 (DQ15) 0B: Normal 1B: Invert	
NOTE 1 The default value for the Read Training Pattern Invert DQU7:0 (DQ15:8) register setting is: 0x00.				

3.5.32 MR30 (MA[7:0]=1E_H) - Read LFSR Assignments

MR30 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
LFSR Assignment DQL7/ DQU7	LFSR Assignment DQL6/ DQU6	LFSR Assignment DQL5/ DQU5	LFSR Assignment DQL4/ DQU4	LFSR Assignment DQL3/ DQU3	LFSR Assignment DQL2/ DQU2	LFSR Assignment DQL1/ DQU1	LFSR Assignment DQL0/ DQU0

Function	Register Type	Operand	Data	Notes
LFSR Assignment DQL0/DQU0	R/W	OP[0]	0B: Read Pattern Data0/ LFSR0 1B: Read Pattern Data1/ LFSR1	1
LFSR Assignment DQL1/DQU1	R/W	OP[1]	0B: Read Pattern Data0/ LFSR0 1B: Read Pattern Data1/ LFSR1	
LFSR Assignment DQL2/DQU2	R/W	OP[2]	0B: Read Pattern Data0/ LFSR0 1B: Read Pattern Data1/ LFSR1	
LFSR Assignment DQL3/DQU3	R/W	OP[3]	0B: Read Pattern Data0/ LFSR0 1B: Read Pattern Data1/ LFSR1	
LFSR Assignment DQL4/DQU4	R/W	OP[4]	0B: Read Pattern Data0/ LFSR0 1B: Read Pattern Data1/ LFSR1	
LFSR Assignment DQL5/DQU5	R/W	OP[5]	0B: Read Pattern Data0/ LFSR0 1B: Read Pattern Data1/ LFSR1	
LFSR Assignment DQL6/DQU6	R/W	OP[6]	0B: Read Pattern Data0/ LFSR0 1B: Read Pattern Data1/ LFSR1	
LFSR Assignment DQL7/DQU7	R/W	OP[7]	NOTE 10B: Read Pattern Data0/LFSR0 1B: Read Pattern Data1/LFSR1	
NOTE 1 The default value for the Read LFSR Assignments register setting is: 0xFE.				

3.5.33 MR31 (MA[7:0]=1F_H) - Read Training Pattern Address

MR31 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
Read Training Pattern Address							

Function	Register Type	Operand	Data	Notes
Read Training Pattern Address	R	OP[7:0]	This MR address is reserved. There are no specific register fields associated with this address. In response to the MRR to this address the DRAM shall send the BL16 read training pattern. All 8 bits associated with this MR address are reserved.	

3.5.34 MR32 (MA[7:0]=20H) - CK and CS ODT

MR32 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
RFU	CA_ODT Strap Value	CS ODT			CK ODT		

Function	Register Type	Operand	Data	Notes
CK ODT	R	OP[2:0]	000 _B : RTT_OFF (Disable) Group A default 001 _B : RZQ/0.5 (480) 010 _B : RZQ/1 (240) 011 _B : RZQ/2 (120) 100 _B : RZQ/3 (80) 101 _B : RZQ/4 (60) 110 _B : RFU 111 _B : RZQ/6 (40) Group B default	1, 2
CS ODT	R	OP[5:3]	000 _B : RTT_OFF (Disable) Group A default 001 _B : RZQ/0.5 (480) 010 _B : RZQ/1 (240) 011 _B : RZQ/2 (120) 100 _B : RZQ/3 (80) 101 _B : RZQ/4 (60) 110 _B : RFU 111 _B : RZQ/6 (40) Group B default	1, 2
CA_ODT Strap Value	R	OP[6]	0 _B : Strap Configured to Group A 1 _B : Strap Configured to Group B	3
RFU	RFU	OP[7]	RFU	

NOTE 1 This mode register is programmed via an explicit MPC command only.
 NOTE 2 CK ODT RZQ/2 (120 Ω) and CS ODT RZQ/2 (120 Ω) are only supported at >5200 MT/s data rates.
 NOTE 3 Strapping for ODT on Command and Address. The DRAM applies to Group A settings if the CA_ODT pin is connected to VSS and applies Group B settings if the pin is connected to VDD. This MR is used to confirm the DRAM's setting for that config.

3.5.35 MR33 (MA[7:0]=21H) - CA and DQS_PARK ODT**MR33 Register Information**

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
RFU		DQS_RTT_PARK			CA ODT		

Function	Register Type	Operand	Data	Notes
CA ODT	R	OP[2:0]	000B: RTT_OFF (Disable) Group A default 001B: RZQ/0.5 (480) 010B: RZQ/1 (240) 011B: RZQ/2 (120) 100B: RZQ/3 (80) Group B default 101B: RZQ/4 (60) 110B: RFU 111B: RZQ/6 (40)	1,2
DQS_RTT_PARK	R	OP[5:3]	000B: RTT_OFF default 001B: RZQ (240) 010B: RZQ/2 (120) 011B: RZQ/3 (80) 100B: RZQ/4 (60) 101B: RZQ/5 (48) 110B: RZQ/6 (40) 111B: RZQ/7 (34)	1
RFU	RFU	OP[7:6]	RFU	

NOTE 1 This mode register is programmed via an explicit MPC command only.

NOTE 2 CA ODT RZQ/2 (120Ω) is only supported at >5200 MT/s data rates.

3.5.36 MR34 (MA[7:0]=22_H) - RTT_PARK and RTT_WR

MR34 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
RFU		RTT_WR			RTT_PARK		

Function	Register Type	Operand	Data	Notes
RTT_PARK	R	OP[2:0]	000 _B : RTT_OFF default 001 _B : RZQ (240) 010 _B : RZQ/2 (120) 011 _B : RZQ/3 (80) 100 _B : RZQ/4 (60) 101 _B : RZQ/5 (48) 110 _B : RZQ/6 (40) 111 _B : RZQ/7 (34)	1
RTT_WR	R/W	OP[5:3]	000 _B : RTT_OFF 001 _B : RZQ (240) default 010 _B : RZQ/2 (120) 011 _B : RZQ/3 (80) 100 _B : RZQ/4 (60) 101 _B : RZQ/5 (48) 110 _B : RZQ/6 (40) 111 _B : RZQ/7 (34)	
RFU	RFU	OP[7:6]	RFU	
NOTE 1 This mode register is programmed via an explicit MPC command only.				

3.5.37 MR35 (MA[7:0]=23_H) - RTT_NOM_WR and RTT_NOM_RD**MR35 Register Information**

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
RFU		RTT_NOM_RD			RTT_NOM_WR		

Function	Register Type	Operand	Data	Notes
RTT_NOM_WR	R/W	OP[2:0]	000B: RTT_OFF 001B: RZQ (240) 010B: RZQ/2 (120) 011B: RZQ/3 (80) default 100B: RZQ/4 (60) 101B: RZQ/5 (48) 110B: RZQ/6 (40) 111B: RZQ/7 (34)	
RTT_NOM_RD	R/W	OP[5:3]	000B: RTT_OFF 001B: RZQ (240) 010B: RZQ/2 (120) 011B: RZQ/3 (80) default 100B: RZQ/4 (60) 101B: RZQ/5 (48) 110B: RZQ/6 (40) 111B: RZQ/7 (34)	
RFU	RFU	OP[7:6]	RFU	

3.5.38 MR36 (MA[7:0]=24_H) - RTT Loopback

MR36 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
RFU					RTT Loopback		

Function	Register Type	Operand	Data	Notes
RTT Loopback	R/W	OP[2:0]	000B: RTT_OFF Default 001B: RFU 010B: RFU 011B: RFU 100B: RFU 101B: RZQ/5 (48) 110B: RFU 111B: RFU	1, 2
RFU	RFU	OP[7:3]	RFU	

NOTE 1 When Loopback is disabled, both LBDQS and LBDQ pins are either at HiZ or Termination Mode this configuration. When Loopback is enabled, it is in driver mode.

3.5.39 MR37 (MA[7:0]= 25_H) - ODTL Write Control Offset**MR37 Register Information**

This byte is setup to allow the host controller to push out or pull in the Write RTT enable time (tODTLon_WR) or the Write RTT disable time (tODTLoff_WR) outside of the default setting. The default state is based on internal DRAM design and is defined in Table 183 in the ODT Configuration section. The DRAM is not responsible for inappropriate states set by the host controller using this register.

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
RFU		ODTLoff_WR_Offset			ODTLon_WR_Offset		

Function	Register Type	Operand	Data	Notes
ODTLon_WR_Offset	R/W	OP[2:0]	000B: RFU 001B: -4 Clocks 010B: -3 Clocks 011B: -2 Clocks 100B: -1 Clock - Default 101B: 0 Clock 110B: +1 Clock 111B: +2 Clocks	1
ODTLoff_WR_Offset	R/W	OP[5:3]	000B: RFU 001B: +4 Clocks 010B: +3 Clocks 011B: +2 Clocks 100B: +1 Clock 101B: 0 Clock - Default 110B: -1 Clock 111B: -2 Clocks	1
RFU	RFU	OP[7:6]	RFU	

NOTE 1 The Write ODTLon -4 offset and ODTLoff +4 offset is not allowed for 1980-2100 data rates.

3.5.40 MR38 (MA[7:0]=26_H) - ODTL NT Write Control Offset

MR38 Register Information

This byte is setup to allow the host controller to push out or pull in the Non-Target Write RTT enable time (tODTLon_WR_NT) or the Non-Target Write RTT disable time (tODTLoff_WR_NT) outside of the default setting. The default state is based on internal DRAM design and is defined in Table 183 in the ODT Configuration section. The DRAM is not responsible for inappropriate states set by the host controller using this register.

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
RFU		ODTLoff_WR_NT_Offset			ODTLon_WR_NT_Offset		

Function	Register Type	Operand	Data	Notes
ODTLon_WR_NT_Offset	R/W	OP[2:0]	000B: RFU 001B: -4 Clocks 010B: -3 Clocks 011B: -2 Clocks 100B: -1 Clock - Default 101B: 0 Clock 110B: +1 Clock 111B: +2 Clocks	1
ODTLoff_WR_NT_Offset	R/W	OP[5:3]	000B: RFU 001B: +4 Clocks 010B: +3 Clocks 011B: +2 Clocks 100B: +1 Clock 101B: 0 Clock - Default 110B: -1 Clock 111B: -2 Clocks	1
RFU	RFU	OP[7:6]	RFU	

NOTE 1 The Write NT ODTLon -4 offset and ODTLoff +4 offset is not allowed for 1980-2100 data rates.

3.5.41 MR39 (MA[7:0]=27_H) - ODTL NT Read Control Offset**MR39 Register Information**

This byte is setup to allow the host controller to push out or pull in the Non-Target Read RTT enable time (tODTLon_RD_NT) or the Non-Target Read RTT disable time (tODTLoff_RD_NT) outside of the default setting. The default state is based on internal DRAM design and is defined in Table 183 in the ODT Configuration section. The DRAM is not responsible for inappropriate states set by the host controller using this register.

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
RFU		ODTLoff_RD_NT_Offset			ODTLon_RD_NT_Offset		

Function	Register Type	Operand	Data	Notes
ODTLon_RD_NT_Offset	R/W	OP[2:0]	000 _B : RFU 001 _B : RFU 010 _B : -3 Clocks 011 _B : -2 Clocks 100 _B : -1 Clock - Default 101 _B : 0 Clock 110 _B : +1 Clock 111 _B : RFU	
ODTLoff_RD_NT_Offset	R/W	OP[5:3]	000 _B : RFU 001 _B : RFU 010 _B : +3 Clocks 011 _B : +2 Clocks 100 _B : +1 Clock 101 _B : 0 Clock - Default 110 _B : -1 Clock 111 _B : RFU	
RFU	RFU	OP[7:6]	RFU	

3.5.42 MR40 (MA[7:0]=28_H) - Read DQS Offset Timing

MR40 Register Information

This byte is used for configuring the DRAM to support different HOST receiver designs.

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
RFU					Read DQS offset timing		

Function	Register Type	Operand	Data	Notes
Read DQS offset timing	R/W	OP[2:0]	000 _B : 0 Clock (DEFAULT) 001 _B : 1 Clock 010 _B : 2 Clocks 011 _B : 3 Clocks 100 _B : RFU 101 _B : RFU 110 _B : RFU 111 _B : RFU	
RFU	RFU	OP[7:3]	RFU	

NOTE 1 When operating at low speed (CL ≤ 30), tRPRE + Read DQS Offset ≥ 5 Clocks cannot be supported.

When CL ≤ 30	tRPRE \ DQS Offset	0	1	2	3
	1	O	O	O	O
	2	O	O	O	X
	3	O	O	X	X
	4	O	X	X	X

3.5.43 MR41 (MA[7:0]=29_H) - RFU

MR41 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
RFU							

Function	Register Type	Operand	Data	Notes

3.5.44 MR42 (MA[7:0]=2A_H) - DCA Types Supported

MR42 Register Information

This byte is used for configuring DCA

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DCA Training Assist Mode II (Optional)				DCA Training Assist Mode I		DCA Types Supported	

MR42 Table

Function	Register Type	Operand	Data	Notes
DCA Types Supported	R	OP[1:0]	00 _B : Device does not support DCA 01 _B : Device supports DCA for single/two-phase internal clock(s) 10 _B : Device supports DCA for 4-phase internal clocks 11 _B : RFU	
DCA Training Assist Mode I	R/W	OP[3:2]	00 _B : Disable (default) 01 _B : MRR (or Read) synchronized with IBCLK is blocked 10 _B : MRR (or Read) synchronized with ICLK is blocked 11 _B : RFU	1, 2, 3, 4, 5, 6, 9
DCA Training Assist Mode II (Optional)	R/W	OP[6:4]	000 _B : Disable (default) 001 _B : DQ fires 0001 data pattern 010 _B : DQ fires 0011 data pattern 011 _B : DQ fires 0111 data pattern 100 _B : DQ fires 1000 data pattern 101 _B : DQ fires 1100 data pattern 110 _B : DQ fires 1110 data pattern 111 _B : RFU	3, 4, 5, 7, 8, 9, 10, 11
	SR/W	OP[7]	DRAM Status Read (SR) 0 _B : DCA Training Assist Mode II is not supported 1 _B : DCA Training Assist Mode II is supported Host Write (W): 0 _B : MRR based (Default) 1 _B : Continuous Burst output	

MR42 Table (cont'd)

NOTE 1	When "MRR (or Read) synchronized with IBCLK is blocked" is set by MR42 OP[3:2]=01b, DQs caused by MRR (or Read) synchronized with IBCLK are driven HIGH.
NOTE 2	When "MRR (or Read) synchronized with ICLK is blocked" is set by MR42 OP[3:2]=10b, DQs caused by MRR (or Read) synchronized with ICLK are driven HIGH.
NOTE 3	DQS_t/DQS_c output normal toggling waveforms meaning that DQS_t/DQS_c are not affected by the settings of DCA Assist Mode MR42 OP[3:2].
NOTE 4	The CRC function is not supported during DCA Training Assist Mode.
NOTE 5	DCA Training Assist Mode is only supported by DRAMs with DCAs that have 4-phase internal clocks
NOTE 6	If MR42:OP[3:2] is set to either 01B or 10B, odd-gap READ or odd-gap Mode Register Read Pattern commands should follow the tMRR timing spec.
NOTE 7	The MRR based mode of DCA Training Assist Mode II is enabled when the host sets MR42 OP[6:4]≠000 & OP[7]=0. DRAM will provide selected predefined pattern when the host issues MRR to MR42 command to the DRAM. For MRR, issuing any MRR other than to MR42 is invalid and may result in unexpected DQ behavior, requiring the host to issue a Reset to exit MRR based mode and restart the training with an MRR to MR42. The returned data will be a predefined pattern instead of the contents of a mode register. The timing of the read data return is the same as for an MRR command, including the operation of the strobes (DQS_t, DQS_c). The host can issue MRW to MR42:OP[6:4]≠000 _B to change data pattern during MRR based mode. The host needs to issue MRW to MR42:OP[6:4]=000 _B to exit the MRR based mode.
NOTE 8	Continuous burst output mode is available for DCA Training Assist Mode II and is enabled with MRW to MR42 OP[6:4]≠000 _B & OP[7]=1. Once this mode is enabled, MRR to MR42 will start the pattern output and will automatically continue to output the appropriate pattern until it is stopped by either a system reset or issuing an MRW MR42 OP[7]=0 command that reverts it to the "MRR based (Default)" mode. Issuing any MRR other than to MR42 is invalid and may result in unexpected DQ behavior, requiring the host to issue a Reset to exit continuous burst output mode and restart the training with an MRR to MR42. Once the MR42 OP[7]=0 "MRR based (Default)" is registered by the DRAM, it will stop all pattern traffic by tCont_Exit. Since there is no min time for tCont_Exit, the DRAM may stop the pattern prior to tCont_Exit, potentially truncating any current burst pattern. To ensure that the DRAM's state-machine doesn't get into some meta-stability while turning off the output pattern, the host must issue a second MR42 OP[7]=0 "MRR based (Default)" after waiting tMRR, which will then start tCont_Exit_delay. After tCont_Exit_delay has expired, any other valid command is then legal. Issuing any MRW other than MRW to MR 42 OP[7]=0 during Continuous Burst Mode after predefined pattern output started is invalid and may result in unexpected DQ behavior, requiring the host to issue a Reset to exit continuous burst output mode and restart the training with an MRR to MR 42
NOTE 9	The DCA Training Assist Mode I and II can not be enabled at the same time by the host. That is, while the DCA Training Assist Mode II is enabled, the DCA Training Assist Mode I shall be disabled by Host and vice versa.
NOTE 10	Depending on relationship of internal clocks and MRR command, DRAM may start outputting from the 3rd bit of data pattern. The data pattern is defined with ICLK bases (1st=ICLK, 2nd=QCLK, 3rd=IBCLK and 4th=QBCLK), but MRR burst may start from IBCLK.
NOTE 11	In DCA Assist Mode II, MRR to MR42 outputs the repetitive data pattern through all DQ bits for a 16-bit burst. For example, if MR42:OP[7:4]=0010B, and MRR to MR42 is issued, all DQs output the same 0011001100110011 data pattern.

3.5.45 MR43 (MA[7:0]=2B_H) - DCA Settings 1

MR43 Register Information

This byte is used for configuring DCA

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
Sign Bit for OP[6:4]	DCA for IBCLK in 4-phase clocks			Sign Bit for OP[2:0]	DCA for single/two-phase clock(s) or QCLK in 4-phase clocks		

Function	Register Type	Operand	Data	Notes
DCA for single/two-phase clock(s) or QCLK in 4-phase clocks	W	OP[2:0]	000 _B : DCA step +0 (default) 001 _B : DCA step +1 010 _B : DCA step +2 011 _B : DCA step +3 100 _B : DCA step +4 101 _B : DCA step +5 110 _B : DCA step +6 111 _B : DCA step +7	1
Sign Bit for OP[2:0]	W	OP[3]	0 _B : Positive Offset (default) 1 _B : Negative Offset	1
DCA for IBCLK in 4-phase clocks	W	OP[6:4]	000 _B : DCA step +0 (default) 001 _B : DCA step +1 010 _B : DCA step +2 011 _B : DCA step +3 100 _B : DCA step +4 101 _B : DCA step +5 110 _B : DCA step +6 111 _B : DCA step +7	2
Sign Bit for OP[6:4]	W	OP[7]	0 _B : Positive Offset (default) 1 _B : Negative Offset	2
NOTE 1 These settings can only be applied if MR42:OP[1:0] = 01 _B or 10 _B .				
NOTE 2 These settings can only be applied if MR42:OP[1:0] = 10 _B .				

3.5.46 MR44 (MA[7:0]=2C_H) - DCA Settings 2

MR44 Register Information

This byte is used for configuring DCA

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
RFU				Sign Bit for QBCLK in 4-phase clocks	DCA for QBCLK in 4-phase clocks		

Function	Register Type	Operand	Data	Notes
DCA for QBCLK in 4-phase clocks	W	OP[2:0]	000B: DCA step +0 (default) 001B: DCA step +1 010B: DCA step +2 011B: DCA step +3 100B: DCA step +4 101B: DCA step +5 110B: DCA step +6 111B: DCA step +7	1
Sign Bit for QBCLK in 4-phase clocks	W	OP[3]	0B: Positive Offset (default) 1B: Negative Offset	1
RFU	RFU	OP[7:4]	RFU	

NOTE 1 These settings can only be applied if MR42 OP[1:0]=10_B.

3.5.47 MR45 (MA[7:0]=2D_H) - DQS Interval Control

MR45 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQS Interval Timer Run Time							

Function	Register Type	Operand	Data	Notes
DQS Interval Timer Run Time	W	OP[7:0]	0000 0000B: DQS interval timer stop via MPC Command (Default) 0000 0001B: DQS timer stops automatically at 16th clocks after timer start 0000 0010B: DQS timer stops automatically at 32nd clocks after timer start 0000 0011B: DQS timer stops automatically at 48th clocks after timer start 0000 0100B: DQS timer stops automatically at 64th clocks after timer start ----- Thru ----- 0011 1111B: DQS timer stops automatically at (63X16)th clocks after timer start 01XX XXXXB: DQS timer stops automatically at 2048th clocks after timer start 10XX XXXXB: DQS timer stops automatically at 4096th clocks after timer start 11XX XXXXB: DQS timer stops automatically at 8192nd clocks after timer start	1, 2
NOTE 1 MPC command with OP[7:0]=0000 0110 _B (Stop DQS Interval Oscillator) stops DQS interval timer in case of MR45:OP[7:0] = 00000000 _B .				
NOTE 2 MPC command with OP[7:0]=0000 0110 _B (Stop DQS Interval Oscillator) is illegal with non-zero values in MR45:OP[7:0].				

3.5.48 MR46 (MA[7:0]=2E_H) - DQS Osc Count - LSB

MR46 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQS Oscillator Count - LSB							

Function	Register Type	Operand	Data	Notes
DQS Oscillator Count - LSB	R	OP[7:0]	0 - 255 LSB DRAM DQS Oscillator Count	
NOTE 1 MR46 reports the LSB bits of the DRAM DQS Oscillator count. The DRAM DQS Oscillator count value is used to train DQS to the DQ data valid window. The value reported by the DRAM in this mode register can be used by the memory controller to periodically adjust the phase of DQS relative to DQ.				
NOTE 2 Both MR46 and MR47 must be read (MRR) and combined to get the value of the DQS Oscillator count.				
NOTE 3 A new MPC [Start DQS Oscillator] should be issued to reset the contents of MR46/MR47.				

3.5.49 MR47 (MA[7:0]=2F_H) - DQS Osc Count - MSB

MR47 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQS Oscillator Count - MSB							

Function	Register Type	Operand	Data	Notes
DQS Oscillator Count - MSB	R	OP[7:0]	0 - 255 MSB DRAM DQS Oscillator Count	
NOTE 1 MR47 reports the MSB bits of the DRAM DQS Oscillator count. The DRAM DQS Oscillator count value is used to train DQS to the DQ data valid window. The value reported by the DRAM in this mode register can be used by the memory controller to periodically adjust the phase of DQS relative to DQ.				
NOTE 2 Both MR46 and MR47 must be read (MRR) and combined to get the value of the DQS Oscillator count.				
NOTE 3 A new MPC [Start DQS Oscillator] should be issued to reset the contents of MR46/MR47.				

3.5.50 MR48 (MA[7:0]=30_H) - Write Pattern Mode

MR48 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
Write Pattern Mode							

Function	Register Type	Operand	Data	Notes
DQL0/DQU0	R/W	OP[0]	Valid	1, 2
DQL1/DQU1	R/W	OP[1]		
DQL2/DQU2	R/W	OP[2]		
DQL3/DQU3	R/W	OP[3]		
DQL4/DQU4	R/W	OP[4]		
DQL5/DQU5	R/W	OP[5]		
DQL6/DQU6	R/W	OP[6]		
DQL7/DQU7	R/W	OP[7]		
NOTE 1 OP[7:0] can be independently programmed with either “0” or “1”.				
NOTE 2 Default is all zeros for OP[7:0]				

3.5.51 MR50 (MA[7:0]=32_H) - Write CRC Settings

MR50 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
RFU	RFU	Write CRC auto-disable status	Write CRC auto-disable enable	Write CRC error status	Write CRC enable upper nibble	Write CRC enable lower nibble	Read CRC enable

Function	Register Type	Operand	Data	Notes
Read CRC enable	R/W	OP[0]	0B: Disable (Default) 1B: Enable	
Write CRC enable lower nibble	R/W	OP[1]	0B: Disable (Default) 1B: Enable	1
Write CRC enable upper nibble	R/W	OP[2]	0B: Disable (Default) 1B: Enable	1
Write CRC error status	R/W	OP[3]	0B: Clear 1B: Error	2
Write CRC auto-disable enable	R/W	OP[4]	0B: Disable (Default) 1B: Enable	
Write CRC auto-disable status	R/W	OP[5]	0B: Not triggered 1B: Triggered	
RFU	RFU	OP[6]	RFU	
RFU	RFU	OP[7]	RFU	

NOTE 1 When at least one of the two write CRC enable bits is set to '1' in x8, the timing of write CRC enable mode is applied to the entire device (i.e. both nibbles). When write CRC is enabled in one nibble and disabled in the other nibble in x8, then the DRAM does not check CRC errors on the disabled nibble, and hence the ALERT_n signal and any internal status bit related to CRC error is not impacted by the disabled nibble.

NOTE 2 The host shall disable Write CRC, if it was enabled, prior to entering Write Leveling Training mode.

3.5.52 MR51 (MA[7:0]=33_H) - Write CRC Auto-Disable Threshold**MR51 Register Information**

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
RFU	bit[6]	bit[5]	bit[4]	bit[3]	bit[2]	bit[1]	bit[0]

Function	Register Type	Operand	Data	Notes
Write CRC auto-disable threshold	R/W	OP[6:0]	0000000B: 0 ... 1111111B: 127	
RFU	RFU	OP[7]	RFU	

3.5.53 MR52 (MA[7:0]=34_H) - Write CRC Auto-Disable Window**MR52 Register Information**

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
RFU	bit[6]	bit[5]	bit[4]	bit[3]	bit[2]	bit[1]	bit[0]

Function	Register Type	Operand	Data	Notes
Write CRC auto-disable window	R/W	OP[6:0]	0000000B: 0 ... 1111111B: 127	
RFU	RFU	OP[7]	RFU	

3.5.54 MR53 (MA[7:0]=35_H) - Loopback**MR53 Register Information**

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
Loopback Output Mode	Loopback Select Phase		Loopback Output Select				

Function	Register Type	Operand	Data	Notes
Loopback Output Select	R/W	OP[4:0]	00000B: Loopback Disabled (Default) 00001B: Loopback DML (X8 and X16 only) 00010B: Loopback DMU (X16 only) 00011B: Vendor Specific 00100B: Vendor Specific 00101B: RFU ...thru 01111B: RFU 10000B: Loopback DQL0 10001B: Loopback DQL1 10010B: Loopback DQL2 10011B: Loopback DQL3 10100B: Loopback DQL4 (X8 and X16 only) 10101B: Loopback DQL5 (X8 and X16 only) 10110B: Loopback DQL6 (X8 and X16 only) 10111B: Loopback DQL7 (X8 and X16 only) 11000B: Loopback DQU0 (X16 only) 11001B: Loopback DQU1 (X16 only) 11010B: Loopback DQU2 (X16 only) 11011B: Loopback DQU3 (X16 only) 11100B: Loopback DQU4 (X16 only) 11101B: Loopback DQU5 (X16 only) 11110B: Loopback DQU6 (X16 only) 11111B: Loopback DQU7 (X16 only)	1, 2, 5
Loopback Select Phase	R/W	OP[6:5]	00B: Loopback Select Phase A 01B: Loopback Select Phase B (4-way and 2-way interleave only) 10B: Loopback Select Phase C (4-way interleave only) 11B: Loopback Select Phase D (4-way interleave only)	3
Loopback Output Mode	R/W	OP[7]	0B: Normal Output (Default) 1B: Write Burst Output	4
NOTE 1 When Loopback is disabled, both LBDQS and LBDQ pins are either at HiZ or Termination Mode per MR36:OP[2:0]. Loopback Termination default value is 48 ohms NOTE 2 When Loopback is enabled, both LBDQS and LBDQ pins are in driver mode using default RON of 34 ohms NOTE 3 Phase A through D selects which bit in the multiplexer is being selected for Loopback output NOTE 4 This configures the DRAM Loopback output to either send data out every time the DQS toggles in Normal Output Mode, or to only send data out when enabled by the Write command, so that only write burst data is send out via Loopback. NOTE 5 The DM function shall be enabled (MR5:OP[5]=1) when loopback output from the DML or DMU pin is selected for Loopback measurement (MR53:OP[4:0]=00001 _B or 00010 _B).				

3.5.55 MR54 (MA[7:0]=36H) - hPPR Resources

With hPPR, DDR5 can correct one Row address per Bank Group and the Electrical-fuse cannot be switched back to un-fused states once it is programmed. The controller should prevent unintended the hPPR mode entry and repair. (i.e. During the Command/Address training period). Entry into hPPR is through a register enable, ACT command is used to transmit the bank and row address of the row to be replaced in DRAM. After tRCD time, a WR command is used to select the individual DRAM through the DQ bits and to transfer the repair address to the DRAM. After program time, and PRE, the hPPR mode can be exited and normal operation can resume.

MR54 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
hPPR Resource BG1 Bank 3	hPPR Resource BG1 Bank 2	hPPR Resource BG1 Bank 1	hPPR Resource BG1 Bank 0	hPPR Resource BG0 Bank 3	hPPR Resource BG0 Bank 2	hPPR Resource BG0 Bank 1	hPPR Resource BG0 Bank 0

Function	Register Type	Operand	Data	Notes
hPPR Resource BG0 Bank 0	R	OP[0]	0B: hPPR Resource is not available 1B: hPPR Resource is available	3
hPPR Resource BG0 Bank 1		OP[1]	0B: hPPR Resource is not available 1B: hPPR Resource is available	3
hPPR Resource BG0 Bank 2		OP[2]	0B: hPPR Resource is not available 1B: hPPR Resource is available	1, 3
hPPR Resource BG0 Bank 3		OP[3]	0B: hPPR Resource is not available 1B: hPPR Resource is available	1, 3
hPPR Resource BG1 Bank 0		OP[4]	0B: hPPR Resource is not available 1B: hPPR Resource is available	3
hPPR Resource BG1 Bank 1		OP[5]	0B: hPPR Resource is not available 1B: hPPR Resource is available	3
hPPR Resource BG1 Bank 2		OP[6]	0B: hPPR Resource is not available 1B: hPPR Resource is available	1, 3
hPPR Resource BG1 Bank 3		OP[7]	0B: hPPR Resource is not available 1B: hPPR Resource is available	1, 3
NOTE 1	Don't care for 8 Gb.			
NOTE 2	Don't care for x16. (This note is not used in the table above but left for consistency across MR54-57 hPPR Resources).			
NOTE 3	MR14:OP[3:0] applies to CID[3:0] for 3DS-DDR5 and must be setup to indicate which slice in the 3DS DDR5 Stack is referenced in the MR54 through MR57 hPPR resource information.			

3.5.56 MR55 (MA[7:0]=37H) - hPPR Resources

MR55 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
hPPR Resource BG3 Bank 3	hPPR Resource BG3 Bank 2	hPPR Resource BG3 Bank 1	hPPR Resource BG3 Bank 0	hPPR Resource BG2 Bank 3	hPPR Resource BG2 Bank 2	hPPR Resource BG2 Bank 1	hPPR Resource BG2 Bank 0

Function	Register Type	Operand	Data	Notes
hPPR Resource BG2 Bank 0	R	OP[0]	0B: hPPR Resource is not available 1B: hPPR Resource is available	3
hPPR Resource BG2 Bank 1		OP[1]	0B: hPPR Resource is not available 1B: hPPR Resource is available	3
hPPR Resource BG2 Bank 2		OP[2]	0B: hPPR Resource is not available 1B: hPPR Resource is available	1, 3
hPPR Resource BG2 Bank 3		OP[3]	0B: hPPR Resource is not available 1B: hPPR Resource is available	1, 3
hPPR Resource BG3 Bank 0		OP[4]	0B: hPPR Resource is not available 1B: hPPR Resource is available	3
hPPR Resource BG3 Bank 1		OP[5]	0B: hPPR Resource is not available 1B: hPPR Resource is available	3
hPPR Resource BG3 Bank 2		OP[6]	0B: hPPR Resource is not available 1B: hPPR Resource is available	1, 3
hPPR Resource BG3 Bank 3		OP[7]	0B: hPPR Resource is not available 1B: hPPR Resource is available	1, 3
NOTE 1 Don't care for 8 Gb.				
NOTE 2 Don't care for x16. (This note is not used in the table above but left for consistency across MR54-57 hPPR Resources).				
NOTE 3 MR14:OP[3:0] applies to CID[3:0] for 3DS-DDR5 and must be setup to indicate which slice in the 3DS DDR5 Stack is referenced in the MR54 through MR57 hPPR resource information.				

3.5.57 MR56 (MA[7:0]=38H) - hPPR Resources**MR56 Register Information**

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
hPPR Resource BG5 Bank 3	hPPR Resource BG5 Bank 2	hPPR Resource BG5 Bank 1	hPPR Resource BG5 Bank 0	hPPR Resource BG4 Bank 3	hPPR Resource BG4 Bank 2	hPPR Resource BG4 Bank 1	hPPR Resource BG4 Bank 0

Function	Register Type	Operand	Data	Notes
hPPR Resource BG4 Bank 0	R	OP[0]	0B: hPPR Resource is not available 1B: hPPR Resource is available	2, 3
hPPR Resource BG4 Bank 1		OP[1]	0B: hPPR Resource is not available 1B: hPPR Resource is available	2, 3
hPPR Resource BG4 Bank 2		OP[2]	0B: hPPR Resource is not available 1B: hPPR Resource is available	1, 2, 3
hPPR Resource BG4 Bank 3		OP[3]	0B: hPPR Resource is not available 1B: hPPR Resource is available	1, 2, 3
hPPR Resource BG5 Bank 0		OP[4]	0B: hPPR Resource is not available 1B: hPPR Resource is available	2, 3
hPPR Resource BG5 Bank 1		OP[5]	0B: hPPR Resource is not available 1B: hPPR Resource is available	2, 3
hPPR Resource BG5 Bank 2		OP[6]	0B: hPPR Resource is not available 1B: hPPR Resource is available	1, 2, 3
hPPR Resource BG5 Bank 3		OP[7]	0B: hPPR Resource is not available 1B: hPPR Resource is available	1, 2, 3
NOTE 1 Don't care for 8Gb.				
NOTE 2 Don't care for x16.				
NOTE 3 MR14:OP[3:0] applies to CID[3:0] for 3DS-DDR5 and must be setup to indicate which slice in the 3DS DDR5 Stack is referenced in the MR54 through MR57 hPPR resource information.				

3.5.58 MR57 (MA[7:0]=39H) - hPPR Resources

MR57 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
hPPR Resource BG7 Bank 3	hPPR Resource BG7 Bank 2	hPPR Resource BG7 Bank 1	hPPR Resource BG7 Bank 0	hPPR Resource BG6 Bank 3	hPPR Resource BG6 Bank 2	hPPR Resource BG6 Bank 1	hPPR Resource BG6 Bank 0

Function	Register Type	Operand	Data	Notes
hPPR Resource BG6 Bank 0	R	OP[0]	0B: hPPR Resource is not available 1B: hPPR Resource is available	2, 3
hPPR Resource BG6 Bank 1		OP[1]	0B: hPPR Resource is not available 1B: hPPR Resource is available	2, 3
hPPR Resource BG6 Bank 2		OP[2]	0B: hPPR Resource is not available 1B: hPPR Resource is available	1, 2, 3
hPPR Resource BG6 Bank 3		OP[3]	0B: hPPR Resource is not available 1B: hPPR Resource is available	1, 2, 3
hPPR Resource BG7 Bank 0		OP[4]	0B: hPPR Resource is not available 1B: hPPR Resource is available	2, 3
hPPR Resource BG7 Bank 1		OP[5]	0B: hPPR Resource is not available 1B: hPPR Resource is available	2, 3
hPPR Resource BG7 Bank 2		OP[6]	0B: hPPR Resource is not available 1B: hPPR Resource is available	1, 2, 3
hPPR Resource BG7 Bank 3		OP[7]	0B: hPPR Resource is not available 1B: hPPR Resource is available	1, 2, 3
NOTE 1 Don't care for 8 Gb.				
NOTE 2 Don't care for x16.				
NOTE 3 MR14:OP[3:0] applies to CID[3:0] for 3DS-DDR5 and must be setup to indicate which slice in the 3DS DDR5 Stack is referenced in the MR54 through MR57 hPPR resource information.				

3.5.59 MR58 (MA[7:0]=3A_H) - Refresh Management

MR58 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
RAAMMT[2:0]			RAAIMT[3:0]				RFM Required

Function	Register Type	Operand	Data	Notes
RFM Required	R	OP[0]	0B: Refresh Management not required 1B: Refresh Management required	1
Rolling Accumulated ACT Initial Management Threshold (RAAIMT)	R	OP[4:1]	0000B- 0011B: RFU 0100B: 32 (Normal), 16 (FGR) 0101B: 40 (Normal), 20 (FGR) ... 1001B: 72 (Normal), 36 (FGR) 1010B: 80 (Normal), 40 (FGR) 1011B-1111B: RFU	1, 2
Rolling Accumulated ACT Maximum Management Threshold (RAAMMT)	R	OP[7:5]	000B-010B: RFU 011B: 3x (Normal), 6x (FGR) 100B: 4x (Normal), 8x (FGR) 101B: 5x (Normal), 10x (FGR) 110B: 6x (Normal), 12x (FGR) 111B: RFU	1, 2
NOTE 1 Refresh Management settings are vendor specific by the MR settings.				
NOTE 2 Only applicable if the Refresh Management Required bit is set to "1" (MR58 OP[0]=1)				

3.5.60 MR59 (MA[7:0]=3B_H) - DRFM, ARFM, RFM RAA Counter

MR59 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
RFM RAA Counter		ARFM		BRC Support Level	Bounded Refresh Configuration		DRFE

Function	Register Type	Operand	Data	Note
DRFM Enable (DRFE)	SR/W	OP[0]	DRAM Status Read (SR): 0B: DRFM not implemented (Default) 1B: DRFM implemented Host Write (W): 0B: DRFM disable (Default) 1B: DRFM enable	3 (Host Write)
Bounded Refresh Configuration (BRC)	R/W	OP[2:1]	00B: BRC 2 (default) 01B: BRC 3 10B: BRC 4 11B: RFU	3
BRC Support Level	R	OP[3]	0B: BRC2, 3, 4 (Default) 1B: BRC2 Only	3
Adaptive RFM (ARFM)	R/W	OP[5:4]	00B: Default - RAAIMT, RAAMMT, RAADEC 01B: Level A - RAAIMT-A, RAAMMT-A, RAADEC-A 10B: Level B - RAAIMT-B, RAAMMT-B, RAADEC-B 11B: Level C - RAAIMT-C, RAAMMT-C, RAADEC-C	1
RAA Counter Decrement per REF Command	R	OP[7:6]	00B: RAAIMT 01B: RAAIMT * 0.5 10B: RFU 11B: RFU	1, 2

NOTE 1 Refresh Management settings are vendor specific by the MR settings.

NOTE 2 Only applicable if the Refresh Management Required bit is set to "1" (MR58 OP[0]=1).

NOTE 3 Only applicable if the DRFM function is supported (Status Read MR59:OP[0]=1).

3.5.61 MR60 Partial Array Self Refresh

PASR has been deprecated starting with spec working revision 1.90, JESD79-5C-v1.30. All MR60 bits will behave as RFU on devices that do not support PASR.

MR60 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
PASR Segment Mask							

Function	Register Type	Operand	Data	Notes
Segment 0 (000)	W	OP0	0=Normal, 1=Masked	
Segment 1 (001)	W	OP1	0=Normal, 1=Masked	
Segment 2 (010)	W	OP2	0=Normal, 1=Masked	
Segment 3 (011)	W	OP3	0=Normal, 1=Masked	
Segment 4 (100)	W	OP4	0=Normal, 1=Masked	
Segment 5 (101)	W	OP5	0=Normal, 1=Masked	
Segment 6 (110)	W	OP6	0=Normal, 1=Masked	Must be 0 for 24Gbit and 48 Gbit devices.
Segment 7 (111)	W	OP7	0=Normal, 1=Masked	Must be 0 for 24Gbit and 48 Gbit devices.

3.5.62 MR61 (MA[7:0]=3D_H) - Package Output Driver Test Mode

MR61 Register Information

This MR is used for the characterization of the DRAM package. Refer to the Package Output Driver Test Mode function for more details.

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
RSVD			Package Output Driver Test Mode				

Function	Register Type	Operand	Data	Notes
Package Output Driver Test Mode	W	OP[4:0]	00000B: Package Test Disabled (Default) 00001B: Package Test DML 00010B: Package Test DMU (X16 only) 00011B: RFU 00100B: RFU 00101B: RFU ...thru 01111B: RFU 10000B: Package Test DQL0 10001B: Package Test DQL1 10010B: Package Test DQL2 10011B: Package Test DQL3 10100B: Package Test DQL4 (X8 and X16 only) 10101B: Package Test DQL5 (X8 and X16 only) 10110B: Package Test DQL6 (X8 and X16 only) 10111B: Package Test DQL7 (X8 and X16 only) 11000B: Package Test DQU0 (X16 only) 11001B: Package Test DQU1 (X16 only) 11010B: Package Test DQU2 (X16 only) 11011B: Package Test DQU3 (X16 only) 11100B: Package Test DQU4 (X16 only) 11101B: Package Test DQU5 (X16 only) 11110B: Package Test DQU6 (X16 only) 11111B: Package Test DQU7 (X16 only)	
RSVD	W	OP[7:5]	Must be programmed to 000	

3.5.63 MR62 (MA[7:0]=3E_H) - Vendor Specified

MR62 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
Vendor Specified							

Function	Register Type	Operand	Data	Notes

3.5.64 MR63 (MA[7:0]=3F_H) - DRAM Scratch Pad

MR63 Register Information

This MR is used by the host controller to read back Control Words from the RCD. Control Words are the RCD equivalent of the DRAM MR registers. The DRAM implements MR63 as a simple read/write register, writable via an MRW to address 3Fh, and readable via an MRR to address 3Fh.

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DRAM Scratch Pad							

Function	Register Type	Operand	Data	Notes
DRAM Scratch Pad	R/W	OP[7:0]	Any value is valid	1

NOTE 1 The contents of this register have no function in the DRAM. Details for this function can be found in the DDR5 RCD01 Specification.

The following data are just for reference and are not part of the DRAM specification.

3.5.64.1 RCD Control Word Usage Example

The method to read an RCD Control Word is as follows:

- The host controller writes to the RCD's CW Read Pointer, which selects the RCD control word to be read.
- The host controller then does an MRW to DRAM MR63. This MRW passes through the RCD to the DRAMs, but is modified by the RCD. The RCD will detect this write to MR63 and replace the data from the host controller with the contents of the RCD register pointed to by the CW Read Pointer.
- The host controller will then read the DRAM's MR63, which now contains the value from the desired RCD control word. All DRAMs in the rank will return this same value to the host controller

3.5.65 MR64 (MA[7:0]=40_H) - NVRAM Paging (RFU)

MR64 Register Information

This MR is reserved for NVRAM paging.

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
RFU							

Function	Register Type	Operand	Data	Notes

3.5.66 MR65 (MA[7:0]=41_H) - Serial Number 1

MR65 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
Serial Number 1							

Function	Register Type	Operand	Data	Notes
Serial Number 1	R	OP[7:0]	Serial Number 1	1, 2

NOTE 1 The serial number of 0x00 in all bytes is not allowed

NOTE 2 This byte definition is optional feature, but expect mandatory implementation upon next DDR5 DRAM die release.

3.5.67 MR66 (MA[7:0]=42_H) - Serial Number 2

MR66 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
Serial Number 2							

Function	Register Type	Operand	Data	Notes
Serial Number 2	R	OP[7:0]	Serial Number 2	1, 2

NOTE 1 The serial number of 0x00 in all bytes is not allowed

NOTE 2 This byte definition is optional feature, but expect mandatory implementation upon next DDR5 DRAM die release.

3.5.68 MR67 (MA[7:0]=43_H) - Serial Number 3

MR67 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
Serial Number 3							

Function	Register Type	Operand	Data	Notes
Serial Number 3	R	OP[7:0]	Serial Number 3	1, 2
NOTE 1 The serial number of 0x00 in all bytes is not allowed				
NOTE 2 This byte definition is optional feature, but expect mandatory implementation upon next DDR5 DRAM die release.				

3.5.69 MR68 (MA[7:0]=44_H) - Serial Number 4

MR68 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
Serial Number 4							

Function	Register Type	Operand	Data	Notes
Serial Number 4	R	OP[7:0]	Serial Number 4	1, 2
NOTE 1 The serial number of 0x00 in all bytes is not allowed				
NOTE 2 This byte definition is optional feature, but expect mandatory implementation upon next DDR5 DRAM die release.				

3.5.70 MR69 (MA[7:0]=45_H) - Serial Number 5

MR69 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
Serial Number 5							

Function	Register Type	Operand	Data	Notes
Serial Number 5	R	OP[7:0]	Serial Number 5	1, 2
NOTE 1 The serial number of 0x00 in all bytes is not allowed				
NOTE 2 This byte definition is optional feature, but expect mandatory implementation upon next DDR5 DRAM die release.				

3.5.71 MR70 (MA[7:0]=46_H) thru MR75 (MA[7:0]=4B_H)

MR70 thru MR75 Register Information

These Mode Registers are used when optional feature PRAC is enabled. Detailed Mode Register information can be found from Section 16.10.

3.5.72 Mode Register Definitions for DFE

The following mode registers are used to configure the Decision Feedback Equalization (DFE), Per Bit Duty Cycle Adjuster and Per Bit VrefDQ features of the DRAM. The Mode Registers MA[7:0]=70-FF_H are organized in a way such that mode registers for programming of DFE, DCA & VrefDQ configuration per DQ or DM are grouped together. For example:

DQL0 starts at MA[7:0]=80_H, MR128
DQL1 starts at MA[7:0]=88_H, MR136
:
DQU6 starts at MA[7:0]=F0_H, MR240
DQU7 starts at MA[7:0]=F8_H, MR248

Looking further into the 8-bit binary encoding, MA[6:3] is defined as a direct mapping for DQL0 to DQU7, i.e.,
MA[7:0]=1000:0XXXb for DQ0,
MA[7:0]=1000:1XXXb for DQ1,
:
MA[7:0]=1111:0XXXb for DQU6
MA[7:0]=1111:1XXXb for DQU7.

Table 29 — Visual Representation of DFE, per Bit DCA, and per Bit VrefDQ Mode Register Mapping

[illegible]

3.5.73 MR103 (MA[7:0]=67_H) - DQSL_t DCA for IBCLK and QCLK**MR103 Register Information**

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQSL _t IBCLK Sign	RFU	DQSL _t DCA for IBCLK		DQSL _t QCLK sign	RFU	DQSL _t DCA for QCLK	

Function	Register Type	Operand	Data	Notes
DQSL _t DCA for QCLK	W	OP[1:0]	00 _B : disable (Default) 01 _B : step +1 10 _B : step +2 11 _B : step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[2]		
DQSL _t QCLK sign	W	OP[3]	0 _B : positive (Default) 1 _B : negative	
DQSL _t DCA for IBCLK	W	OP[5:4]	00 _B : disable (Default) 01 _B : step +1 10 _B : step +2 11 _B : step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[6]		
DQSL _t IBCLK sign	W	OP[7]	0 _B : positive (Default) 1 _B : negative	

3.5.74 MR104 (MA[7:0]=68_H) - DQSL_t DCA for QBCLK**MR104 Register Information**

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
RFU				DQSL _t QBCLK Sign	RFU	DQSL _t DCA for QBCLK	

Function	Register Type	Operand	Data	Notes
DQSL _t DCA for QBCLK	W	OP[1:0]	00 _B : disable (Default) 01 _B : step +1 10 _B : step +2 11 _B : step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[2]		
DQSL _t QBCLK sign	W	OP[3]	0 _B : positive (Default) 1 _B : negative	
RFU		OP[7:4]		

3.5.75 MR105 (MA[7:0]=69_H) - DQSL_c DCA for IBCLK and QCLK

MR105 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQSL_c IBCLK Sign	RFU	DQSL_c DCA for IBCLK		DQSL_c QCLK sign	RFU	DQSL_c DCA for QCLK	

Function	Register Type	Operand	Data	Notes
DQSL_c DCA for QCLK	W	OP[1:0]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[2]		
DQSL_c QCLK sign	W	OP[3]	0B: positive (Default) 1B: negative	
DQSL_c DCA for IBCLK	W	OP[5:4]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[6]		
DQSL_c IBCLK sign	W	OP[7]	0B: positive (Default) 1B: negative	

3.5.76 MR106 (MA[7:0]=6A_H) - DQSL_c DCA for QBCLK

MR106 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
RFU				DQSL_c QBCLK Sign	RFU	DQSL_c DCA for QBCLK	

Function	Register Type	Operand	Data	Notes
DQSL_c DCA for QBCLK	W	OP[1:0]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[2]		
DQSL_c QBCLK sign	W	OP[3]	0B: positive (Default) 1B: negative	
RFU		OP[7:4]		

3.5.77 MR107 (MA[7:0]=6B_H) - DQSU_t DCA for IBCLK and QCLK**MR107 Register Information**

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQSU_t IBCLK Sign	RFU	DQSU_t DCA for IBCLK		DQSU_t QCLK sign	RFU	DQSU_t DCA for QCLK	

Function	Register Type	Operand	Data	Notes
DQSU_t DCA for QCLK	W	OP[1:0]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[2]		
DQSU_t QCLK sign	W	OP[3]	0B: positive (Default) 1B: negative	
DQSU_t DCA for IBCLK	W	OP[5:4]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[6]		
DQSU_t IBCLK sign	W	OP[7]	0B: positive (Default) 1B: negative	

3.5.78 MR108 (MA[7:0]=6C_H) - DQSU_t DCA for QBCLK**MR108 Register Information**

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
RFU				DQSU_t QBCLK Sign	RFU	DQSU_t DCA for QBCLK	

Function	Register Type	Operand	Data	Notes
DQSU_t DCA for QBCLK	W	OP[1:0]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[2]		
DQSU_t QBCLK sign	W	OP[3]	0B: positive (Default) 1B: negative	
RFU		OP[7:4]		

3.5.79 MR109 (MA[7:0]=6D_H) - DQSU_c DCA for IBCLK and QCLK

MR109 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQSU_c IBCLK Sign	RFU	DQSU_c DCA for IBCLK		DQSU_c QCLK sign	RFU	DQSU_c DCA for QCLK	

Function	Register Type	Operand	Data	Notes
DQSU_c DCA for QCLK	W	OP[1:0]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[2]		
DQSU_c QCLK sign	W	OP[3]	0B: positive (Default) 1B: negative	
DQSU_c DCA for IBCLK	W	OP[5:4]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[6]		
DQSU_c IBCLK sign	W	OP[7]	0B: positive (Default) 1B: negative	

3.5.80 MR110 (MA[7:0]=6E_H) - DQSU_c DCA for QBCLK

MR110 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
RFU				DQSU_c QBCLK Sign	RFU	DQSU_c DCA for QBCLK	

Function	Register Type	Operand	Data	Notes
DQSU_c DCA for QBCLK	W	OP[1:0]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[2]		
DQSU_c QBCLK sign	W	OP[3]	0B: positive (Default) 1B: negative	
RFU		OP[7:4]		

3.5.81 MR111 (MA[7:0]=6F_H) - DFE Global Settings**MR111 Register Information**

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
RFU			Global DFE Tap-4 Enable	Global DFE Tap-3 Enable	Global DFE Tap-2 Enable	Global DFE Tap-1 Enable	Global DFE Gain Enable

Function	Register Type	Operand	Data	Notes
Global DFE Gain Enable	R/W	OP[0]	0B: DFE Gain Enabled (DEFAULT) 1B: DFE Gain Disabled	1, 2
Global DFE Tap-1 Enable	R/W	OP[1]	0B: DFE Tap-1 Enabled (DEFAULT) 1B: DFE Tap-1 Disabled	1, 2
Global DFE Tap-2 Enable	R/W	OP[2]	0B: DFE Tap-2 Enabled (DEFAULT) 1B: DFE Tap-2 Disabled	1, 2
Global DFE Tap-3 Enable	R/W	OP[3]	0B: DFE Tap-3 Enabled (DEFAULT) 1B: DFE Tap-3 Disabled	1, 2
Global DFE Tap-4 Enable	R/W	OP[4]	0B: DFE Tap-4 Enabled (DEFAULT) 1B: DFE Tap-4 Disabled	1, 2
RFU	RFU	OP[7:5]	RFU	

NOTE 1 This bit applies to all DM and DQ pins.

NOTE 2 Setting MR111:OP[4:0] = 1111_B disables the DFE.

3.5.82 MR112 (MA[7:0]=70_H) thru MR248 (MA[7:0]=F8_H) - DFE Gain Bias

MR112 Register Information

This definition covers registers for DML, DMU, DQL[7:0], and DQU[7:0] for DFE Gain Bias. The MRs are positioned every 8 MRs (MR112, MR120, MR128, etc.) until all pins are covered. Refer to Mode Register Assignment table for details.

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
RFU				Sign Bit Gain Bias	DFE Gain Bias		

Function	Register Type	Operand	Data	Notes
DFE Gain Bias	R/W	OP[2:0]	000B: DFE Gain Bias Step 0 (Default) 001B: DFE Gain Bias Step 1 010B: DFE Gain Bias Step 2 011B: DFE Gain Bias Step 3 100B: RFU 101B: RFU 111B: RFU	1, 2, 3, 4
Sign Bit Gain Bias	R/W	OP[3]	0B: Positive DFE Gain Bias (Default) 1B: Negative DFE Gain Bias	4
RFU	RFU	OP[7:4]	RFU	

NOTE 1 Refer to the DDR5 DFE specification for information on Step Size, Step Size Tolerance, and Range values
NOTE 2 The step size and step values are related to the min/max ranges specified in the DFE Gain Adjustment and Tap Coefficient tables
NOTE 3 The number of step size, step values and range are speed dependent
NOTE 4 Setting all DFE Gain Bias bits (MR112, MR120, MR128, etc.) OP[3:0]=0000_B and all DFE Tap 1-4 Bias bits (MR113, MR114, MR115, MR116, etc.) OP[7:0]=00000000_B disables DFE.

3.5.83 MR113 (MA[7:0]=71_H) thru MR249 (MA[7:0]=F9_H) - DFE Tap-1**MR113 Register Information**

This definition covers registers for DML, DMU, DQL[7:0], and DQU[7:0] for DFE Tap-1. The MRs are positioned every 8 MRs (MR113, MR121, MR129, etc.) until all pins are covered. Refer to Mode Register Assignment table for details.

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
Enable/Disable DFE Tap-1	Sign Bit DFE Tap-1 Bias	DFE Tap-1 Bias Programming					

Function	Register Type	Operand	Data	Notes
DFE Tap-1 Bias	R/W	OP[5:0]	000000B: DFE Tap-1 Bias Step 0 (Default) 000001B: DFE Tap-1 Bias Step 1 000010B: DFE Tap-1 Bias Step 2 000011B: DFE Tap-1 Bias Step 3 000100B: DFE Tap-1 Bias Step 4 000101B: DFE Tap-1 Bias Step 5 : 100110B: DFE Tap-1 Bias Step 38 100111B: DFE Tap-1 Bias Step 39 101000B: DFE Tap-1 Bias Step 40 101001B: RFU : 111111B: RFU	1, 2, 3, 4
Sign Bit DFE Tap-1 Bias	R/W	OP[6]	0B: Positive DFE Tap-1 Bias (Default) 1B: Negative DFE Tap-1 Bias	4
Enable/Disable DFE Tap-1	R/W	OP[7]	0B: DFE Tap-1 Disable (Default) 1B: DFE Tap-1 Enable	4

NOTE 1 Refer to the DDR5 DFE specification for information on Step Size, Step Size Tolerance, and Range values

NOTE 2 The step size and step values are related to the min/max ranges specified in the DFE Gain Adjustment and Tap Coefficient tables

NOTE 3 The number of step size, step values and range are speed dependent

NOTE 4 Setting all DFE Gain Bias bits (MR112, MR120, MR128, etc.) OP[3:0]=0000_B and all DFE Tap 1-4 Bias bits (MR113, MR114, MR115, MR116, etc.) OP[7:0]=00000000_B disables DFE.

3.5.84 MR114 (MA[7:0]=72_H) thru MR250 (MA[7:0]=FA_H) - DFE Tap-2**MR114 Register Information**

This definition covers registers for DML, DMU, DQL[7:0], and DQU[7:0] for DFE Tap-2. The MRs are positioned every 8 MRs (MR114, MR122, MR130, etc.) until all pins are covered. Refer to Mode Register Assignment table for details.

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
Enable/Disable DFE Tap-2	Sign Bit DFE Tap-2 Bias	DFE Tap-2 Bias Programming					

Function	Register Type	Operand	Data	Notes
DFE Tap-2 Bias	R/W	OP[5:0]	000000B: DFE Tap-2 Bias Step 0 (Default) 000001B: DFE Tap-2 Bias Step 1 000010B: DFE Tap-2 Bias Step 2 000011B: DFE Tap-2 Bias Step 3 000100B: DFE Tap-2 Bias Step 4 000101B: DFE Tap-2 Bias Step 5 : 001101B: DFE Tap-2 Bias Step 13 001110B: DFE Tap-2 Bias Step 14 001111B: DFE Tap-2 Bias Step 15 010000B: RFU : 111111B: RFU	1, 2, 3, 4
Sign Bit DFE Tap-2 Bias	R/W	OP[6]	0B: Positive DFE Tap-2 Bias (Default) 1B: Negative DFE Tap-2 Bias	4
Enable/Disable DFE Tap-2	R/W	OP[7]	0B: DFE Tap-2 Disable (Default) 1B: DFE Tap-2 Enable	4
NOTE 1 Refer to the DDR5 DFE specification for information on Step Size, Step Size Tolerance, and Range values NOTE 2 The step size and step values are related to the min/max ranges specified in the DFE Gain Adjustment and Tap Coefficient tables NOTE 3 The number of step size, step values and range are speed dependent NOTE 4 Setting all DFE Gain Bias bits (MR112, MR120, MR128, etc.) OP[3:0]=0000 _B and all DFE Tap 1-4 Bias bits (MR113, MR114, MR115, MR116, etc.) OP[7:0]=00000000 _B disables DFE.				

3.5.85 MR115 (MA[7:0]=73_H) thru MR251 (MA[7:0]=FB_H) - DFE Tap-3**MR115 Register Information**

This definition covers registers for DML, DMU, DQL[7:0], and DQU[7:0] for DFE Gain Bias. The MRs are positioned every 8 MRs (MR115, MR123, MR131, etc.) until all pins are covered. Refer to Mode Register Assignment table for details.

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
Enable/Disable DFE Tap-3	Sign Bit DFE Tap-3 Bias	DFE Tap-3 Bias Programming					

Function	Register Type	Operand	Data	Notes
DFE Tap-3 Bias	R/W	OP[5:0]	000000B: DFE Tap-3 Bias Step 0 (Default) 000001B: DFE Tap-3 Bias Step 1 000010B: DFE Tap-3 Bias Step 2 000011B: DFE Tap-3 Bias Step 3 000100B: DFE Tap-3 Bias Step 4 000101B: DFE Tap-3 Bias Step 5 : 001010B: DFE Tap-3 Bias Step 10 001011B: DFE Tap-3 Bias Step 11 001100B: DFE Tap-3 Bias Step 12 001101B: RFU : 111111B: RFU	1, 2, 3, 4
Sign Bit DFE Tap-3 Bias	R/W	OP[6]	0B: Positive DFE Tap-3 Bias (Default) 1B: Negative DFE Tap-3 Bias	4
Enable/Disable DFE Tap-3	R/W	OP[7]	0B: DFE Tap-3 Disable (Default) 1B: DFE Tap-3 Enable	4
NOTE 1 Refer to the DDR5 DFE specification for information on Step Size, Step Size Tolerance, and Range values NOTE 2 The step size and step values are related to the min/max ranges specified in the DFE Gain Adjustment and Tap Coefficient tables NOTE 3 The number of step size, step values and range are speed dependent NOTE 4 Setting all DFE Gain Bias bits (MR112, MR120, MR128, etc.) OP[3:0]=0000 _B and all DFE Tap 1-4 Bias bits (MR113, MR114, MR115, MR116, etc.) OP[7:0]=00000000 _B disables DFE.				

3.5.86 MR116 (MA[7:0]=74_H) thru MR252 (MA[7:0]=FC_H) - DFE Tap-4

MR116 Register Information

This definition covers registers for DML, DMU, DQL[7:0], and DQU[7:0] for DFE Tap-4. The MRs are positioned every 8 MRs (MR116, MR124, MR132, etc.) until all pins are covered. Refer to Mode Register Assignment table for details.

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
Enable/Disable DFE Tap-4	Sign Bit DFE Tap-4 Bias	DFE Tap-4 Bias Programming					

Function	Register Type	Operand	Data	Notes
DFE Tap-4 Bias	R/W	OP[5:0]	000000B: DFE Tap-4 Bias Step 0 (Default) 000001B: DFE Tap-4 Bias Step 1 000010B: DFE Tap-4 Bias Step 2 000011B: DFE Tap-4 Bias Step 3 000100B: DFE Tap-4 Bias Step 4 000101B: DFE Tap-4 Bias Step 5 000110B: DFE Tap-4 Bias Step 6 000111B: DFE Tap-4 Bias Step 7 001000B: DFE Tap-4 Bias Step 8 001001B: DFE Tap-4 Bias Step 9 001010B: RFU : 111111B: RFU	1, 2, 3, 4
Sign Bit DFE Tap-4 Bias	R/W	OP[6]	0B: Positive DFE Tap-4 Bias (Default) 1B: Negative DFE Tap-4 Bias	4
Enable/Disable DFE Tap-4	R/W	OP[7]	0B: DFE Tap-4 Disable (Default) 1B: DFE Tap-4 Enable	4
NOTE 1 Refer to the DDR5 DFE specification for information on Step Size, Step Size Tolerance, and Range values NOTE 2 The step size and step values are related to the min/max ranges specified in the DFE Gain Adjustment and Tap Coefficient tables NOTE 3 The number of step size, step values and range are speed dependent NOTE 4 Setting all DFE Gain Bias bits (MR112, MR120, MR128, etc.) OP[3:0]=0000 _B and all DFE Tap 1-4 Bias bits (MR113, MR114, MR115, MR116, etc.) OP[7:0]=00000000 _B disables DFE.				

3.5.87 MR117 (MA[7:0]=75_H) - RFU

3.5.88 MR118 (MA[7:0]=76_H) - DML VrefDQ Offset**MR118 Register Information**

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DML VREFDQ sign	DML VREFDQ Offset			RFU			

Function	Register Type	Operand	Data	Notes
RFU		OP[3:0]		
DML VREFDQ Offset	W	OP[6:4]	000B: disable (Default) 001B: step +1 010B: step +2 011B: step +3 100B ~ 111B: RFU	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
DML VREFDQ sign	W	OP[7]	0B: positive (Default) 1B: negative	

3.5.89 MR126 (MA[7:0]=7E_H) - DMU VrefDQ Offset**MR126 Register Information**

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DMU VREFDQ sign	DMU VREFDQ Offset			RFU			

Function	Register Type	Operand	Data	Notes
RFU		OP[3:0]		
DMU VREFDQ Offset	W	OP[6:4]	000B: disable (Default) 001B: step +1 010B: step +2 011B: step +3 100B ~ 111B: RFU	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
DMU VREFDQ sign	W	OP[7]	0B: positive (Default) 1B: negative	

3.5.90 MR133 (MA[7:0]=85_H) - DQL0 DCA for IBCLK and QCLK

MR133 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQL0 IBCLK Sign	RFU	DQL0 DCA for IBCLK		DQL0 QCLK sign	RFU	DQL0 DCA for QCLK	

Function	Register Type	Operand	Data	Notes
DQL0 DCA for QCLK	W	OP[1:0]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[2]		
DQL0 QCLK sign	W	OP[3]	0B: positive (Default) 1B: negative	
DQL0 DCA for IBCLK	W	OP[5:4]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[6]		
DQL0 IBCLK sign	W	OP[7]	0B: positive (Default) 1B: negative	

3.5.91 MR134 (MA[7:0]=86_H) - DQL0 DCA for QBCLK and DQL0 VrefDQ Offset

MR134 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQL0 VREFDQ sign	DQL0 VREFDQ Offset			DQL0 QBCLK Sign	RFU	DQL0 DCA for QBCLK	

Function	Register Type	Operand	Data	Notes
DQL0 DCA for QBCLK	W	OP[1:0]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[2]		
DQL0 QBCLK sign	W	OP[3]	0B: positive (Default) 1B: negative	
DQL0 VREFDQ Offset	W	OP[6:4]	000B: disable (Default) 001B: step +1 010B: step +2 011B: step +3 100B ~ 111B: RFU	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
DQL0 VREFDQ sign	W	OP[7]	0B: positive (Default) 1B: negative	

3.5.92 MR141 (MA[7:0]=8D_H) - DQL1 DCA for IBCLK and QCLK**MR141 Register Information**

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQL1 IBCLK Sign	RFU	DQL1 DCA for IBCLK		DQL1 QCLK sign	RFU	DQL1 DCA for QCLK	

Function	Register Type	Operand	Data	Notes
DQL1 DCA for QCLK	W	OP[1:0]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[2]		
DQL1 QCLK sign	W	OP[3]	0B: positive (Default) 1B: negative	
DQL1 DCA for IBCLK	W	OP[5:4]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[6]		
DQL1 IBCLK sign	W	OP[7]	0B: positive (Default) 1B: negative	

3.5.93 MR142 (MA[7:0]=8E_H) - DQL1 DCA for QBCLK and DQL1 VrefDQ Offset**MR142 Register Information**

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQL1 VREFDQ sign	DQL1 VREFDQ Offset			DQL1 QBCLK Sign	RFU	DQL1 DCA for QBCLK	

Function	Register Type	Operand	Data	Notes
DQL1 DCA for QBCLK	W	OP[1:0]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[2]		
DQL1 QBCLK sign	W	OP[3]	0B: positive (Default) 1B: negative	
DQL1 VREFDQ Offset	W	OP[6:4]	000B: disable (Default) 001B: step +1 010B: step +2 011B: step +3 100B ~ 111B: RFU	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
DQL1 VREFDQ sign	W	OP[7]	0B: positive (Default) 1B: negative	

3.5.94 MR149 (MA[7:0]=95_H) - DQL2 DCA for IBCLK and QCLK

MR149 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQL2 IBCLK Sign	RFU	DQL2 DCA for IBCLK		DQL2 QCLK sign	RFU	DQL2 DCA for QCLK	

Function	Register Type	Operand	Data	Notes
DQL2 DCA for QCLK	W	OP[1:0]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[2]		
DQL2 QCLK sign	W	OP[3]	0B: positive (Default) 1B: negative	
DQL2 DCA for IBCLK	W	OP[5:4]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[6]		
DQL2 IBCLK sign	W	OP[7]	0B: positive (Default) 1B: negative	

3.5.95 MR150 (MA[7:0]=96_H) - DQL2 DCA for QBCLK and DQL2 VrefDQ Offset

MR150 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQL2 VREFDQ sign	DQL2 VREFDQ Offset			DQL2 QBCLK Sign	RFU	DQL2 DCA for QBCLK	

Function	Register Type	Operand	Data	Notes
DQL2 DCA for QBCLK	W	OP[1:0]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[2]		
DQL2 QBCLK sign	W	OP[3]	0B: positive (Default) 1B: negative	
DQL2 VREFDQ Offset	W	OP[6:4]	000B: disable (Default) 001B: step +1 010B: step +2 011B: step +3 100B ~ 111B: RFU	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
DQL2 VREFDQ sign	W	OP[7]	0B: positive (Default) 1B: negative	

3.5.96 MR157 (MA[7:0]=9D_H) - DQL3 DCA for IBCLK and QCLK**MR157 Register Information**

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQL3 IBCLK Sign	RFU	DQL3 DCA for IBCLK		DQL3 QCLK sign	RFU	DQL3 DCA for QCLK	

Function	Register Type	Operand	Data	Notes
DQL3 DCA for QCLK	W	OP[1:0]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[2]		
DQL3 QCLK sign	W	OP[3]	0B: positive (Default) 1B: negative	
DQL3 DCA for IBCLK	W	OP[5:4]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[6]		
DQL3 IBCLK sign	W	OP[7]	0B: positive (Default) 1B: negative	

3.5.97 MR158 (MA[7:0]=9E_H) - DQL3 DCA for QBCLK and DQL3 VrefDQ Offset**MR158 Register Information**

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQL3 VREFDQ sign	DQL3 VREFDQ Offset			DQL3 QBCLK Sign	RFU	DQL3 DCA for QBCLK	

Function	Register Type	Operand	Data	Notes
DQL3 DCA for QBCLK	W	OP[1:0]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[2]		
DQL3 QBCLK sign	W	OP[3]	0B: positive (Default) 1B: negative	
DQL3 VREFDQ Offset	W	OP[6:4]	000B: disable (Default) 001B: step +1 010B: step +2 011B: step +3 100B ~ 111B: RFU	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
DQL3 VREFDQ sign	W	OP[7]	0B: positive (Default) 1B: negative	

3.5.98 MR165 (MA[7:0]=A5_H) - DQL4 DCA for IBCLK and QCLK

MR165 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQL4 IBCLK Sign	RFU	DQL4 DCA for IBCLK		DQL4 QCLK sign	RFU	DQL4 DCA for QCLK	

Function	Register Type	Operand	Data	Notes
DQL4 DCA for QCLK	W	OP[1:0]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[2]		
DQL4 QCLK sign	W	OP[3]	0B: positive (Default) 1B: negative	
DQL4 DCA for IBCLK	W	OP[5:4]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[6]		
DQL4 IBCLK sign	W	OP[7]	0B: positive (Default) 1B: negative	

3.5.99 MR166 (MA[7:0]=A6_H) - DQL4 DCA for QBCLK and DQL4 VrefDQ Offset

MR166 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQL4 VREFDQ sign	DQL3 VREFDQ Offset			DQL3 QBCLK Sign	RFU	DQL3 DCA for QBCLK	

Function	Register Type	Operand	Data	Notes
DQL4 DCA for QBCLK	W	OP[1:0]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[2]		
DQL4 QBCLK sign	W	OP[3]	0B: positive (Default) 1B: negative	
DQL4 VREFDQ Offset	W	OP[6:4]	000B: disable (Default) 001B: step +1 010B: step +2 011B: step +3 100B ~ 111B: RFU	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
DQL4 VREFDQ sign	W	OP[7]	0B: positive (Default) 1B: negative	

3.5.100 MR173 (MA[7:0]=AD_H) - DQL5 DCA for IBCLK and QCLK**MR173 Register Information**

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQL5 IBCLK Sign	RFU	DQL5 DCA for IBCLK		DQL5 QCLK sign	RFU	DQL5 DCA for QCLK	

Function	Register Type	Operand	Data	Notes
DQL5 DCA for QCLK	W	OP[1:0]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[2]		
DQL5 QCLK sign	W	OP[3]	0B: positive (Default) 1B: negative	
DQL5 DCA for IBCLK	W	OP[5:4]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[6]		
DQL5 IBCLK sign	W	OP[7]	0B: positive (Default) 1B: negative	

3.5.101 MR174 (MA[7:0]=AE_H) - DQL5 DCA for QBCLK and DQL5 VrefDQ Offset**MR174 Register Information**

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQL5 VREFDQ sign	DQL5 VREFDQ Offset			DQL5 QBCLK Sign	RFU	DQL5 DCA for QBCLK	

Function	Register Type	Operand	Data	Notes
DQL5 DCA for QBCLK	W	OP[1:0]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[2]		
DQL5 QBCLK sign	W	OP[3]	0B: positive (Default) 1B: negative	
DQL5 VREFDQ Offset	W	OP[6:4]	000B: disable (Default) 001B: step +1 010B: step +2 011B: step +3 100B ~ 111B: RFU	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
DQL5 VREFDQ sign	W	OP[7]	0B: positive (Default) 1B: negative	

3.5.102 MR181 (MA[7:0]=B5_H) - DQL6 DCA for IBCLK and QCLK

MR181 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQL6 IBCLK Sign	RFU	DQL6 DCA for IBCLK		DQL6 QCLK sign	RFU	DQL6 DCA for QCLK	

Function	Register Type	Operand	Data	Notes
DQL6 DCA for QCLK	W	OP[1:0]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[2]		
DQL6 QCLK sign	W	OP[3]	0B: positive (Default) 1B: negative	
DQL6 DCA for IBCLK	W	OP[5:4]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[6]		
DQL6 IBCLK sign	W	OP[7]	0B: positive (Default) 1B: negative	

3.5.103 MR182 (MA[7:0]=B6_H) - DQL6 DCA for QBCLK and DQL6 VrefDQ Offset

MR182 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQL6 VREFDQ sign	DQL6 VREFDQ Offset			DQL6 QBCLK Sign	RFU	DQL6 DCA for QBCLK	

Function	Register Type	Operand	Data	Notes
DQL6 DCA for QBCLK	W	OP[1:0]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[2]		
DQL6 QBCLK sign	W	OP[3]	0B: positive (Default) 1B: negative	
DQL6 VREFDQ Offset	W	OP[6:4]	000B: disable (Default) 001B: step +1 010B: step +2 011B: step +3 100B ~ 111B: RFU	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
DQL6 VREFDQ sign	W	OP[7]	0B: positive (Default) 1B: negative	

3.5.104 MR189 (MA[7:0]=BD_H) - DQL7 DCA for IBCLK and QCLK**MR189 Register Information**

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQL7 IBCLK Sign	RFU	DQL7 DCA for IBCLK		DQL7 QCLK sign	RFU	DQL7 DCA for QCLK	

Function	Register Type	Operand	Data	Notes
DQL7 DCA for QCLK	W	OP[1:0]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[2]		
DQL7 QCLK sign	W	OP[3]	0B: positive (Default) 1B: negative	
DQL7 DCA for IBCLK	W	OP[5:4]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[6]		
DQL7 IBCLK sign	W	OP[7]	0B: positive (Default) 1B: negative	

3.5.105 MR190 (MA[7:0]=BE_H) - DQL7 DCA for QBCLK and DQL7 VrefDQ Offset**MR190 Register Information**

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQL7 VREFDQ sign	DQL7 VREFDQ Offset			DQL7 QBCLK Sign	RFU	DQL7 DCA for QBCLK	

Function	Register Type	Operand	Data	Notes
DQL7 DCA for QBCLK	W	OP[1:0]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[2]		
DQL7 QBCLK sign	W	OP[3]	0B: positive (Default) 1B: negative	
DQL7 VREFDQ Offset	W	OP[6:4]	000B: disable (Default) 001B: step +1 010B: step +2 011B: step +3 100B ~ 111B: RFU	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
DQL7 VREFDQ sign	W	OP[7]	0B: positive (Default) 1B: negative	

3.5.106 MR197 (MA[7:0]=C5_H) - DQU0 DCA for IBCLK and QCLK

MR197 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQU0 IBCLK Sign	RFU	DQU0 DCA for IBCLK		DQU0 QCLK sign	RFU	DQU0 DCA for QCLK	

Function	Register Type	Operand	Data	Notes
DQU0 DCA for QCLK	W	OP[1:0]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[2]		
DQU0 QCLK sign	W	OP[3]	0B: positive (Default) 1B: negative	
DQU0 DCA for IBCLK	W	OP[5:4]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[6]		
DQU0 IBCLK sign	W	OP[7]	0B: positive (Default) 1B: negative	

3.5.107 MR198 (MA[7:0]=C6_H) - DQU0 DCA for QBCLK and DQU0 VrefDQ Offset

MR198 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQU0 VREFDQ sign	DQU0 VREFDQ Offset			DQU0 QBCLK Sign	RFU	DQU0 DCA for QBCLK	

Function	Register Type	Operand	Data	Notes
DQU0 DCA for QBCLK	W	OP[1:0]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[2]		
DQU0 QBCLK sign	W	OP[3]	0B: positive (Default) 1B: negative	
DQU0 VREFDQ Offset	W	OP[6:4]	000B: disable (Default) 001B: step +1 010B: step +2 011B: step +3 100B ~ 111B: RFU	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
DQU0 VREFDQ sign	W	OP[7]	0B: positive (Default) 1B: negative	

3.5.108 MR205 (MA[7:0]=CD_H) - DQU1 DCA for IBCLK and QCLK**MR205 Register Information**

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQU1 IBCLK Sign	RFU	DQU1 DCA for IBCLK		DQU1 QCLK sign	RFU	DQU1 DCA for QCLK	

Function	Register Type	Operand	Data	Notes
DQU1 DCA for QCLK	W	OP[1:0]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[2]		
DQU1 QCLK sign	W	OP[3]	0B: positive (Default) 1B: negative	
DQU1 DCA for IBCLK	W	OP[5:4]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[6]		
DQU1 IBCLK sign	W	OP[7]	0B: positive (Default) 1B: negative	

3.5.109 MR206 (MA[7:0]=CE_H) - DQU1 DCA for QBCLK and DQU1 VrefDQ Offset**MR206 Register Information**

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQU1 VREFDQ sign	DQU1 VREFDQ Offset			DQU1 QBCLK Sign	RFU	DQU1 DCA for QBCLK	

Function	Register Type	Operand	Data	Notes
DQU1 DCA for QBCLK	W	OP[1:0]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[2]		
DQU1 QBCLK sign	W	OP[3]	0B: positive (Default) 1B: negative	
DQU1 VREFDQ Offset	W	OP[6:4]	000B: disable (Default) 001B: step +1 010B: step +2 011B: step +3 100B ~ 111B: RFU	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
DQU1 VREFDQ sign	W	OP[7]	0B: positive (Default) 1B: negative	

3.5.110 MR213 (MA[7:0]=D5_H) - DQU2 DCA for IBCLK and QCLK

MR213 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQU2 IBCLK Sign	RFU	DQU2 DCA for IBCLK		DQU2 QCLK sign	RFU	DQU2 DCA for QCLK	

Function	Register Type	Operand	Data	Notes
DQU2 DCA for QCLK	W	OP[1:0]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[2]		
DQU2 QCLK sign	W	OP[3]	0B: positive (Default) 1B: negative	
DQU2 DCA for IBCLK	W	OP[5:4]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[6]		
DQU2 IBCLK sign	W	OP[7]	0B: positive (Default) 1B: negative	

3.5.111 MR214 (MA[7:0]=D6_H) - DQU2 DCA for QBCLK and DQU2 VrefDQ Offset

MR214 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQU2 VREFDQ sign	DQU2 VREFDQ Offset			DQU2 QBCLK Sign	RFU	DQU2 DCA for QBCLK	

Function	Register Type	Operand	Data	Notes
DQU2 DCA for QBCLK	W	OP[1:0]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[2]		
DQU2 QBCLK sign	W	OP[3]	0B: positive (Default) 1B: negative	
DQU2 VREFDQ Offset	W	OP[6:4]	000B: disable (Default) 001B: step +1 010B: step +2 011B: step +3 100B ~ 111B: RFU	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
DQU2 VREFDQ sign	W	OP[7]	0B: positive (Default) 1B: negative	

3.5.112 MR221 (MA[7:0]=DD_H) - DQU3 DCA for IBCLK and QCLK**MR221 Register Information**

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQU3 IBCLK Sign	RFU	DQU3 DCA for IBCLK		DQU3 QCLK sign	RFU	DQU3 DCA for QCLK	

Function	Register Type	Operand	Data	Notes
DQU3 DCA for QCLK	W	OP[1:0]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[2]		
DQU3 QCLK sign	W	OP[3]	0B: positive (Default) 1B: negative	
DQU3 DCA for IBCLK	W	OP[5:4]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[6]		
DQU3 IBCLK sign	W	OP[7]	0B: positive (Default) 1B: negative	

3.5.113 MR222 (MA[7:0]=DE_H) - DQU3 DCA for QBCLK and DQU3 VrefDQ Offset**MR222 Register Information**

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQU3 VREFDQ sign	DQU3 VREFDQ Offset			DQU3 QBCLK Sign	RFU	DQU3 DCA for QBCLK	

Function	Register Type	Operand	Data	Notes
DQU3 DCA for QBCLK	W	OP[1:0]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[2]		
DQU3 QBCLK sign	W	OP[3]	0B: positive (Default) 1B: negative	
DQU3 VREFDQ Offset	W	OP[6:4]	000B: disable (Default) 001B: step +1 010B: step +2 011B: step +3 100B ~ 111B: RFU	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
DQU3 VREFDQ sign	W	OP[7]	0B: positive (Default) 1B: negative	

3.5.114 MR229 (MA[7:0]=E5_H) - DQU4 DCA for IBCLK and QCLK

MR229 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQU4 IBCLK Sign	RFU	DQU4 DCA for IBCLK		DQU4 QCLK sign	RFU	DQU4 DCA for QCLK	

Function	Register Type	Operand	Data	Notes
DQU4 DCA for QCLK	W	OP[1:0]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[2]		
DQU4 QCLK sign	W	OP[3]	0B: positive (Default) 1B: negative	
DQU4 DCA for IBCLK	W	OP[5:4]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[6]		
DQU4 IBCLK sign	W	OP[7]	0B: positive (Default) 1B: negative	

3.5.115 MR230 (MA[7:0]=E6_H) - DQU4 DCA for QBCLK and DQU4 VrefDQ Offset

MR230 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQU4 VREFDQ sign	DQU4 VREFDQ Offset			DQU4 QBCLK Sign	RFU	DQU4 DCA for QBCLK	

Function	Register Type	Operand	Data	Notes
DQU4 DCA for QBCLK	W	OP[1:0]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[2]		
DQU4 QBCLK sign	W	OP[3]	0B: positive (Default) 1B: negative	
DQU4 VREFDQ Offset	W	OP[6:4]	000B: disable (Default) 001B: step +1 010B: step +2 011B: step +3 100B ~ 111B: RFU	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
DQU4 VREFDQ sign	W	OP[7]	0B: positive (Default) 1B: negative	

3.5.116 MR237 (MA[7:0]=ED_H) - DQU5 DCA for IBCLK and QCLK**MR237 Register Information**

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQU5 IBCLK Sign	RFU	DQU5 DCA for IBCLK		DQU5 QCLK sign	RFU	DQU5 DCA for QCLK	

Function	Register Type	Operand	Data	Notes
DQU5 DCA for QCLK	W	OP[1:0]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[2]		
DQU5 QCLK sign	W	OP[3]	0B: positive (Default) 1B: negative	
DQU5 DCA for IBCLK	W	OP[5:4]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[6]		
DQU5 IBCLK sign	W	OP[7]	0B: positive (Default) 1B: negative	

3.5.117 MR238 (MA[7:0]=EE_H) - DQU5 DCA for QBCLK and DQU5 VrefDQ Offset**MR238 Register Information**

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQU5 VREFDQ sign	DQU5 VREFDQ Offset			DQU5 QBCLK Sign	RFU	DQU5 DCA for QBCLK	

Function	Register Type	Operand	Data	Notes
DQU5 DCA for QBCLK	W	OP[1:0]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[2]		
DQU5 QBCLK sign	W	OP[3]	0B: positive (Default) 1B: negative	
DQU5 VREFDQ Offset	W	OP[6:4]	000B: disable (Default) 001B: step +1 010B: step +2 011B: step +3 100B ~ 111B: RFU	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
DQU5 VREFDQ sign	W	OP[7]	0B: positive (Default) 1B: negative	

3.5.118 MR245 (MA[7:0]=F5_H) - DQU6 DCA for IBCLK and QCLK

MR245 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQU6 IBCLK Sign	RFU	DQU6 DCA for IBCLK		DQU6 QCLK sign	RFU	DQU6 DCA for QCLK	

Function	Register Type	Operand	Data	Notes
DQU6 DCA for QCLK	W	OP[1:0]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[2]		
DQU6 QCLK sign	W	OP[3]	0B: positive (Default) 1B: negative	
DQU6 DCA for IBCLK	W	OP[5:4]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[6]		
DQU6 IBCLK sign	W	OP[7]	0B: positive (Default) 1B: negative	

3.5.119 MR246 (MA[7:0]=F6_H) - DQU6 DCA for QBCLK and DQU6 VrefDQ Offset

MR246 Register Information

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQU6 VREFDQ sign	DQU6 VREFDQ Offset			DQU6 QBCLK Sign	RFU	DQU6 DCA for QBCLK	

Function	Register Type	Operand	Data	Notes
DQU6 DCA for QBCLK	W	OP[1:0]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[2]		
DQU6 QBCLK sign	W	OP[3]	0B: positive (Default) 1B: negative	
DQU6 VREFDQ Offset	W	OP[6:4]	000B: disable (Default) 001B: step +1 010B: step +2 011B: step +3 100B ~ 111B: RFU	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
DQU6 VREFDQ sign	W	OP[7]	0B: positive (Default) 1B: negative	

3.5.120 MR253 (MA[7:0]=FD_H) - DQU7 DCA for IBCLK and QCLK**MR253 Register Information**

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQU7 IBCLK Sign	RFU	DQU7 DCA for IBCLK		DQU7 QCLK sign	RFU	DQU7 DCA for QCLK	

Function	Register Type	Operand	Data	Notes
DQU7 DCA for QCLK	W	OP[1:0]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[2]		
DQU7 QCLK sign	W	OP[3]	0B: positive (Default) 1B: negative	
DQU7 DCA for IBCLK	W	OP[5:4]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[6]		
DQU7 IBCLK sign	W	OP[7]	0B: positive (Default) 1B: negative	

3.5.121 MR254 (MA[7:0]=FE_H) - DQU7 DCA for QBCLK and DQU7 VrefDQ Offset**MR254 Register Information**

OP[7]	OP[6]	OP[5]	OP[4]	OP[3]	OP[2]	OP[1]	OP[0]
DQU7 VREFDQ sign	DQU7 VREFDQ Offset			DQU7 QBCLK Sign	RFU	DQU7 DCA for QBCLK	

Function	Register Type	Operand	Data	Notes
DQU7 DCA for QBCLK	W	OP[1:0]	00B: disable (Default) 01B: step +1 10B: step +2 11B: step +3	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
RFU		OP[2]		
DQU7 QBCLK sign	W	OP[3]	0B: positive (Default) 1B: negative	
DQU7 VREFDQ Offset	W	OP[6:4]	000B: disable (Default) 001B: step +1 010B: step +2 011B: step +3 100B ~ 111B: RFU	Range: -3 ~ +3 LSB Codes Step Size: 1LSB
DQU7 VREFDQ sign	W	OP[7]	0B: positive (Default) 1B: negative	

3.5.122 Undefined Mode Registers Spaced in DFE, per Bit DCA, and per Bit VrefDQ Section

MR117
MR119
MR125
MR127
MR135
MR143
MR155
MR159
MR167
MR175
MR183
MR191
MR199
MR207
MR215
MR223
MR231
MR239
MR247
MR255

There are currently no plans to utilize these MR addresses.

4 DDR5 SDRAM Command Description and Operation

4.1 Command Truth Table

- (a) Notes 1, 2, and 14 apply to the entire Command truth table
 (b) To improve command decode time, the table has been optimized to orient all 1-cycle commands together and all 2-cycle commands together; allowing CA1 to be used to identify the difference between a 1-cycle and a 2-cycle command.

[BG=Bank Group Address, BA=Bank Address, R=Row Address, C=Column Address, MRA=Mode Register Address, OP=Op Code, CID=Chip ID, CW=Control Word, X=Don't Care, V=Valid].

Table 30 — Command Truth Table

Function	Abbreviation	CS_n	CA Pins														NOTES
			CA0	CA1	CA2	CA3	CA4	CA5	CA6	CA7	CA8	CA9	CA10	CA11	CA12	CA13	
Activate	ACT	L	L	L	R0	R1	R2	R3	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2	11, 17, 20
		H	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13	R14	R15	R16	CID3/R17	
RFU	RFU	L	H	L	L	L	L	V	V	V	V	V	V	V	V	V	
		H	V	V	V	V	V	V	V	V	V	V	V	V	V	V	
RFU	RFU	L	H	L	L	L	H	V	V	V	V	V	V	V	V	V	
		H	V	V	V	V	V	V	V	V	V	V	V	V	V	V	
Write Pattern	WRP	L	H	L	L	H	L	H	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2	11, 15, 18, 19, 20
		H	V	C3	C4	C5	C6	C7	C8	C9	C10	V	H	H	V	CID3	
Write Pattern w/ Auto Precharge	WRPA	L	H	L	L	H	L	H	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2	11, 15, 18, 19, 20
		H	V	C3	C4	C5	C6	C7	C8	C9	C10	V or DRFM=L	AP=L	H	V	CID3	
RFU	RFU	L	H	L	L	H	H	V	V	V	V	V	V	V	V	V	
		H	V	V	V	V	V	V	V	V	V	V	V	V	V	V	
Mode Register Write	MRW	L	H	L	H	L	L	MRA0	MRA1	MRA2	MRA3	MRA4	MRA5	MRA6	MRA7	V	8, 11, 13, 20
		H	OP0	OP1	OP2	OP3	OP4	OP5	OP6	OP7	V	V	CW	V	V	V	
Mode Register Read	MRR	L	H	L	H	L	H	MRA0	MRA1	MRA2	MRA3	MRA4	MRA5	MRA6	MRA7	V	8, 13, 21, 20
		H	L	L	V	V	V	V	V	V	V	V	CW	V	V	V	
Write	WR	L	H	L	H	H	L	BL*=L	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2	8, 12, 15, 19, 20
		H	V	C3	C4	C5	C6	C7	C8	C9	C10	V	H	WR Partial=L	V	CID3	
Write w/Auto Precharge	WRA	L	H	L	H	H	L	BL*=L	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2	8, 12, 15, 19, 20
		H	V	C3	C4	C5	C6	C7	C8	C9	C10	V or DRFM=L	AP=L	WR Partial=L	V	CID3	
Read	RD	L	H	L	H	H	H	BL*=L	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2	8, 15, 19, 20
		H	C2	C3	C4	C5	C6	C7	C8	C9	C10	V	H	V	V	CID3	
Read w/Auto Precharge	RDA	L	H	L	H	H	H	BL*=L	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2	8, 15, 19, 20
		H	C2	C3	C4	C5	C6	C7	C8	C9	C10	V or DRFM=L	AP=L	V	V	CID3	
VrefCA Command	VrefCA	L	H	H	L	L	L	OP0	OP1	OP2	OP3	OP4	OP5	OP6	L	V	
VrefCS Command	VrefCS	L	H	H	L	L	L	OP0	OP1	OP2	OP3	OP4	OP5	OP6	H	V	
Refresh All	REFab	L	H	H	L	L	H	CID3	V	V	V or RIR	V or H	L	CID0	CID1	CID2	3, 23, 24
Refresh Management All	RFMab	L	H	H	L	L	H	CID3 or DRFM=L	V	V	V	L	L	CID0	CID1	CID2	3, 20
Refresh Same Bank	REFsb	L	H	H	L	L	H	CID3	BA0	BA1	V or RIR	V or H	H	CID0	CID1	CID2	4, 20, 23, 24
Refresh Management Same Bank	RFMsb	L	H	H	L	L	H	CID3 or DRFM=L	BA0	BA1	V	L	H	CID0	CID1	CID2	4, 20
Precharge All	PREab	L	H	H	L	H	L	CID3	V	V	V	V	L	CID0	CID1	CID2	5, 20
Precharge Same Bank	PREsb	L	H	H	L	H	L	CID3	BA0	BA1	V	V	H	CID0	CID1	CID2	6, 20
Precharge	PREpb	L	H	H	L	H	H	CID3 or DRFM=L	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2	7, 20
RFU	RFU	L	H	H	H	L	L	V	V	V	V	V	V	V	V	V	
Self Refresh Entry	SRE	L	H	H	H	L	H	V	V	V	V	H	L	V	V	V	9
Self Refresh Entry w/Frequency Change	SREF	L	H	H	H	L	H	V	V	V	V	L	L	V	V	V	9
Power Down Entry	PDE	L	H	H	H	L	H	V	V	V	V	V	H	ODT=L	V	V	10, 16

Table 30 — Command Truth Table (cont'd)

Function	Abbrevia- tion	CS_n	CA Pins														NOTES
			CA0	CA1	CA2	CA3	CA4	CA5	CA6	CA7	CA8	CA9	CA10	CA11	CA12	CA13	
MPC	MPC	L	H	H	H	H	L	OP0	OP1	OP2	OP3	OP4	OP5	OP6	OP7	V	20
NOP	NOP	L	H	H	H	H	H	V	V	V	V	V	V	V	V	V	26
Power Down Exit	PDX	L	H	H	H	H	H	V	V	V	V	V	V	V	V	V	
Deselect	DES	H	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
Reserved		L	L	H	Reserved												
NOTE 1 V means H or L (a defined logic level). X means don't care in which case the signal may be floated.																	
NOTE 2 Bank group addresses BG[2:0] and Bank addresses BA[1:0] determine which bank is to be operated upon in a specific bank group.																	
NOTE 3 The Refresh All and Refresh Management All commands are applied to all banks in all bank groups. CA6 and CA7 are required to be valid ("V").																	
NOTE 4 The Refresh Same Bank and Refresh Management Same Bank commands refresh the same bank in all bank group bits. The bank bits, BA0 and BA1 on CA6 and CA7, respectively, specify the bank within each bank group.																	
NOTE 5 The Precharge All command applies to all open banks in all bank groups.																	
NOTE 6 The Precharge Same Bank command applies to the same bank in all bank groups. The bank bits specify the bank within each bank group.																	
NOTE 7 The Precharge command applies to a single bank as specified by bank address and bank group bits.																	
NOTE 8 CS_n=LOW during the 2nd cycle of a two cycle command controls ODT in non-target ranks for WR, RD and MRR commands.																	
NOTE 9 The SRE command places the DRAM in self refresh state																	
NOTE 10 The PDE command places the DRAM in power down state																	
NOTE 11 Two cycle commands with no ODT control (ACT, MRW, WRP). DRAM does not execute the command if it receives CS as LOW on 2nd cycle.																	
NOTE 12 WR command with WR_Partial (WR_P) = Low indicates a partial write command. This is to help DRAM start an internal read for 'read modify write'.																	
NOTE 13 If CW=Low during the MRW command then DRAM should execute the command, Mode Register will be written. If CW=HIGH then DRAM ignores the MRW command, and the Mode Register is not changed. If CW=Low, DRAM should execute the MRR command. If CW=High, DRAM may or may not execute MRR command.																	
NOTE 14 CID[3:0] bits are used for 3DS stacking support. When CID[3:0] bits are not used, they are required to be Valid ("V").																	
NOTE 15 If CA5:BL*=L, the command places the DRAM into the alternate Burst mode described by MR0[1:0] instead of the default Burst Length 16 mode.																	
NOTE 16 ODT=L is defined to allow On Die Termination (ODT) to persist when the device is in Power Down Mode.																	
NOTE 17 CID3/R17 is a multi-mode pin allowing for either 16H 3DS stacking with the CID3 bit usage or R17 for high bit density monolithic usage. These usages are mutually exclusive.																	
NOTE 18 Write Pattern only supports BL16 and BL32.																	
NOTE 19 When CID3 is not used, its required to be Valid ("V").																	
NOTE 20 In the case of a DRAM where the density or stacking doesn't require CA[13] the ball location for that function (considering the state of MIR) shall be connected to VDDQ, and the DRAM shall decode CA[13]=L so that the proper selection of die and RA is provided.																	
NOTE 21 CA[0:1] =[L:L] on the second cycle for burst ordering.																	
NOTE 22 When host issue MRR with CRC enabled, data comes out with CRC bit.																	
NOTE 23 If the Refresh Management Required bit is "0" (MR58 OP[0]=0), CA9 is only required to be valid ("V") for a REF command, and the DRAM will treat a RFM command as a REF command. If MR58 OP[0]=1, a REF command requires CA9=H.																	
NOTE 24 If the Refresh Interval Rate indicator bit is disabled (MR4:OP[3]=0) by the host, CA8 is only required to be valid ("V"). If the Refresh Interval Rate indicator bit is enabled (MR4:OP[3]=1) by the host, the host is required to set CA8=H for REF commands issued at the 1x refresh interval rate and CA8=L for REF commands issued at the 2x refresh interval rate. If the host issues 2x REF commands with CA8=L while MR4:OP[3]=1, but the Refresh rate requirement is 1x as determined internally by the DRAM, the DRAM may internally align to the 1x refresh rate.																	
NOTE 25 Command Truth Table bits such as BG2, BA1, R16, R17, C10, and CID[3:0] which are unused based on a device's density, configuration width, and stacking options, the CA decode is defined as VALID when in these unused states.																	
NOTE 26 Unlike DES, NOP is considered a <i>valid command</i> , and timing from a preceding valid command must satisfy any associated command timings.																	

4.1.1 2-Cycle Command Cancel

DDR5 DRAM commands ACT, WRP, WRPA and MRW are 2-cycle commands without associated ODT control requirements. The DRAM will not execute these 2-cycle commands if the CS_n is LOW on the 2nd cycle (command cancel). If the RCD detects a parity error on the 2nd cycle of two-cycle command, the CS_n will remain LOW for both 1st and 2nd cycle of the command. If the command is either Read, Write or MRR, then it will be converted to non-target termination command in the DRAM. If the command is either ACT, WRP, WRPA or MRW, then the erroneous command will be canceled in the DRAM. Command cancel is not intended by the host rather it is a result of CA parity error detected by the RCD. So the relationship between canceled command and the next valid command shall not be illegal. For example, MRR cannot be issued after canceled ACT even with tCMD_cancel satisfied. In that case, the host is supposed to issue PRE first before issuing MRR.

4.1.1 2-Cycle Command Cancel (cont'd)

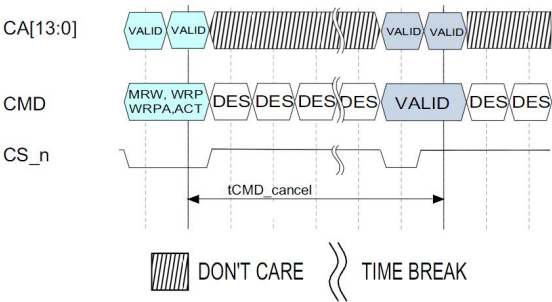


Figure 11 — Command Cancel Timing

Table 31 — Command Cancel Timing

Parameter	Symbol	DDR5 3200 ~ 6400		DDR5 6800 ~ 8800		Unit
		Min	Max	Min	Max	
Command cancel timing for ACT, WRP, WRPA, MRW when CS_n is low on 2nd cycle	tCMD_cancel	8	-	8	-	nCK

4.2 Burst Length, Type, and Order

Accesses within a given burst is currently limited to only sequential, interleaved is not supported. The ordering of accesses within a burst is determined by the burst length and the starting column address as shown in Table . The burst length is defined by bits OP[1:0] of Mode Register MR0. Burst length options include BC8 OTF, BL16, BL32 (optional) and BL32 OTF.

Table 32 — Burst Type and Burst Order for READ

Burst Length	Burst Type	C3	C2	C1	C0	Read Burst Cycle and Burst Address Sequence															
						1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
BC8	SEQ	0	0	V	V	0	1	2	3	4	5	6	7	T	T	T	T	T	T	T	T
		0	1	V	V	4	5	6	7	0	1	2	3	T	T	T	T	T	T	T	T
		1	0	V	V	8	9	A	B	C	D	E	F	T	T	T	T	T	T	T	T
		1	1	V	V	C	D	E	F	8	9	A	B	T	T	T	T	T	T	T	T
BL16	SEQ	0	0	V	V	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
		0	1	V	V	4	5	6	7	0	1	2	3	C	D	E	F	8	9	A	B
		1	0	V	V	8	9	A	B	C	D	E	F	0	1	2	3	4	5	6	7
		1	1	V	V	C	D	E	F	8	9	A	B	4	5	6	7	0	1	2	3

Table 33 — Burst Type and Burst Order for WRITE

Burst Length	Burst Type	C3	C2	C1	C0	Write Burst Cycle and Burst Address Sequence															
						1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
BC8	SEQ	0	V	V	V	0	1	2	3	4	5	6	7	X	X	X	X	X	X	X	X
		1	V	V	V	8	9	A	B	C	D	E	F	X	X	X	X	X	X	X	X
BL16	SEQ	V	V	V	V	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F

NOTE 1 T: Output driver for data and strobes are in RTT_PARK.

NOTE 2 V: A valid logic level (0 or 1), but respective buffer input ignores level on input pins.

NOTE 3 X: Don't Care.

4.2.1 Burst Type and Burst Order for Optional BL32 Mode

DDR5 DRAM supports an optional fixed BL32 mode and optional BL32 OTF (On the fly) mode for x4 devices only.

Table 34 — Burst Type and Burst Order for READ BL32

Burst Length	Burst Type	C10	C3	C2	C1	C0	Read Burst Cycle and Burst Address Sequence							
							1-4	5-8	9-12	13-16	17-20	21-24	25-28	29-32
BL32	SEQ	0	0	0	V	V	0-3	4-7	8-B	C-F	10-13	14-17	18-1B	1C-1F
		0	0	1	V	V	4-7	0-3	C-F	8-B	14-17	10-13	1C-1F	18-1B
		0	1	0	V	V	8-B	C-F	0-3	4-7	18-1B	1C-1F	10-13	14-17
		0	1	1	V	V	C-F	8-B	4-7	0-3	1C-1F	18-1B	14-17	10-13
		1	0	0	V	V	10-13	14-17	18-1B	1C-1F	0-3	4-7	8-B	C-F
		1	0	1	V	V	14-17	10-13	1C-1F	18-1B	4-7	0-3	C-F	8-B
		1	1	0	V	V	18-1B	1C-1F	10-13	14-17	8-B	C-F	0-3	4-7
		1	1	1	V	V	1C-1F	18-1B	14-17	10-13	C-F	8-B	4-7	0-3
BL16 in BL32 OTF	SEQ	0	0	0	V	V	0-3	4-7	8-B	C-F	X	X	X	X
		0	0	1	V	V	4-7	0-3	C-F	8-B	X	X	X	X
		0	1	0	V	V	8-B	C-F	0-3	4-7	X	X	X	X
		0	1	1	V	V	C-F	8-B	4-7	0-3	X	X	X	X
		1	0	0	V	V	10-13	14-17	18-1B	1C-1F	X	X	X	X
		1	0	1	V	V	14-17	10-13	1C-1F	18-1B	X	X	X	X
		1	1	0	V	V	18-1B	1C-1F	10-13	14-17	X	X	X	X
		1	1	1	V	V	1C-1F	18-1B	14-17	10-13	X	X	X	X

Table 35 — Burst Type and Burst Order for WRITE BL32

Burst Length	Burst Type	C10	C3	C2	C1	C0	Write Burst Cycle and Burst Address Sequence			
							1-8	9-16	17-24	25-32
BL32	SEQ	0	V	V	V	V	0-7	8-F	17-24	25-32
BL16 in BL32 OTF		0	V	V	V	V	0-7	8-F	X	X
		1	V	V	V	V	10-17	18-1F	X	X

NOTE 1 In case of BL16 in BL32 OTF mode by setting MR0[OP1:0=11], the internal write operation starts eight cycles earlier than for BL32 mode. This means that the starting point for tWR and tWTR shall be pulled in by eight clocks.

NOTE 2 T: Output driver for data and strobes are in high impedance.

NOTE 3 V: A valid logic level (0 or 1), but respective buffer input ignores level on input pins.

NOTE 4 X: Don't Care.

[illegible]

NOTE See Command Truth Table for details

4.4 Programmable Preamble and Postamble

DDR5 shall support programmable preambles and postambles.

4.4.1 Read Preamble and Postamble

DDR5 supports a programmable read preamble and postamble. Read Preamble is configured as 1tCK, 2tCK (two unique modes), 3tCK and 4tCK via MR8:OP[2:0].

Function	Register Type	Operand	Data
Read Preamble Settings	R/W	OP[2:0]	000B: 1 tCK - 10 Pattern 001B: 2 tCK - 0010 Pattern 010B: 2 tCK - 1110 Pattern (DDR4 Style) 011B: 3 tCK - 000010 Pattern 100B: 4 tCK - 00001010 Pattern 101B: Reserved 110B: Reserved 111B: Reserved

Read Postamble is configured as 0.5tCK or 1.5tCK via MR8:OP[6]

NOTE: DQS shall have an option to drive early by x-tCK to accommodate different HOST receiver designs as controlled by the Read DQS Offset in MR40:OP[2:0].

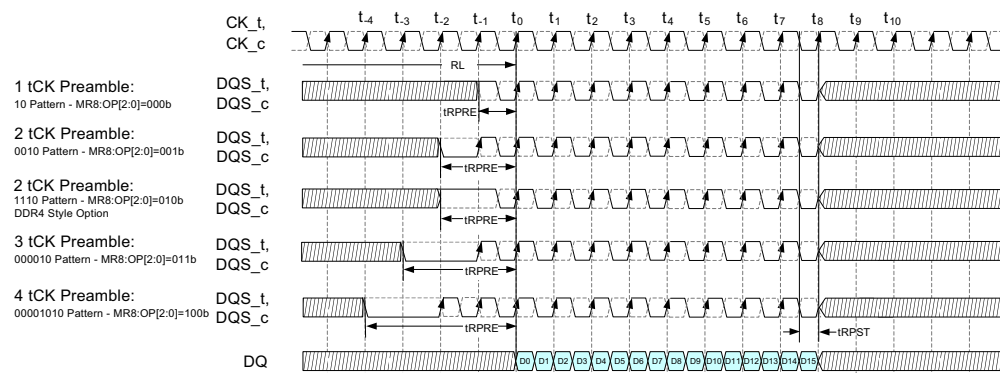


Figure 12 — Example of Read Preamble Modes (Default) with 0.5 tCK Postamble

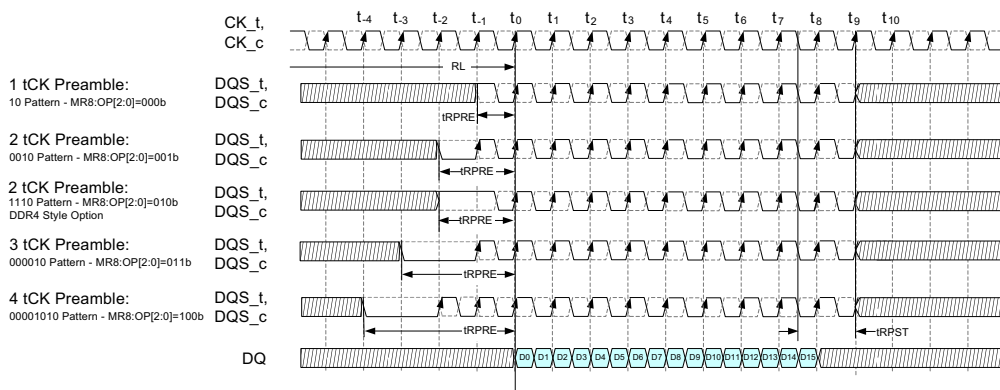


Figure 13 — Example of Read Preamble Modes (Default) with 1.5 tCK Postamble

4.4.1 Read Preamble and Postamble (cont'd)

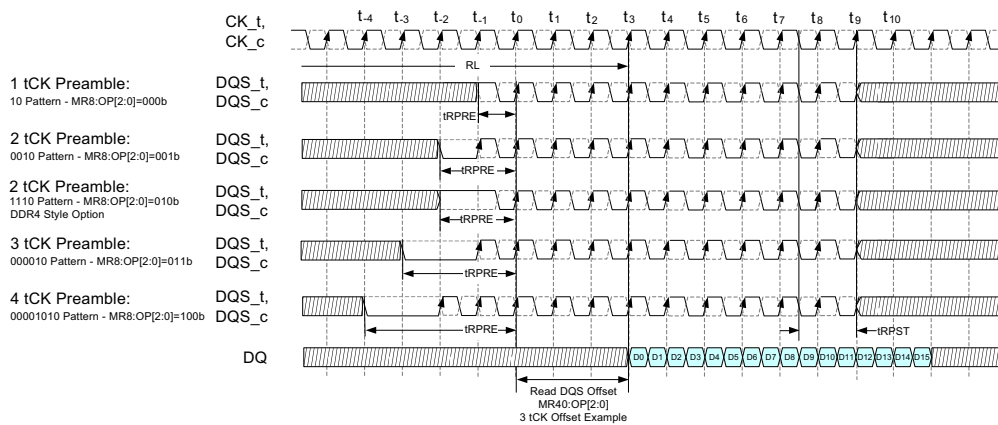


Figure 14 — Example of Read Preamble Modes with 3tCK DQS Offset and with 1.5 tCK Postamble

4.4.2 Write Preamble and Postamble

DDR5 supports a programmable write preamble and postamble.

Write Preamble is configured as 2tCK, 3tCK, and 4tCK via MR8:OP[4:3]

Write Postamble is configured as 0.5tCK or 1.5tCK via MR8:OP[7]

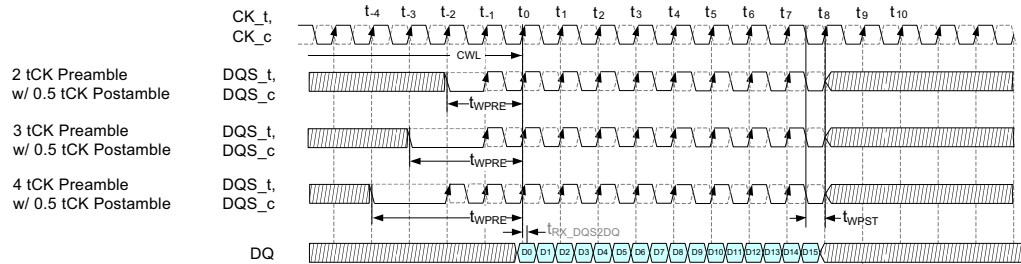


Figure 15 — Example of Write Preamble Modes (Default) with 0.5tCK Postamble

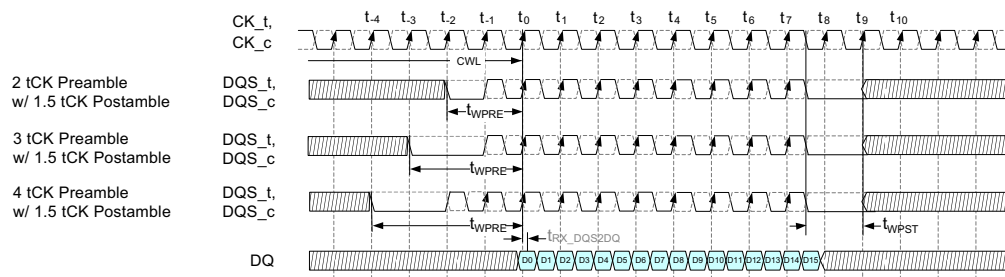
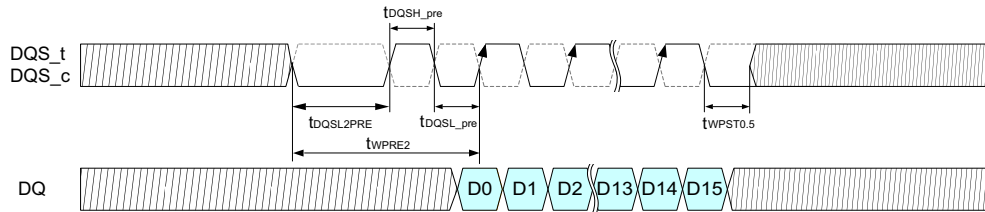


Figure 16 — Example of Write Preamble Modes (Default) with 1.5tCK Postamble

4.4.3 Read and Write Preamble and Postamble Timings

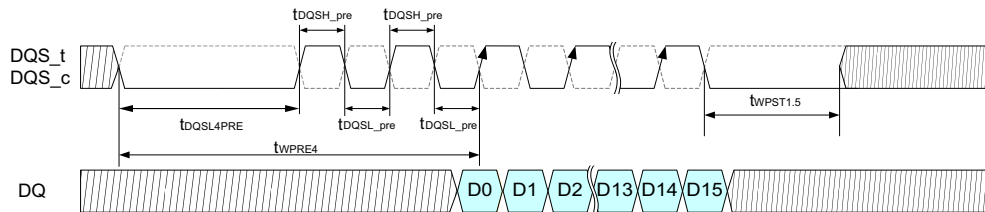
During Read and Write operations, the input receiver strobe shall be aligned with the DQ according to the Preamble and Postamble settings, and the strobe shall meet the specified timing requirements to guarantee enough timing margin by setting the window for the strobe during the Preamble and Postamble time frame. When the DRE is enabled, the DQs shall be high for a minimum of 4-UI prior to the first Write data bit to ensure proper DFE synchronization. Read Preamble is set by MR8:OP[2:0], and Write Preamble is set by MR8:OP[4:3]. Read Postamble is set by MR8:OP[6], and Write Postamble is set by MR8:OP[7].



NOTES:

1. BL=16.
2. tDQSH_pre and tDQSL_pre are shown, and apply to all toggles during the Preamble.
3. 2nd Preamble during Write to Write operation shall follow the same requirement.

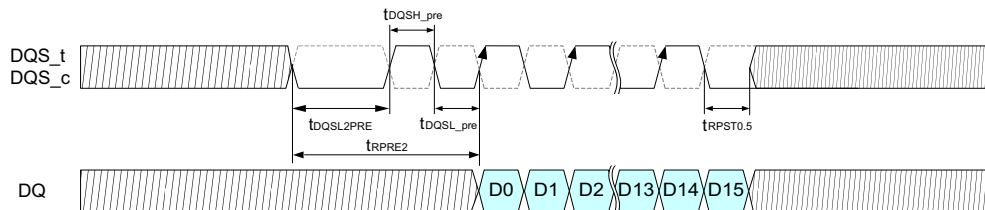
Figure 17 — DQS Timing while 2tCK Write Preamble and 0.5tCK Postamble



NOTES:

1. BL=16
2. tDQSH_pre and tDQSL_pre are shown, and apply to all toggles during the Preamble.
3. 2nd Preamble during Write to Write operation shall follow the same requirement.

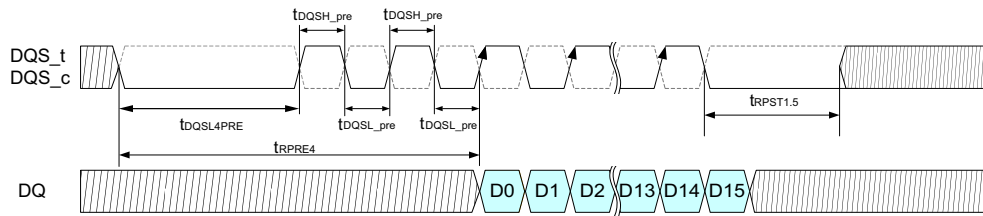
Figure 18 — DQS Timing while 4tCK Write Preamble and 1.5tCK Postamble



NOTES:

1. BL=16.
2. tDQSH_pre and tDQSL_pre are shown, and apply to all toggles during the Preamble.
3. 2nd Preamble during Read to Read operation shall follow the same requirement.

Figure 19 — DQS Timing while 2tCK Read Preamble and 0.5tCK Postamble



1. BL=16.

1. BL=16.
2. tDQSH_pre and tDQSL_pre are shown, and apply to all toggles during the Preamble.
3. 2nd Preamble during Read to Read operation shall follow the same requirement.

Figure 20 — DQS Timing while 4tCK Read Preamble and 1.5tCK Postamble

4.4.3 Read and Write Preamble and Postamble Timings (cont'd)

Table 37 — Strobe Preamble and Postamble Timing Parameters for DDR5-3200 to 4800

Parameter	Symbol	DDR5 3200 - 3600		DDR5 4000 - 4400		DDR5 4800		Unit
		Min	Max	Min	Max	Min	Max	
DQS_t, DQS_c differential READ Preamble (1tCK Preamble)	tRPRE1	0.900	-	-	-	-	-	tCK(avg)
DQS_t, DQS_c differential READ Preamble (2tCK Preamble)	tRPRE2	1.800	-	1.800	-	1.800	-	tCK(avg)
DQS_t, DQS_c differential READ Preamble DDR4 (2tCK Preamble)	tRPRE2_D4	1.800	-	1.800	-	1.800	-	tCK(avg)
DQS_t, DQS_c differential READ Preamble (3tCK Preamble)	tRPRE3	-	-	2.700	-	2.700	-	tCK(avg)
DQS_t, DQS_c differential READ Preamble (4tCK Preamble)	tRPRE4	-	-	-	-	-	-	tCK(avg)
DQS_t, DQS_c differential READ Postamble (0.5tCK Postamble)	tRPST0.5	0.450	-	0.450	-	0.450	-	tCK(avg)
DQS_t, DQS_c differential READ Postamble (1.5tCK Postamble)	tRPST1.5	1.200	-	1.200	-	1.200	-	tCK(avg)
DQS_t, DQS_c differential WRITE Preamble (2tCK Preamble)	tWPRES2	1.800	-	1.800	-	1.800	-	tCK(avg)
DQS_t, DQS_c differential WRITE Preamble (3tCK Preamble)	tWPRES3	-	-	2.700	-	2.700	-	tCK(avg)
DQS_t, DQS_c differential WRITE Preamble (4tCK Preamble)	tWPRES4	-	-	3.600	-	3.600	-	tCK(avg)
DQS_t, DQS_c differential WRITE Postamble (0.5tCK Postamble)	tWPST0.5	0.45	-	0.45	-	0.45	-	tCK(avg)
DQS_t, DQS_c differential WRITE Postamble (1.5tCK Postamble)	tWPST1.5	-	-	1.20	-	1.20	-	tCK(avg)
DQS_t, DQS_c differential high toggle pulse width during DDR4 READ Preamble (2tCK Preamble)	tDQSH2PRE_D4	1.350	-	1.350	-	1.350	-	tCK(avg)
DQS_t, DQS_c differential initial low pulse width during Preamble (2tCK Preamble)	tDQSL2PRE	0.900	-	0.900	-	0.900	-	tCK(avg)
DQS_t, DQS_c differential initial low pulse width during Preamble (3tCK Preamble)	tDQSL3PRE	-	-	1.800	-	1.800	-	tCK(avg)
DQS_t, DQS_c differential initial low pulse width during READ Preamble (4tCK Preamble)	tDQSL4PRE	-	-	-	-	-	-	tCK(avg)
DQS_t, DQS_c differential initial low pulse width during WRITE Preamble (4tCK Preamble)	tDQSL4PRE	-	-	1.800	-	1.800	-	tCK(avg)
DQS_t, DQS_c differential high toggle pulse width during WRITE Preamble	tDQSH_pre	0.395	0.605	0.395	0.605	0.430	0.570	tCK(avg)
DQS_t, DQS_c differential low toggle pulse width during WRITE Preamble	tDQSL_pre	0.395	0.605	0.395	0.605	0.430	0.570	tCK(avg)

NOTES:

1. The 1tCK Read Preamble is valid for the DDR5-3200/3600 speed grades.
2. The 2tCK Read and Write Preamble is valid for the DDR5-3200/3600/4000/4400/4800 speed grades.
3. The 2tCK DDR4 Read Preamble is valid for the DDR5-3200/3600/4000/4400/4800 speed grades.
4. The 3tCK Read and Write Preamble is valid for the DDR5-4000/4400/4800/5200/5600/6000/6400 speed grades.
5. The 4tCK Read and Write Preamble is valid for the DDR5-4000/4400/4800/5200/5600/6000/6400/6800/7200/7600/8000/8400/8800 speed grades.
6. The 0.5tCK Read and Write Postamble is valid for the DDR5-3200/3600/4000/4400/4800 speed grades.
7. The 1.5tCK Read Postamble is valid for the DDR5-3200/3600/4000/4400/4800/5200/5600/6000/6400/6800/7200/7600/8000/8400/8800 speed grades.
8. The 1.5tCK Write Postamble is valid for the DDR5-4000/4400/4800/5200/5600/6000/6400/6800/7200/7600/8000/8400/8800 speed grades.
9. The strobe preamble and postamble timing parameter values shall follow the operating frequency.

4.4.3 Read and Write Preamble and Postamble Timings (cont'd)**Table 38 — Strobe Preamble and Postamble Timing Parameters for DDR5-5200 to 8800**

Parameter	Symbol	DDR5 5200		DDR5 5600 - 6400		DDR5 6800 - 7200		DDR5 7600-8800		Unit
		Min	Max	Min	Max	Min	Max	Min	Max	
DQS_t, DQS_c differential READ Preamble (1tCK Preamble)	tRPRE1	-	-	-	-	-	-	-	-	tCK(avg)
DQS_t, DQS_c differential READ Preamble (2tCK Preamble)	tRPRE2	-	-	-	-	-	-	-	-	tCK(avg)
DQS_t, DQS_c differential READ Preamble DDR4 (2tCK Preamble)	tRPRE2_D4	2.700	-	2.700	-	-	-	-	-	tCK(avg)
DQS_t, DQS_c differential READ Preamble (3tCK Preamble)	tRPRE3	2.700	-	2.700	-	-	-	-	-	tCK(avg)
DQS_t, DQS_c differential READ Preamble (4tCK Preamble)	tRPRE4	-	-	3.600	-	3.600	-	3.600	-	tCK(avg)
DQS_t, DQS_c differential READ Postamble (0.5tCK Preamble)	tRPST0.5	-	-	-	-	-	-	-	-	tCK(avg)
DQS_t, DQS_c differential READ Postamble (1.5tCK Preamble)	tRPST1.5	1.200	-	1.200	-	1.200	-	1.300	-	tCK(avg)
DQS_t, DQS_c differential WRITE Preamble (2tCK Preamble)	tWPRE2	-	-	-	-	-	-	-	-	tCK(avg)
DQS_t, DQS_c differential WRITE Preamble (3tCK Preamble)	tWPRE3	2.700	-	2.700	-	-	-	-	-	tCK(avg)
DQS_t, DQS_c differential WRITE Preamble (4tCK Preamble)	tWPRE4	3.600	-	3.600	-	3.600	-	3.600	-	tCK(avg)
DQS_t, DQS_c differential WRITE Postamble (0.5tCK Postamble)	tWPST0.5	-	-	-	-	-	-	-	-	tCK(avg)
DQS_t, DQS_c differential WRITE Postamble (1.5tCK Postamble)	tWPST1.5	1.200	-	1.200	-	1.200	-	1.200	-	tCK(avg)
DQS_t, DQS_c differential high toggle pulse width during DDR4 READ Preamble (2tCK Preamble)	tDQSH2PRE_E_D4	-	-	-	-	-	-	-	-	tCK(avg)
DQS_t, DQS_c differential initial low pulse width during WRITE Preamble (2tCK Preamble)	tDQSL2PRE	-	-	-	-	-	-	-	-	tCK(avg)
DQS_t, DQS_c differential initial low pulse width during Preamble (3tCK Preamble)	tDQSL3PRE	1.800	-	1.800	-	-	-	-	-	tCK(avg)
DQS_t, DQS_c differential initial low pulse width during READ Preamble (4tCK Preamble)	tDQSL4PRE	-	-	1.800	-	1.800	-	1.800	-	tCK(avg)
DQS_t, DQS_c differential initial low pulse width during WRITE Preamble (4tCK Preamble)	tDQSL4PRE	1.800	-	1.800	-	1.800	-	1.800	-	tCK(avg)
DQS_t, DQS_c differential high toggle pulse width during WRITE Preamble	tDQSH_pre	0.430	0.570	0.430	0.570	0.450	0.550	0.450	0.550	tCK(avg)
DQS_t, DQS_c differential low toggle pulse width during WRITE Preamble	tDQSL_pre	0.430	0.570	0.430	0.570	0.450	0.550	0.450	0.550	tCK(avg)

NOTES:

1. The 1tCK Read Preamble is valid for the DDR5-3200/3600 speed grades.
2. The 2tCK Read and Write Preamble is valid for the DDR5-3200/3600/4000/4400/4800 speed grades.
3. the 2tCK DDR4 Read Preamble is valid for the DDR5-3200/3600/4000/4400/4800 speed grades.
4. The 3tCK Read and Write Preamble is valid for the DDR5-4000/4400/4800/5200/5600/6000/6400 speed grades.
5. The 4tCK Read Preamble is valid for the DDR5-5600/6000/6400/6800/7200/7600/8000/8400/8800 speed grades.
6. The 4tCK Write Preamble is valid for the DDR5-4000/4400/4800/5200/5600/6000/6400/6800/7200/7600/8000/8400/8800 speed grades.
7. The 0.5tCK Read and Write Postamble is valid for the DDR5-3200/3600/4000/4400/4800 speed grades.
8. The 1.5tCK Read Postamble is valid for the DDR5-3200/3600/4000/4400/4800/5200/5600/6000/6400/6800/7200/7600/8000/8400/8800 speed grades.
9. The 1.5tCK Write Postamble is valid for the DDR5-4000/4400/4800/5200/5600/6000/6400/6800/7200/7600/8000/8400/8800 speed grades.
10. The strobe preamble and postamble timing parameter values shall follow the operating frequency.

4.4.4 tWPRE and tRPRE Measurement

tWPRE and tRPRE are measured from a starting point at VswM HIGH or LOW (as defined in the table below) to the differential crossing point of DQS_t/DQS_c corresponding to the first burst bit of data as the ending point. The method is applicable for all programmable Preamble durations.

Examples of the method for measuring the duration of tWPRE and tRPRE are shown in Figure 21:

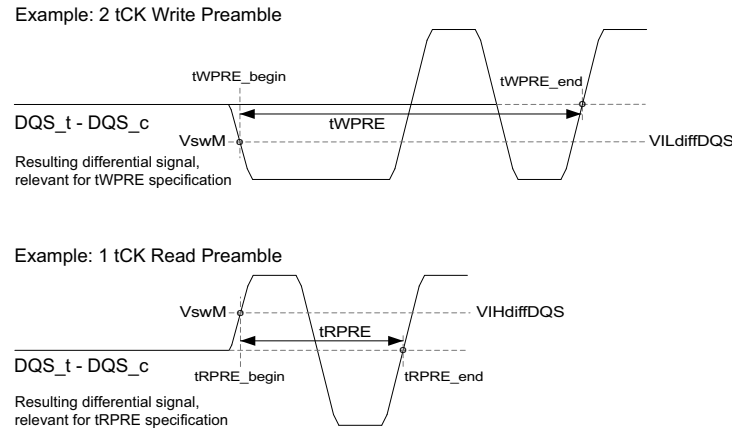


Figure 21 — Method for Measuring Preamble Start and End Points

Table 39 — VswM Reference Voltages for Preamble and Postamble Timing Measurements

Measured Parameter	Reference	Unit
VswM HIGH	VIHdiffDQS	mV
VswM LOW	VILdiffDQS	mV

4.4.5 tWPST and tRPST Measurement

tWPST and tRPST are measured from a starting point at the differential crossing point of DQS_t/DQS_c corresponding to the last burst bit of data to the VswM LOW ending point. The method is applicable for all programmable Postamble durations.

Examples of the method for measuring the duration of tWPST and tRPST are shown in the Figure 22:

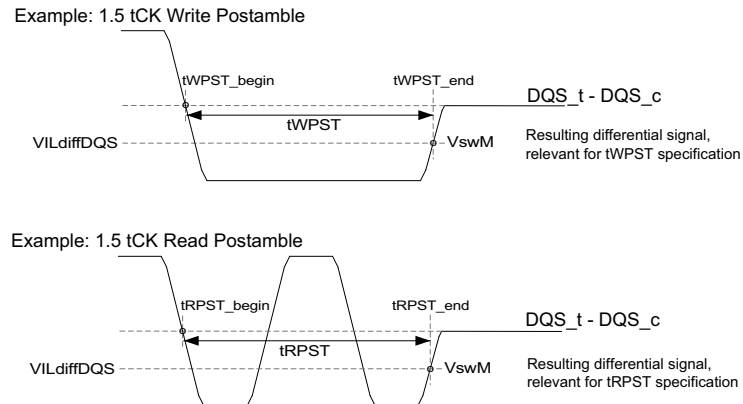


Figure 22 — Method for Measuring Postamble Start and End Points

4.5 Interamble

The DQS strobe for the device requires a preamble prior to the first latching edge (the rising edge of DQS_t with data valid), and it requires a postamble after the last latching edge. The preamble and postamble options are set via Mode Register Write commands. Additionally, the postamble and preamble configured size shall NOT force the HOST to add command gaps in the command interval just to satisfy postamble or preamble settings. (i.e., Preamble=4tCK + Postamble=1.5tCK shall not force tCCD+5).

4.5.1 Read Interamble Timing Diagrams

In Read to Read operations with tCCD=BL/2, postamble for 1st command and preamble for 2nd command shall disappear to create consecutive DQS latching edge for seamless burst operations.

In the case of Read to Read operations with command interval of tCCD+1, tCCD+2, etc., if the postamble and preambles overlap, the toggles take precedence over static preambles.

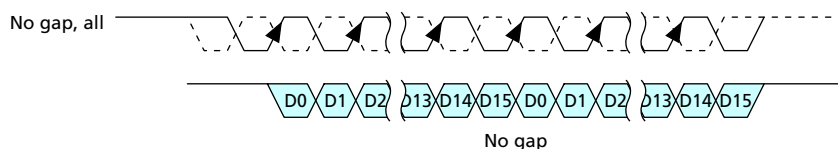


Figure 23 — Example of Seamless Reads Operation: tCCD=Min

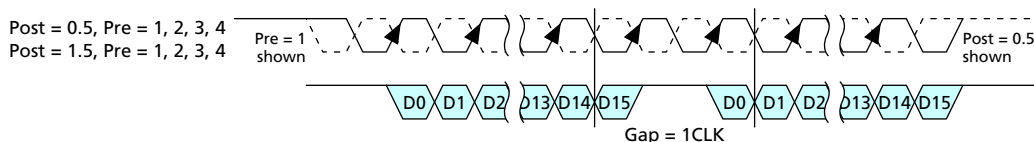


Figure 24 — Example of Consecutive Reads Operation: tCCD=Min+1

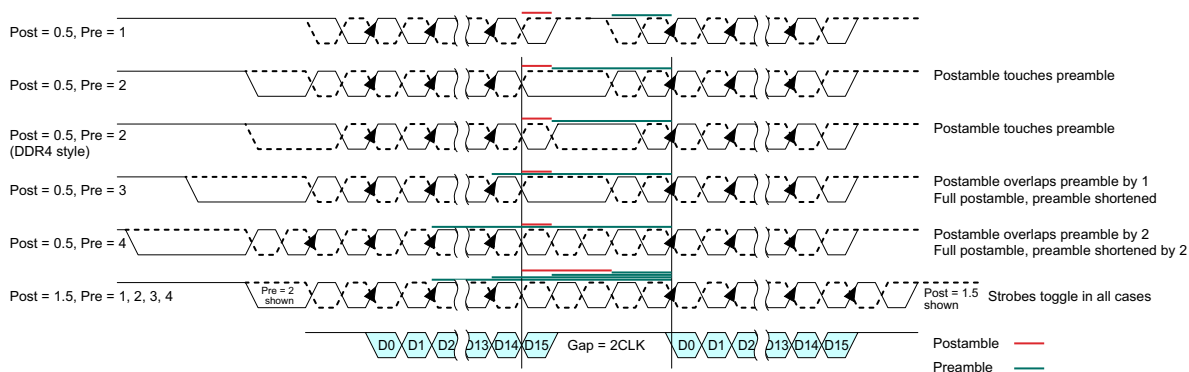


Figure 25 — Example of Consecutive Reads Operation: tCCD=Min+2

4.5.1 Read Interamble Timing Diagrams (cont'd)

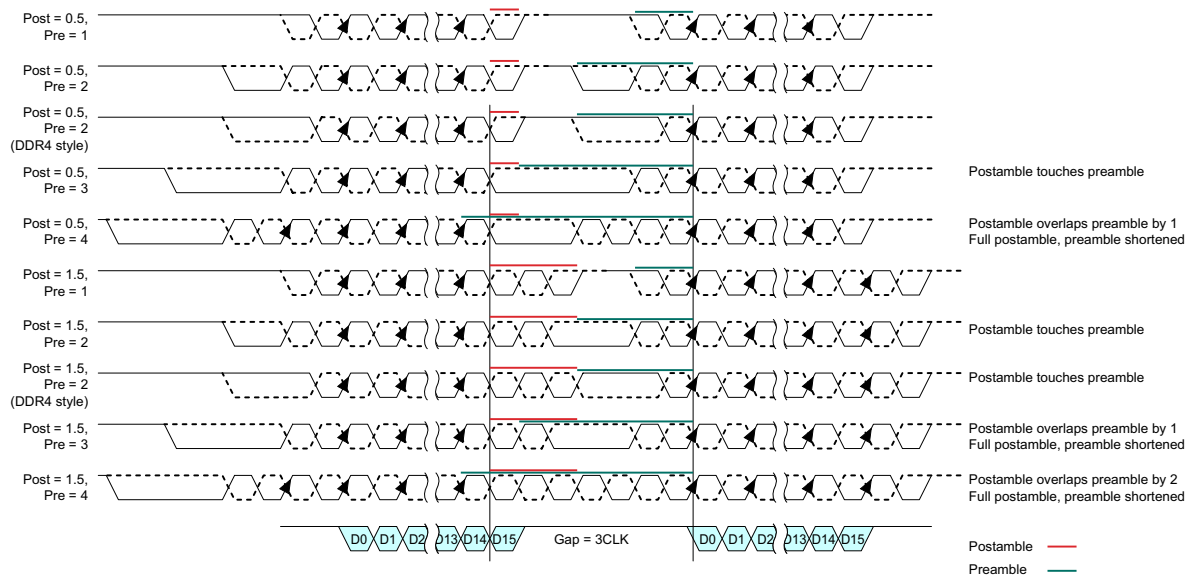


Figure 26 — Example of Consecutive Reads Operation: $t_{CCD} = \text{Min} + 3$

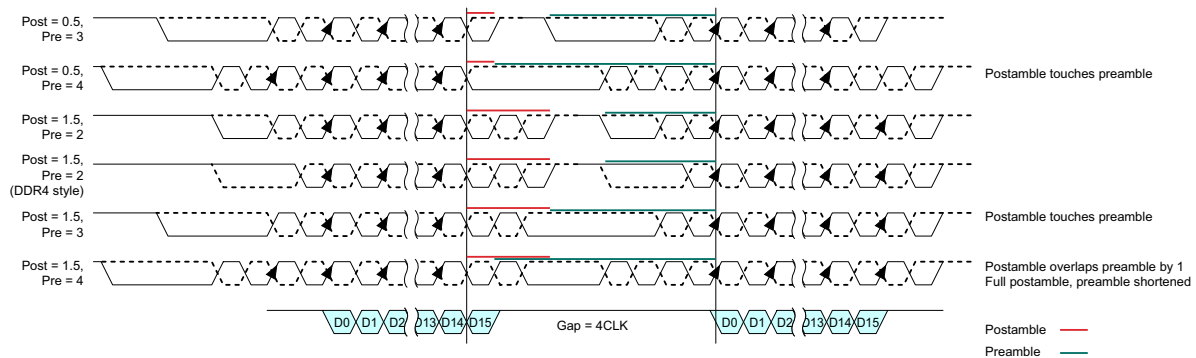


Figure 27 — Example of Consecutive Reads Operation: $t_{CCD} = \text{Min} + 4$

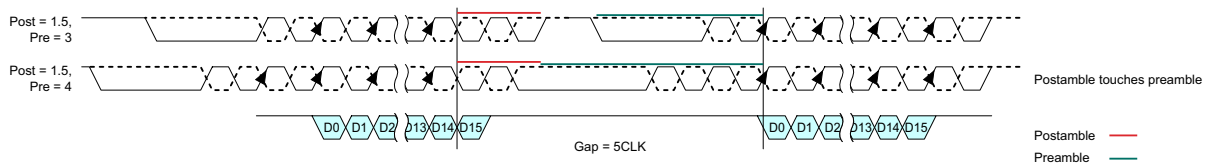


Figure 28 — Example of Consecutive Reads Operation: $t_{CCD} = \text{Min} + 5$

4.5.2 Write Interamble Timing Diagrams

In Write to Write operations with $t_{CCD}=BL/2$, postamble for 1st command and preamble for 2nd command shall disappear to create consecutive DQS latching edge for seamless burst operations.

In the case of Write to Write operations with command interval of $t_{CCD}+1$, $t_{CCD}+2$, etc., if the postamble and preambles overlap, the toggles take precedence over static preambles.

When the DFE is enabled, the DQs shall be high during Interamble for a minimum of 4UI prior to the first Write data bit of the second Write command. If there are 4UI or less and DFE is enabled, the host shall drive DQs high during the full Write Interamble period.

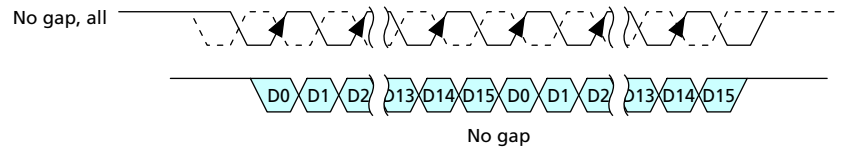


Figure 29 — Example of Seamless Writes Operation: $t_{CCD}=\text{Min}$

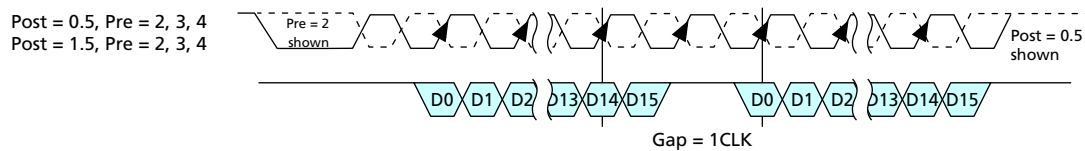


Figure 30 — Example of Consecutive Writes Operation: $t_{CCD}=\text{Min}+1$

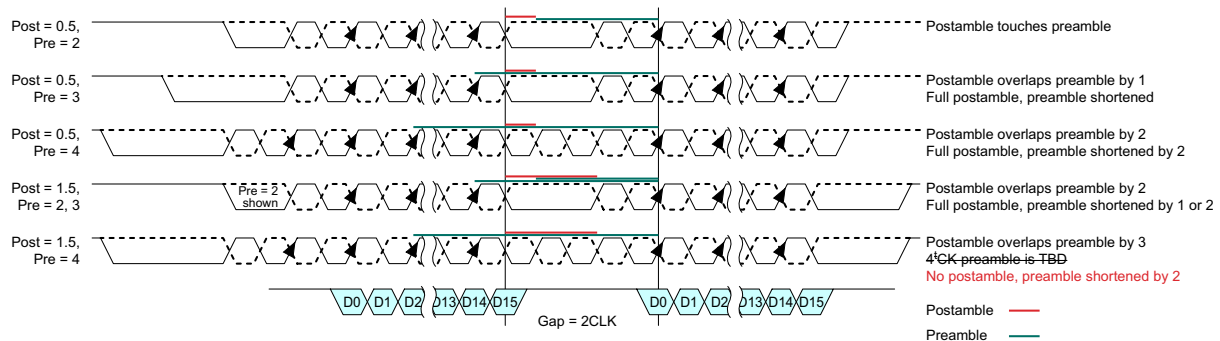


Figure 31 — Example of Consecutive Writes Operation: $t_{CCD}=\text{Min}+2$

4.5.2 Write Interamble Timing Diagrams (cont'd)

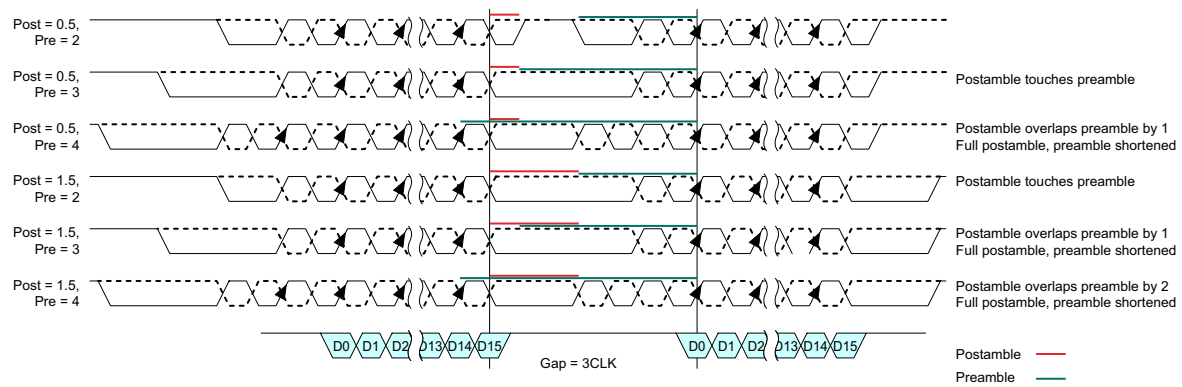


Figure 32 — Example of Consecutive Writes Operation: $t_{CCD} = \text{Min} + 3$

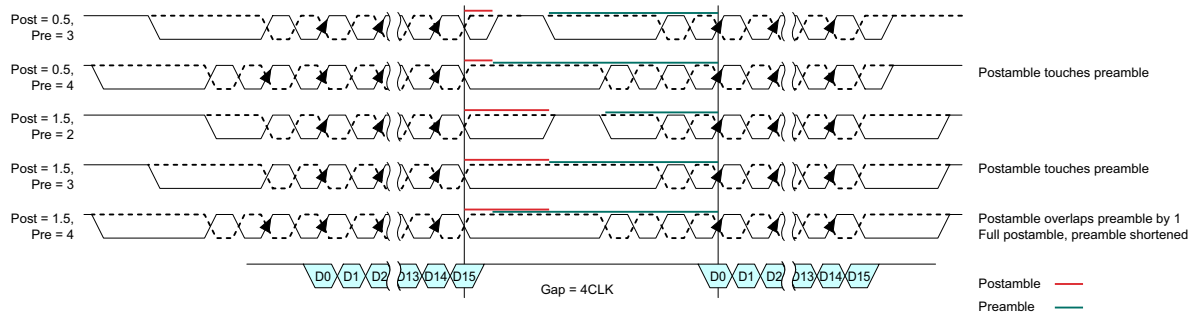


Figure 33 — Example of Consecutive Writes Operation: $t_{CCD} = \text{Min} + 4$

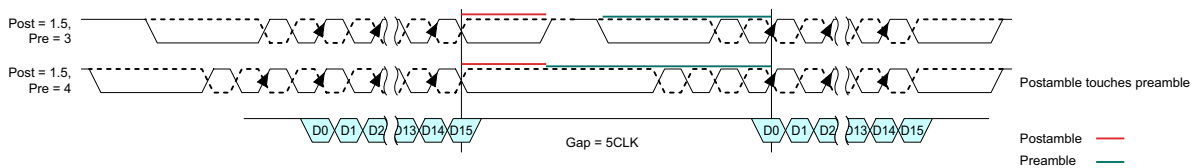


Figure 34 — Example of Consecutive Writes Operation: $t_{CCD} = \text{Min} + 5$

4.6 Activate Command

The ACTIVATE command is used to open (or activate) a row in a particular bank for a subsequent access. The value on the BG[2:0] in X4/8 and BG[1:0] in X16 select the bankgroup; BA[1:0] inputs selects the bank within the bankgroup, and the address provided on the appropriate CA pins for R[17:0] to select the row (see Table 40 below). In the case of a 3DS DDR5 SDRAM device, the CID[3:0] shall also be selected to identify the correct die in the stack. This row remains active (or open) for accesses until a precharge command is issued to that bank or a precharge all command is issued. A bank must be precharged before opening a different row in the same bank.

Bank-to-bank command timing for ACTIVATE commands uses two different timing parameters, depending on whether the banks are in the same or different bank group. tRRD_S (short) is used for timing between banks located in different bank groups. tRRD_L (long) is used for timing between banks located in the same bank group. Consecutive ACTIVATE commands, allowed to be issued at tRRDmin, are restricted to a maximum of four within the time period tFAW (four activate window).

Table 40 — Activate Command (for Reference)

Function	Abbreviation	CS_n	CA Pins														NOTES
			CA0	CA1	CA2	CA3	CA4	CA5	CA6	CA7	CA8	CA9	CA10	CA11	CA12	CA13	
Activate	ACT	L	L	L	R0	R1	R2	R3	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2	1
		H	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13	R14	R15	R16	CID3/ R17	

NOTE 1 See Command Truth Table for details

4.6.1 Non-Binary Density Considerations

An ACT command with row address inputs which violate the 'MSB address bit "HIGH", MSB-1 address bit "LOW"' non-binary density address restriction shall follow all timing and protocol rules as though the ACT were valid.

Any RD or RDA command following an ACT to the invalid address space within the same bank shall drive the DQS strobes with normal timing but shall not output data on the DQs that can be used to learn about data stored in cells with valid addresses. DQ data that coincidentally matches cell array data is permissible (for example: always sending the all 1s and cell data sometimes being all 1s). Consistently exposing data from a previous read or previous activate is not permissible.

Any WR, WRA, WRP or WRPA command following an ACT to the invalid address space within the same bank shall not result in new data being written anywhere within the DRAM.

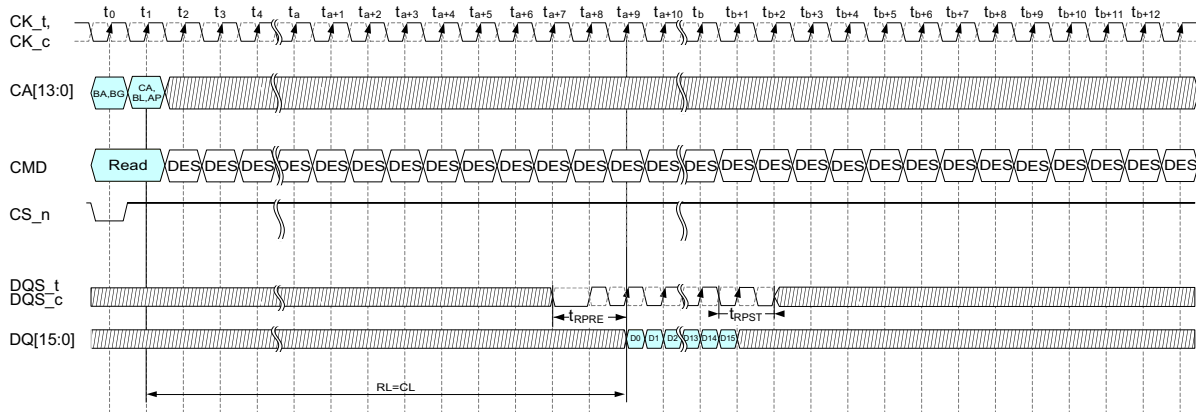
The DRAM will operate normally for Read and Write commands to banks which have pages open for valid rows.

4.7 Read Operation

The Read Operation causes the DRAM to retrieve and output data stored in its array. The Read Operation is initiated by the Read Command during which the beginning column address and bank/group address for the data to be retrieved from the array is provided. The data is driven by the DRAM on its DQ pins RL (CL) cycles after the Read Command along with the proper waveform on the DQS inputs. Read Latency (RL or CL) is defined from the Read command to data and is not affected by the Read DQS offset timing (MR40 OP[2:0]).

4.7.1 READ Burst Operation

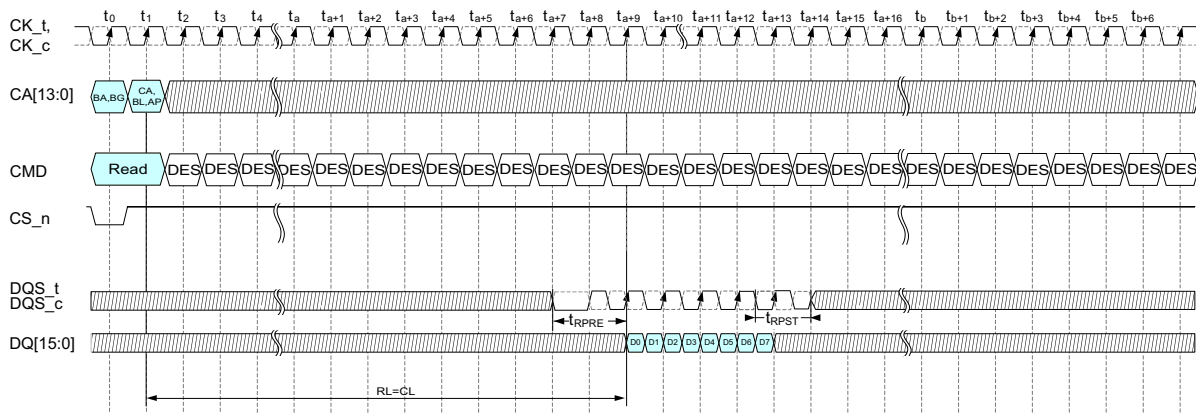
During a READ or WRITE command, DDR5 shall support BC8, BL16, BL32 (optional) and BL32 OTF (optional) during the READ or WRITE. MR0[1:0] is used to select burst operation mode.



NOTES:

1. BL=16, Preamble = 2tCK - 0010 Pattern Preamble, 1.5tCK Postamble
2. DES commands are shown for ease of illustration; other commands may be valid at these times.
3. In this example, Read DQS Offset Timing is set to 0 Clocks.

Figure 35 — READ Burst Operation (BL16)

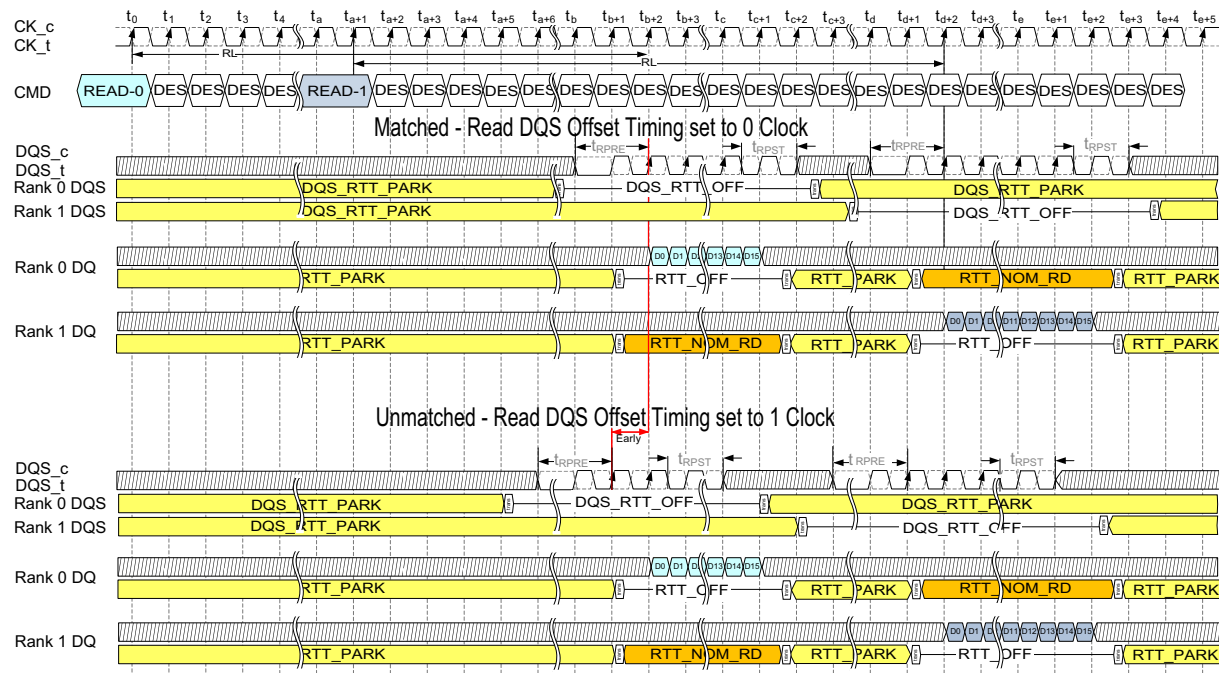


NOTES:

1. BC=8, Preamble = 2tCK - 0010 Pattern Preamble, 1.5tCK Postamble
2. DES commands are shown for ease of illustration; other commands may be valid at these times.
3. In this example, Read DQS Offset Timing is set to 0 Clocks
4. In non-CRC mode, DQS_t and DQS_c stop toggling at the completion of the BC8 data bursts, plus the postamble.

Figure 36 — Read Burst Operation (BC8)

4.7.1 READ Burst Operation (cont'd)



NOTES:

1. BL=16, Preamble = 2tCK - 0010 Pattern Preamble, 1.5tCK Postamble
2. DES commands are shown for ease of illustration; other commands may be valid at these times.
3. Two different examples are shown side by side, the top with the default setting for Read DQS Offset = 0 Clock, the lower with a 1 Clock setting. In the lower case, the DQS is started 1 clock earlier than normal with respect to RL (CL).
4. In both cases, the Data does not move

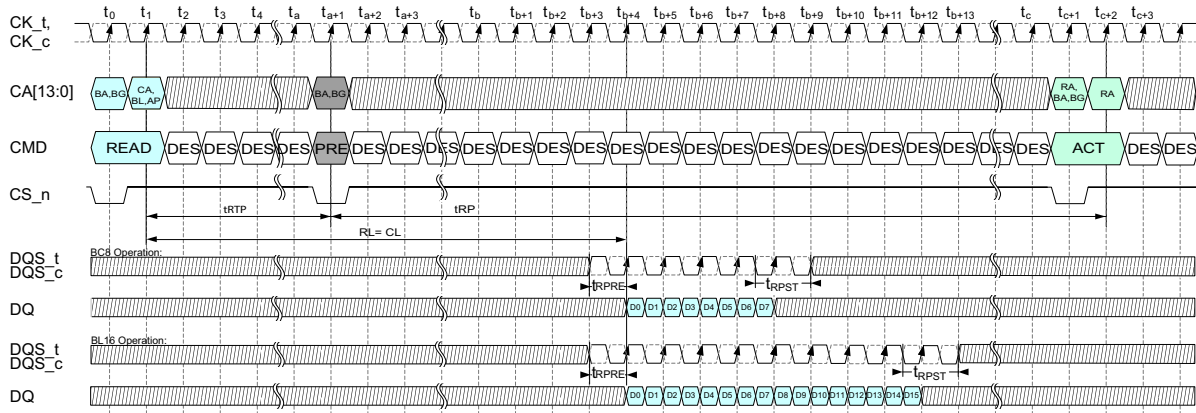
Figure 37 — READ to READ, Different Ranks Operation with Read DQS Offset Usage (BL16)

4.7.2 Burst Read Operation Followed by a Precharge

The minimum external Read command to Precharge command spacing to the same bank is equal to t_{RTP} with t_{RTP} being the Internal Read Command to Precharge Command Delay. Note that the minimum ACT to PRE timing, t_{RAS} , must be satisfied as well. The minimum value for the Internal Read Command to Precharge Command Delay is given by $t_{RTP.min}$. A new bank active command may be issued to the same bank if the following two conditions are satisfied simultaneously:

1. The minimum RAS precharge time ($t_{RP.MIN}$) has been satisfied from the clock at which the precharge begins.
2. The minimum RAS cycle time ($t_{RC.MIN}$) from the previous bank activation has been satisfied.

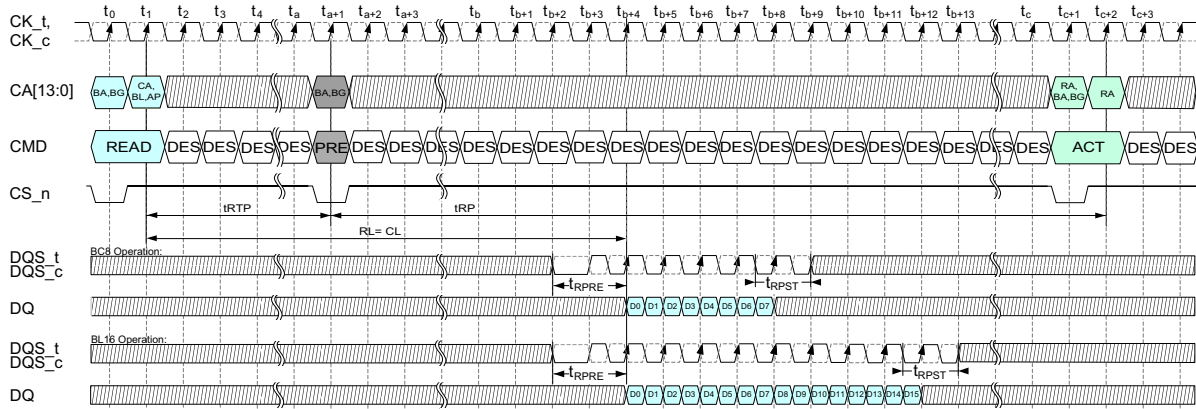
Examples of Read commands followed by Precharge are shown in Figure 38 and Figure 39.



NOTES:

1. BL = 16, 1tCK Preamble, 1.5tCK Postamble
2. DES commands are shown for ease of illustration; other commands may be valid at these times.
3. The example assumes $t_{RAS.MIN}$ is satisfied at Precharge command time(t_{a+1}) and that $t_{RC.MIN}$ is satisfied at the next Active command time(t_{c+2}).

Figure 38 — READ to PRECHARGE with 1tCK Preamble



NOTES:

1. BL = 16, 2tCK - 0010 Pattern Preamble, 1.5tCK Postamble,
2. DES commands are shown for ease of illustration; other commands may be valid at these times.
3. The example assumes $t_{RAS.MIN}$ is satisfied at Precharge command time(t_{a+1}) and that $t_{RC.MIN}$ is satisfied at the next Active command time(t_{c+2}).

Figure 39 — READ to PRECHARGE with 2tCK Preamble

Following parameters shall be defined for CK to read DQS timings.

Speed		DDR5-3200		DDR5-3600		DDR5-4000		DDR5-4400		DDR5-4800		Units	NOTE
Parameter	Symbol	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
DQS_t, DQS_c rising edge output timing location from rising CK_t, CK_c	tDQSCK	-0.240	0.240	-0.252	0.252	-0.270	0.270	-0.286	0.286	-0.300	0.300	tCK	1, 4, 5
DQS_t, DQS_c rising edge output variance window	tDQSCKi	-	0.410	-	0.430	-	0.460	-	0.475	-	0.490	tCK	2, 3, 4, 5, 6
NOTE 1	Measured over full VDD and Temperature spec ranges.												
NOTE 2	Measured for a given DRAM part, and for each DQS_t/DQS_c pair in case of x16 (part variation is excluded).												
NOTE 3	These parameters are verified by design and characterization, and may not be subject to production test.												
NOTE 4	Assume no jitter on input clock signals to the DRAM.												
NOTE 5	Refer to Section 4.7.1 READ Timing Definitions.												
NOTE 6	Measured at a fixed and constant VDD and Temperature condition.												

Speed		DDR5-5200		DDR5-5600		DDR5-6000		DDR5-6400		DDR5-6800		Units	NOTE
Parameter	Symbol	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
DQS_t, DQS_c rising edge output timing location from rising CK_t, CK_c	tDQSCK	-0.313	0.313	-0.325	0.325	-0.337	0.337	-0.348	0.348	-0.359	0.359	tCK	1, 4, 5
DQS_t, DQS_c rising edge output variance window	tDQSCKi	-	0.510	-	0.530	-	0.549	-	0.567	-	0.585	tCK	2, 3, 4, 5, 6
NOTE 1 Measured over full VDD and Temperature spec ranges.													
NOTE 2 Measured for a given DRAM part, and for each DQS_t/DQS_c pair in case of x16 (part variation is excluded).													
NOTE 3 These parameters are verified by design and characterization, and may not be subject to production test.													
NOTE 4 Assume no jitter on input clock signals to the DRAM.													
NOTE 5 Refer to Section 4.7.1 READ Timing Definitions.													
NOTE 6 Measured at a fixed and constant VDD and Temperature condition.													

Speed		DDR5-7200		DDR5-7600		DDR5-8000		DDR5-8400		DDR5-8800		Units	NOTE
Parameter	Symbol	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
DQS_t, DQS_c rising edge output timing location from rising CK_t, CK_c	tDQSCK	-0.370	0.370	-0.380	0.380	-0.390	0.390	-0.400	0.400	-0.410	0.410	tCK	1, 4, 5
DQS_t, DQS_c rising edge output variance window	tDQSCKi	-	0.602	-	0.619	-	0.635	-	0.651	-	0.667	tCK	2, 3, 4, 5, 6
NOTE 1	Measured over full VDD and Temperature spec ranges.												
NOTE 2	Measured for a given DRAM part, and for each DQS_t/DQS_c pair in case of x16 (part variation is excluded).												
NOTE 3	These parameters are verified by design and characterization, and may not be subject to production test.												
NOTE 4	Assume no jitter on input clock signals to the DRAM.												
NOTE 5	Refer to Section 4.7.1 READ Timing Definitions.												
NOTE 6	Measured at a fixed and constant VDD and Temperature condition.												

4.7.2.1 CLK to Read DQS Timing Parameters (cont'd)

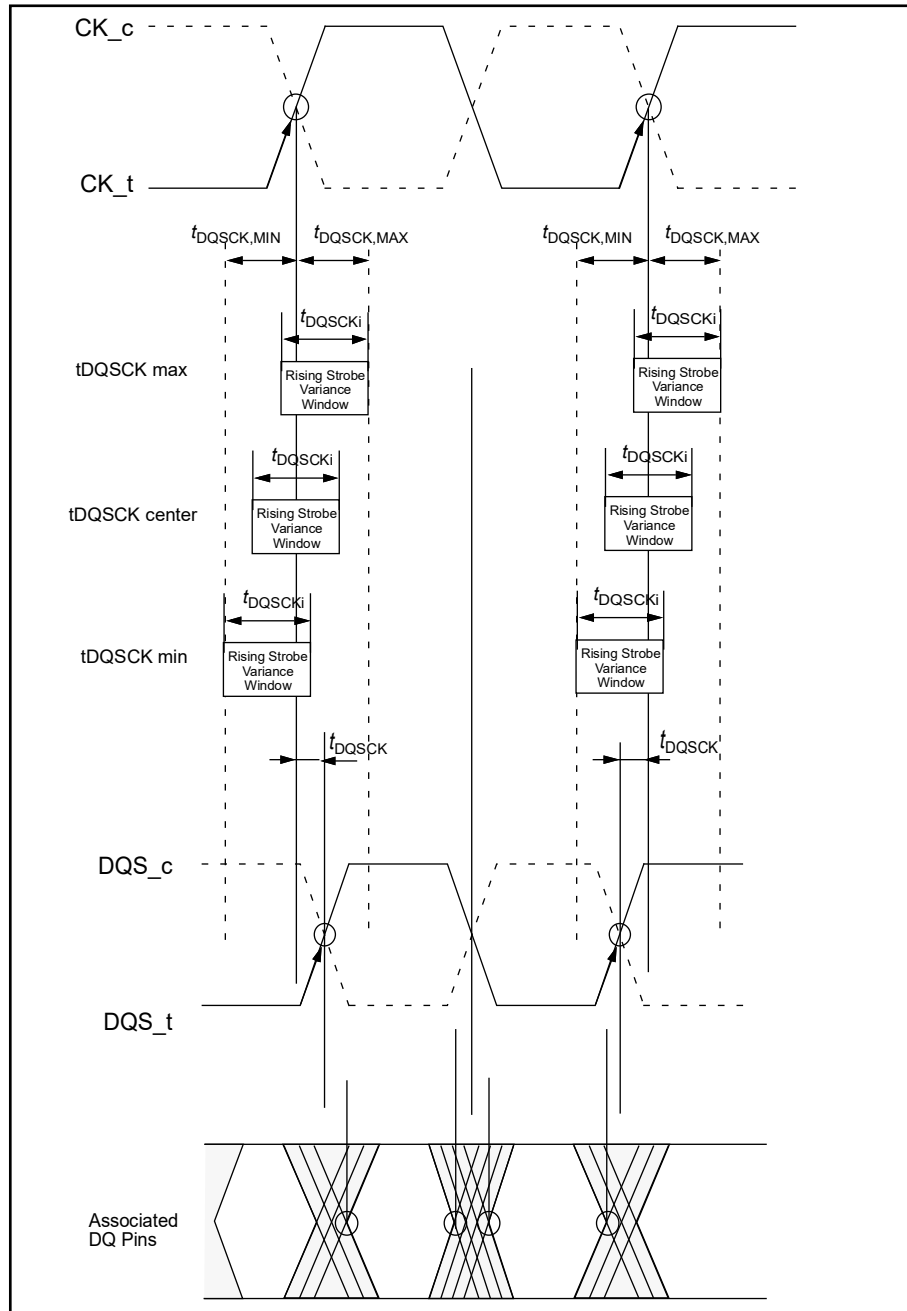


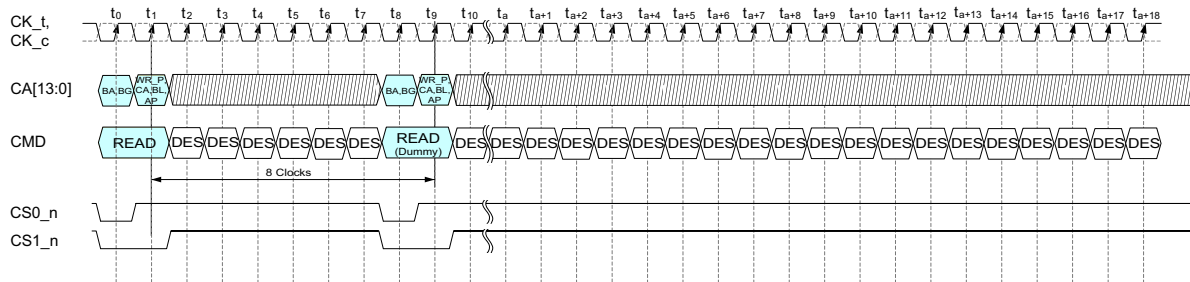
Figure 40 — TDQSCK Timing Definition

4.7.3 Read Burst Operation for Optional BL32 Mode

The following read timing diagrams cover read timings for fixed BL32 BL32 in BL32 OTF mode and BL16 in BL32 OTF mode for x4 devices only.

In these read timing diagrams, for clarity of illustration, CK and DQS are shown aligned. As well, DQS and DQ are shown edge-aligned. Offset between CK and DQS, and between DQS and DQ may be appropriate.

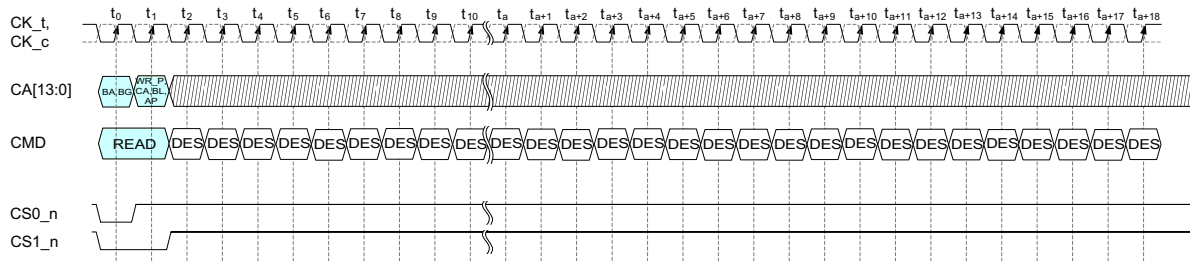
A dummy CAS command is required for second half of the transfer of BL32. If non-target ODT is needed in the system then a dummy ODT command must be issued to the non-target rank for second half of the transfer of BL32.



NOTES:

1. DES commands are shown for ease of illustration; other commands may be valid at these times.
2. A dummy RD command is required for the second half of the transfer with a delay of 8 clocks from the first RD command.
3. The figure also shows a dummy ODT command being issued to non-target rank 1 for the second half of the transfer.
4. C10 is used for burst ordering and can be LOW or HIGH for the first RD command. C10 for the dummy RD command must be the opposite value from the first RD command.
5. DDR5 DRAM supports an optional fixed BL32 mode and optional BL32 OTF (On the fly) mode for x4 devices only.
6. CA bits other than C10 and AP in dummy CAS command are the same as the first CAS command.

Figure 41 — Read Timing for fixed BL32 and BL32 in BL32 OTF Mode

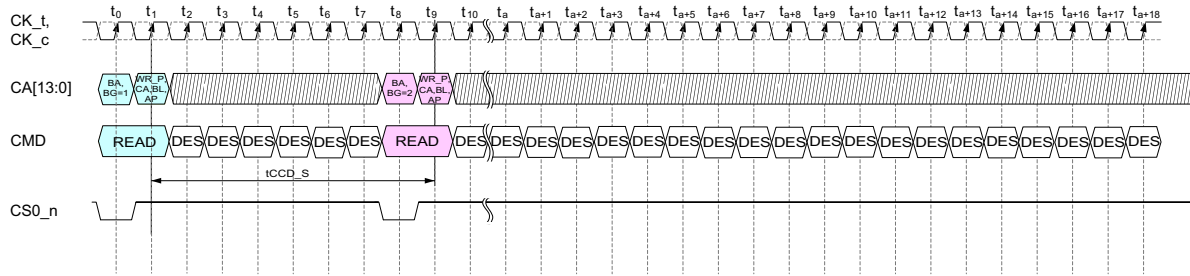


NOTES:

1. Figure shows BL16 read operation when MR0 is programmed to use BL32 OTF mode. In this case, no dummy RD command is required as transfer size is BL16.
2. DES commands are shown for ease of illustration; other commands may be valid at these times including commands to allow data transfer from the same die after transfer of BL16.
3. DDR5 DRAM supports an optional fixed BL32 mode and optional BL32 OTF (On the fly) mode for x4 devices only.

Figure 42 — Read Timings for BL16 in BL32 OTF Mode

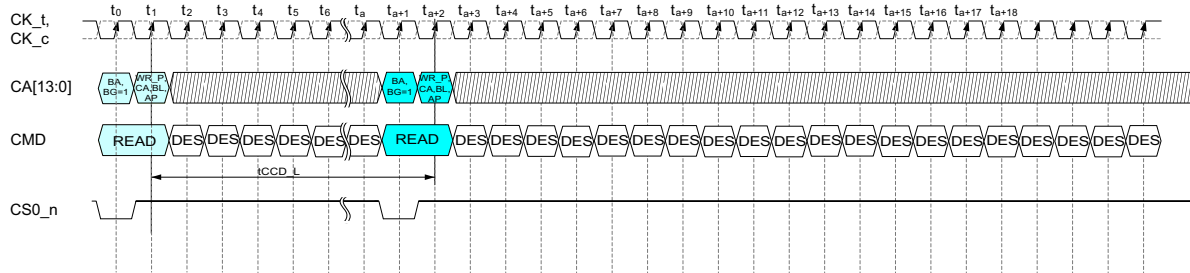
4.7.3 Read Burst Operation for Optional BL32 Mode (cont'd)



NOTES:

1. Figure shows back to back BL16 writes to different bank groups.
2. DDR5 DRAM supports an optional fixed BL32 mode and optional BL32 OTF (On the fly) mode for x4 devices only.

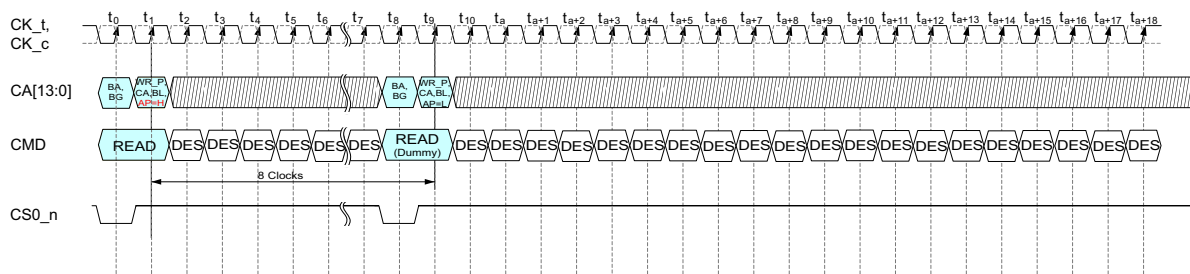
Figure 43 — Read to Read to Different Bank Group for BL16 in BL32 OTF



NOTES::

1. Figure shows back to back BL16 reads to same bank group using a timing of tCCD_L.
2. DDR5 DRAM supports an optional fixed BL32 mode and optional BL32 OTF (On the fly) mode for x4 devices only.

Figure 44 — Read to Read to Same Bank Group for BL16 in BL32 OTF

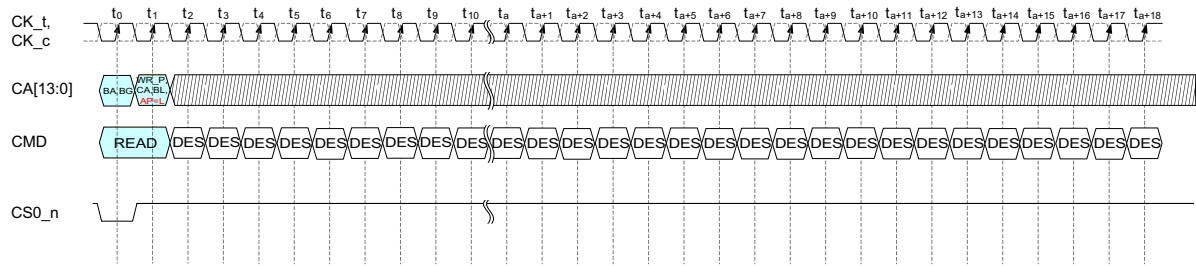


NOTES:

1. AP bit must be set HIGH for first CAS and LOW for dummy CAS command.
2. DDR5 DRAM supports an optional fixed BL32 mode and optional BL32 OTF (On the fly) mode for x4 devices only.
3. CA bits other than C10 and AP in dummy CAS command are the same as the first CAS command.

Figure 45 — Read with Auto-Precharge for Fixed BL32 and BL32 in BL32 OTF Mode

4.7.3 Read Burst Operation for Optional BL32 Mode (cont'd)



- NOTES:
1. AP bit must be set to LOW with the CAS command.
 2. DDR5 DRAM supports an optional fixed BL32 mode and optional BL32 OTF (On the fly) mode for x4 devices only.

Figure 46 — Read with Auto-Precharge for BL16 in BL32 OTF Mode

4.7.4 Read and Write Command Interval

Table 44 — Minimum Read and Write Command Timings

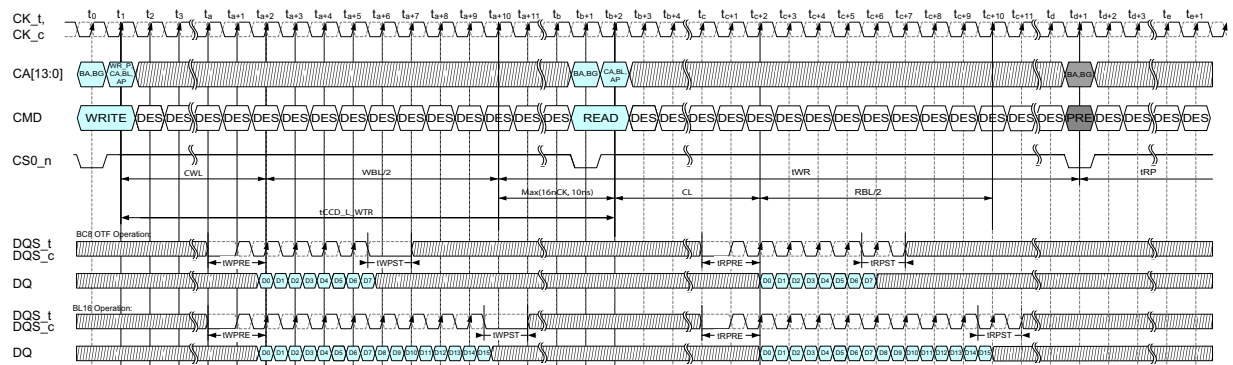
Bank Group	Timing Parameter	DDR5-3200 ~ 8800	Units	Notes
same	Minimum Read to Write	$t_{CCD_L_RTW}$		1, 3, 4
	Minimum Write to Read	$t_{CCD_L_WTR}$		2, 4
	Minimum Write to Read AP, same bank	t_{CCD_WTRA}		2, 4
different	Minimum Read to Write	$t_{CCD_S_RTW}$		1, 3, 4
	Minimum Write to Read	$t_{CCD_S_WTR}$		2, 4

NOTE 1 RBL: Read burst length associated with Read command
RBL = 32 (36 w/ RCRC on) for fixed BL32 and BL32 in BL32 OTF mode
RBL = 16 (18 w/ RCRC on) for fixed BL16 and BL16 in BL32 OTF mode
RBL = 16 (18 w/ RCRC on) for BL16 in BC8 OTF mode and BC8 in BC8 OTF mode

NOTE 2 WBL: Write burst length associated with Write command
WBL = 32 (36 w/ WCRC on) for fixed BL32 and BL32 in BL32 OTF mode
WBL = 16 (18 w/ WCRC on) for fixed BL16 and BL16 in BL32 OTF mode
WBL = 16 (18 w/ WCRC on) for BL16 in BC8 OTF mode and BC8 in BC8 OTF mode

NOTE 3 The following is considered for tRTW equation
1tCK needs to be added due to tDQS2CK
Read DQS offset timing can pull in the tRTW timing
1tCK needs to be added when 1.5tCK postamble

NOTE 4 CWL=CL-2

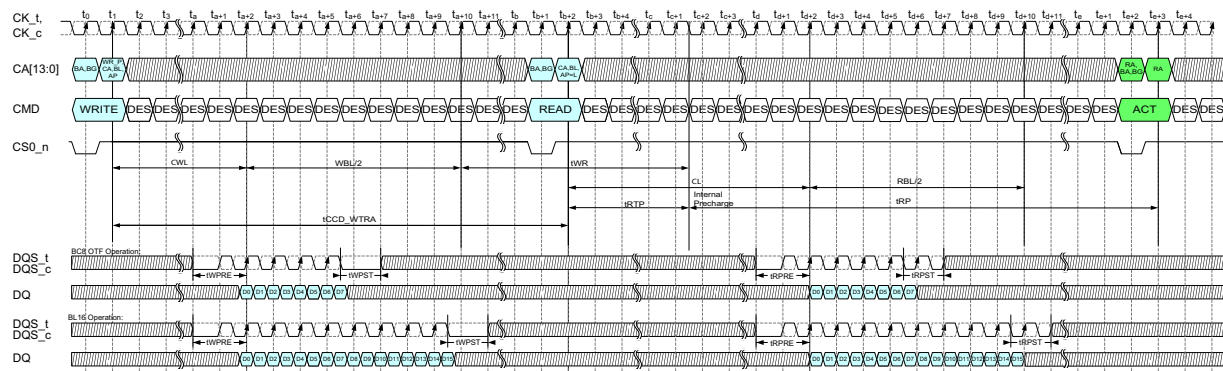


NOTES:

- BC OTF=8 or BL=16, Preamble = 2tCK - 0010 pattern, Postamble = 1.5tCK
- DES commands are shown for ease of illustration; other commands may be valid at these times.
- The write recovery time (tWR) is referenced from the first rising clock edge after the last write data shown at Ta+10.
tWR specifies the last burst write cycle until the precharge command can be issued to the same bank.
- The DQ signal is shown as "Don't Care" before the first Write data bit indicating DFE is disabled. When DFE is enabled, the DQ signal shall be high for a minimum of 4UI prior to the first Write data bit for proper DFE synchronization.

Figure 47 — Timing Diagram for Write to Read

4.7.4 Read and Write Command Interval (cont'd)



NOTES:

1. BC OTF=8 or BL=16, Preamble = 2tCK - 0010 pattern, Postamble = 1.5tCK
2. DES commands are shown for ease of illustration; other commands may be valid at these times.
3. The write recovery time (tWR) is referenced from the first rising clock edge after the last write data shown at Ta+10.
The internal precharge after the Read AutoPrecharge command cannot begin before tWR is satisfied, which is equivalent to tWTRA + tRTP
4. The DQ signal is shown as "Don't Care" before the first Write data bit indicating DFE is disabled. When DFE is enabled, the DQ signal shall be high for a minimum of 4UI prior to the first Write data bit for proper DFE synchronization.

Figure 48 — Timing Diagram for Write to Read AutoPrecharge in Same Bank

4.7.5 Read and Write Command Interval for Optional BL32 Modes

Table 45 — Minimum Read to Read Timings - Same Bank Group

From	To		Units	Notes
	BL16 in BL32 OTF Mode	BL32 in BL32 OTF Mode		
BL16 in BL32 OTF Mode	tCCD_L	tCCD_L		1
BL32 in BL32 OTF Mode	Max(16nCK, 5ns)	Max(16nCK, 5ns)		1
NOTE 1 DDR5 DRAM supports an optional fixed BL32 mode and optional BL32 OTF (On the fly) mode for x4 devices only.				

Table 46 — Minimum Read to Read Timings - Different Bank Group

From	To		Units	Notes
	BL16 in BL32 OTF Mode	BL32 in BL32 OTF Mode		
BL16 in BL32 OTF Mode	tCCD_S	tCCD_S		1
BL32 in BL32 OTF Mode	2*tCCD_S	2*tCCD_S		1
NOTE 1 DDR5 DRAM supports an optional fixed BL32 mode and optional BL32 OTF (On the fly) mode for x4 devices only.				

Table 47 — Minimum Write to Write Timings - Same Bank Group

From	To		Units	Notes
	BL16 in BL32 OTF Mode	BL32 in BL32 OTF Mode		
BL16 in BL32 OTF Mode	tCCD_L_WR	tCCD_L_WR2		1
BL32 in BL32 OTF Mode	8nCK+tCCD_L_WR	9nCK+tCCD_L_WR2		1
NOTE 1 DDR5 DRAM supports an optional fixed BL32 mode and optional BL32 OTF (On the fly) mode for x4 devices only.				

DDR5 x8 and x16 devices will have different write to write same bank group timings, based on whether the second write requires a read-modify-write (RMW), the Burst Length mode of the device set by MR0: OP[1:0], and whether data masking is enabled by MR5:OP[5]. BL16 Partial Writes and BC8 Writes require a RMW on x8/x16 devices. BL16 non-partial writes do not require RMW. See Timing parameters per speed grade for details on parametric timings.

Table 48 — Minimum Write to Write Same Bank Group Timings, x8/x16 Devices

From	To			Units	Notes
	BC8	BL16 Partial Write	BL16 not Partial Write		
BC8 or BL16	tCCD_L_WR	tCCD_L_WR	tCCD_L_WR2		

Table 49 — Minimum Write to Write Timings - Different Bank Group

From	To		Units	Notes
	BL16 in BL32 OTF Mode	BL32 in BL32 OTF Mode		
BL16 in BL32 OTF Mode	tCCD_S_WR	tCCD_S_WR		1
BL32 in BL32 OTF Mode	2*tCCD_S_WR	2*tCCD_S_WR		1
NOTE 1 DDR5 DRAM supports an optional fixed BL32 mode and optional BL32 OTF (On the fly) mode for x4 devices only.				

4.7.6 Read and Write Command Interval for 3DS

Table 50 — Minimum Read and Write Command Timings for x4 3DS

Logical Rank	Bank Group	Parameter Name	Timing Parameter	3DS DDR5 3200 ~ 8800	Units	Note
same	same	tCCD_L_slr	Minimum Read to Read	See Chapter 13, Timing Parameters by Speed Grade for 3DS	nCK	
		tCCD_L_WR_slr	Minimum Write to Write		nCK	
		tCCD_L_RTW_slr	Minimum Read to Write		nCK	
		tCCD_L_WTR_slr	Minimum Write to Read		nCK	
	different	tCCD_S_slr	Minimum Read to Read		nCK	
		tCCD_S_slr	Minimum Write to Write		nCK	
		tCCD_S_RTW_slr	Minimum Read to Write		nCK	
		tCCD_S_WTR_slr	Minimum Write to Read		nCK	
different	same or different	tCCD_S_dlr	Minimum Read to Read		nCK	
		tCCD_S_dlr	Minimum Write to Write		nCK	
		tCCD_S_RTW_dlr	Minimum Read to Write		nCK	
		tCCD_S_WTR_dlr	Minimum Write to Read		nCK	

4.8 Write Operation

The Write Operation stores data to the DRAM. It is initiated by the Write command during which the beginning column address and bank/group address for the data to be written to the array is provided. The data is provided to the DRAM on the DQ inputs CAS Write Latency (CWL) cycles after the Write command along with the proper waveform on the DQS inputs. CAS Write Latency is defined and measured from final cycle of the Write command to the first effective rising DQS (excluding write preamble).

4.8.1 Write Data Mask

One write data mask (DM_n) pin for each byte data group is supported on x8 and x16 DDR5 SDRAMs. The DM_n pin/function is enabled via mode register. For the x4 configuration SDRAM, the DM mode register setting must be disabled. The DM_n pin has identical timings and termination functionality on write operations as the DQ pins, as shown in Figure 43. The DM_n pin is not used for read cycles and the pin should behave like a DQ pin driving high or be terminated to RTT_{PARK}. When the DM function is disabled by MR, the DRAM disables the DM input and output receiver and does not expect nor drive any valid logic level.

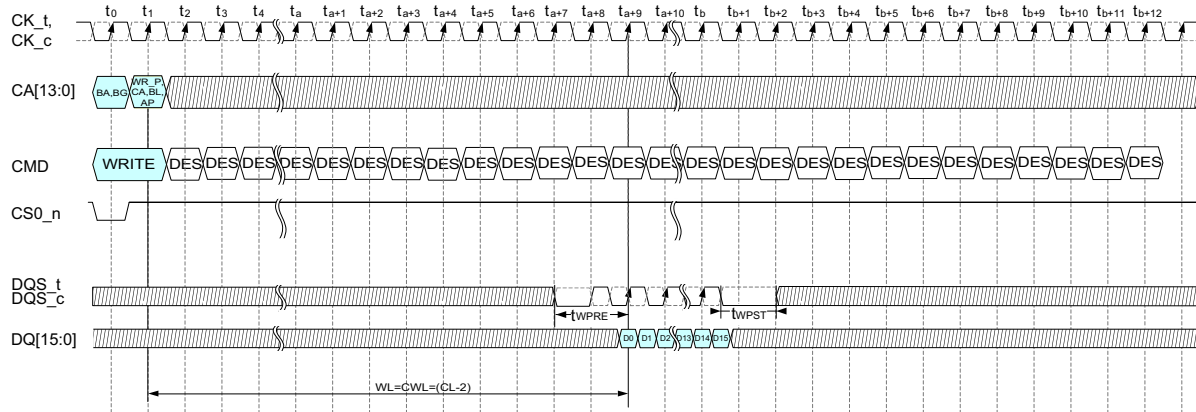
Each data mask burst bit position corresponds to the same bit position in the DQ data burst across the corresponding byte group.

The WR_{Partial} (WR_P) = Low as part of the write command must be used in conjunction with the DM_n data. The WR_{Partial} (WR_P) = Low is to help DRAM start an internal read for 'read modify write' during masked writes. If WR_{Partial} (WR_P) = High during write, then the mask data on DM_n must be high. If DM is disabled, MR5 OP[5] = 0, WR_{Partial} (WR_P) must be "H". DM_n may be high or low.

4.8.2 Write Burst Operation

The following write timing diagrams are to help understand the meaning of each write parameter; the diagrams are just examples. The details of each parameter are defined separately.

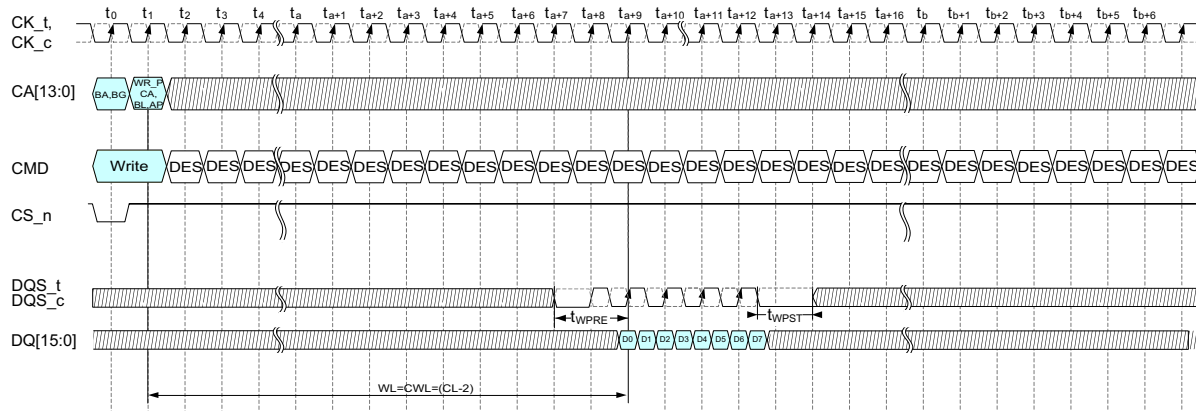
In these write timing diagrams, for clarity of illustration, CK and DQS are shown aligned. As well, DQS and DQ are shown center-aligned. Offset between CK and DQS, and between DQS and DQ may be appropriate.



NOTES:

1. BL=16, 2tCK Preamble, 1.5tCK Postamble
2. DES commands are shown for ease of illustration; other commands may be valid at these times.
3. The DQ signal is shown as "Don't Care" before the first Write data bit indicating DFE is disabled. When DFE is enabled, the DQ signal shall be high for a minimum of 4UI prior to the first Write data bit for proper DFE synchronization.

Figure 49 — WRITE Burst Operation (BL16)

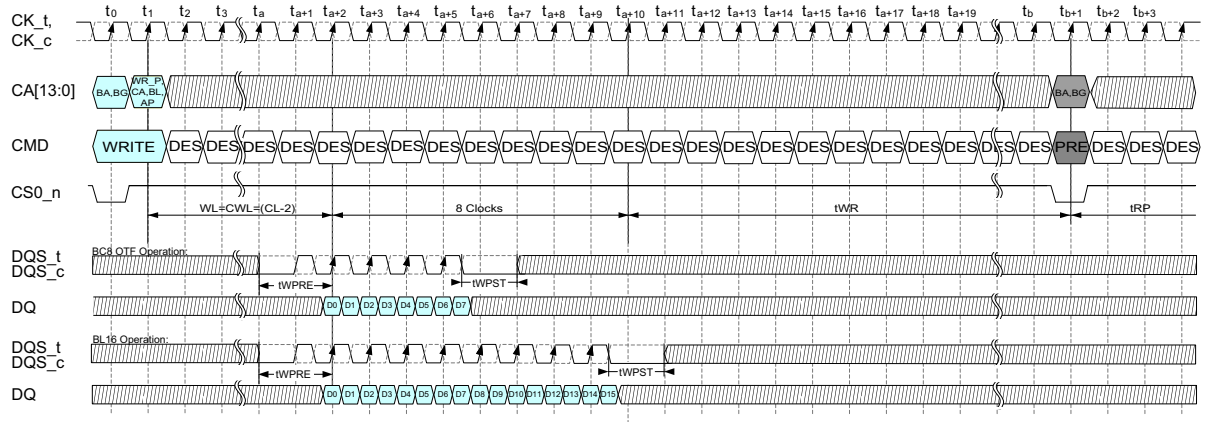


NOTES:

1. BC=8, 2tCK Preamble, 1.5tCK Postamble
2. DES commands are shown for ease of illustration; other commands may be valid at these times.
3. In non-CRC mode, DQS_t and DQS_c stop toggling at the completion of the BC8 data bursts, plus the postamble.
4. The DQ signal is shown as "Don't Care" before the first Write data bit indicating DFE is disabled. When DFE is enabled, the DQ signal shall be high for a minimum of 4UI prior to the first Write data bit for proper DFE synchronization.

Figure 50 — Write Burst Operation (BC8)

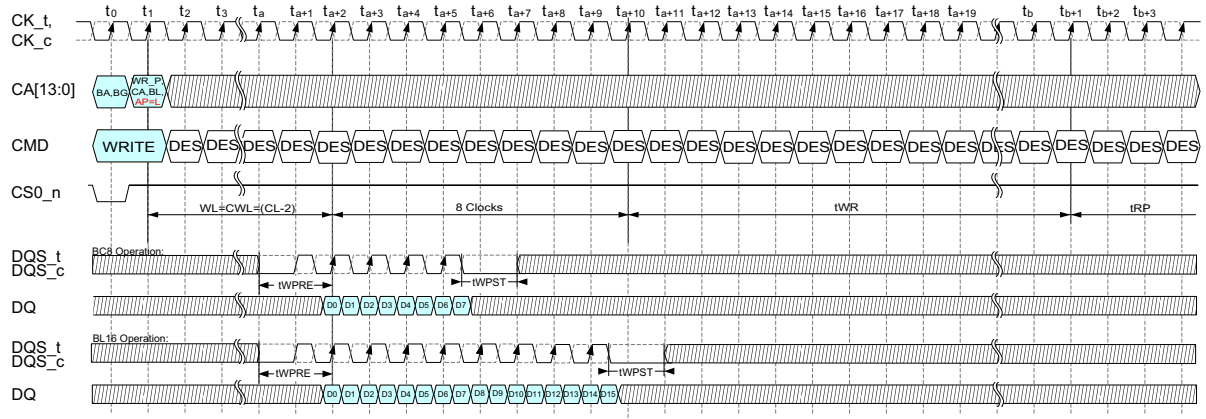
4.8.2 Write Burst Operation (cont'd)



NOTES:

1. BC=8 or BL=16, Preamble = 2tCK - 0010 pattern, Postamble = 1.5tCK
2. DES commands are shown for ease of illustration; other commands may be valid at these times.
3. The write recovery time (tWR) is referenced from the first rising clock edge after the last write data shown at t_a+10 . tWR specifies the last burst write cycle until the precharge command can be issued to the same bank.
4. The DQ signal is shown as "Don't Care" before the first Write data bit indicating DFE is disabled. When DFE is enabled, the DQ signal shall be high for a minimum of 4UI prior to the first Write data bit for proper DFE synchronization.

Figure 51 — WRITE (BL16) to PRECHARGE Operation with 2tCK Preamble



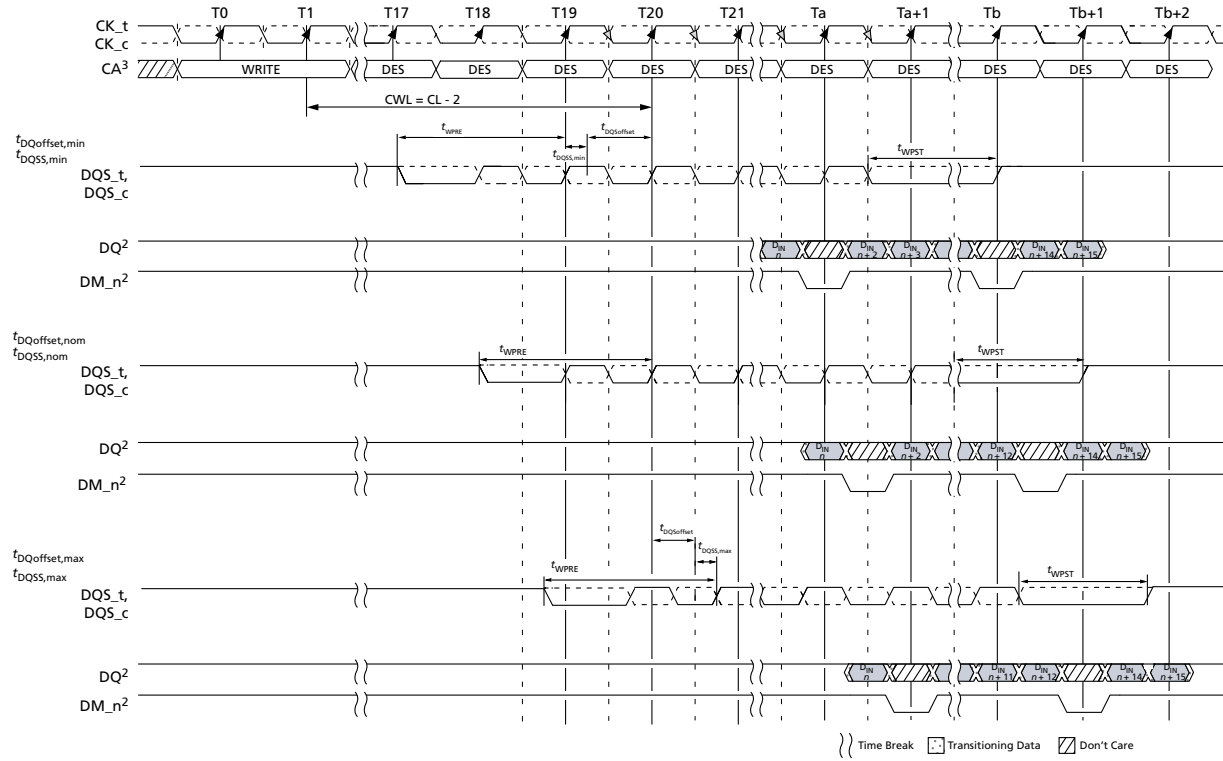
NOTES:

1. BC OTF=8 or BL=16, Preamble = 2tCK - 0010 pattern, Postamble = 1.5tCK
2. DES commands are shown for ease of illustration; other commands may be valid at these times.
3. The write recovery time (WR) is referenced from the first rising clock edge after the last write data shown at t_a+10 . WR specifies the last burst write cycle until the precharge command can be issued to the same bank.
4. The DQ signal is shown as "Don't Care" before the first Write data bit indicating DFE is disabled. When DFE is enabled, the DQ signal shall be high for a minimum of 4UI prior to the first Write data bit for proper DFE synchronization.

Figure 52 — WRITE (BL16) with Auto PRECHARGE Operation and 2tCK Preamble

4.8.3 Write Timing Parameters

The following figure is for example only to enumerate the strobe edges for a particular write burst. For a valid burst, all timing parameters for each edge of a burst must be satisfied.



NOTES:

1. BL=16, Preamble=2CK - 0010 pattern, Postamble=1.5CK,
2. DES commands are shown for ease of illustration, other commands may be valid at these times.
3. t_{DQSS} must be met at each rising clock edge.
4. Figure assumes DRAM internal WL training complete.
5. DQ/DM_n pulse timing and DQS to DQ skew defined by Rx Strobe Jitter Sensitivity Specifications for the respective speed bin.

Figure 53 — DDR5 Write Timing Parameters

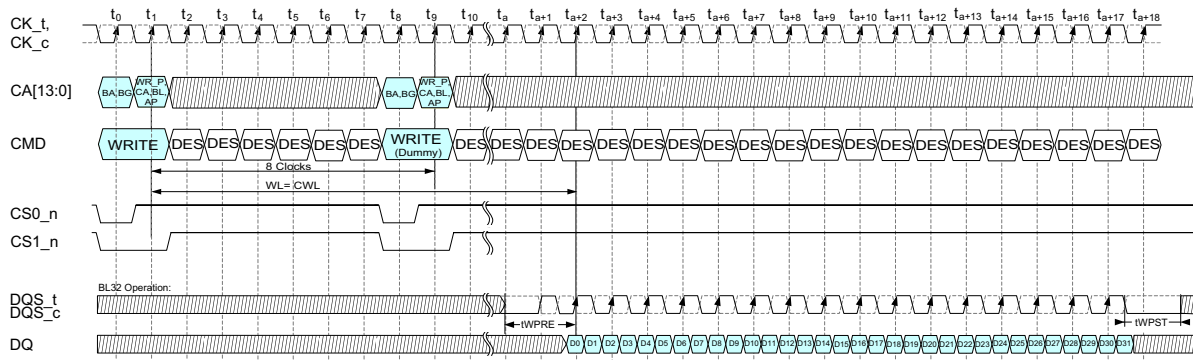
4.8.4 Write Burst Operation for Optional BL32 Mode

The following write timing diagrams cover write timings for fixed BL32, BL32 in BL32 OTF mode and BL16 in BL32 OTF mode for x4 devices only.

In these write timing diagrams, for clarity of illustration, CK and DQS are shown aligned. As well, DQS and DQ are shown center-aligned. Offset between CK and DQS, and between DQS and DQ may be appropriate.

A dummy CAS command is required for second half of the transfer of BL32. If non-target ODT is needed in the system then a dummy ODT command must be issued to the non-target rank for second half of the transfer of BL32.

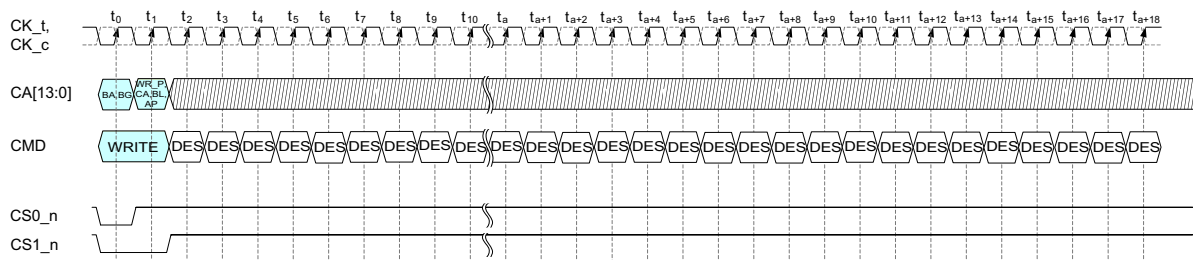
When the DFE is enabled, the DQs shall be high for a minimum of 4UI prior to the first Write data bit to ensure proper DFE synchronization.



NOTES:

1. BL=32, 2tCK Preamble, 1.5tCK Postamble
2. DES commands are shown for ease of illustration; other commands may be valid at these times.
3. A dummy WR command is required for the second half of the transfer with a delay of 8 clocks from the first WR command.
4. The figure also shows a dummy ODT command being issued to non-target rank 1 for the second half of the transfer.
5. C10 is used for burst ordering and shall be LOW for the first WR command, and be HIGH for dummy WR command.
6. DDR5 DRAM supports an optional fixed BL32 mode and optional BL32 OTF (On the fly) mode for x4 devices only.
7. CA bits other than C10 and AP in dummy CAS command are the same as the first CAS command.
8. The DQ signal is shown as "Don't Care" before the first Write data bit indicating DFE is disabled. When DFE is enabled, the DQ signal shall be high for a minimum of 4UI prior to the first Write data bit for proper DFE synchronization.

Figure 54 — Write Timing for Fixed BL32 and BL32 in BL32 OTF Mode

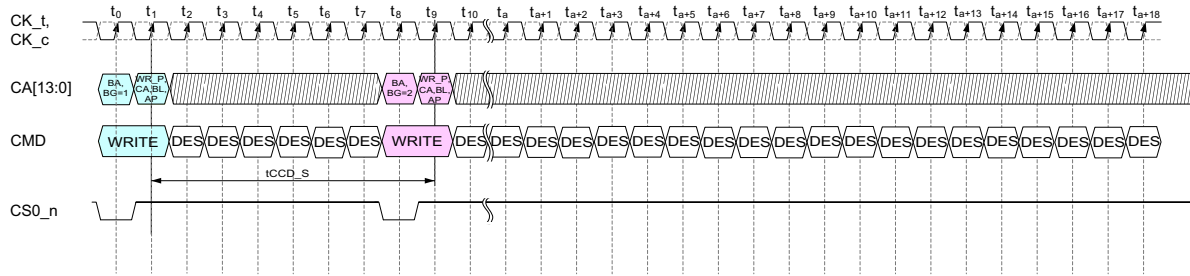


NOTES:

1. Figure shows BL16 write operation when MR0 is programmed to use BL32 OTF mode. In this case, no dummy WR command is required as transfer size is BL16.
2. DES commands are shown for ease of illustration; other commands may be valid at these times including commands to allow data transfer from the same die after transfer of BL16.
3. DDR5 DRAM supports an optional fixed BL32 mode and optional BL32 OTF (On the fly) mode for x4 devices only.

Figure 55 — Write Timings for BL16 in BL32 OTF mode

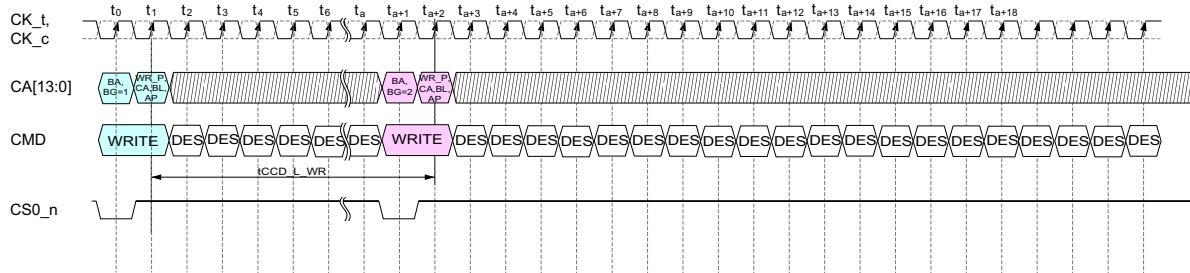
4.8.4 Write Burst Operation for Optional BL32 Mode (cont'd)



NOTES:

1. Figure shows back to back BL16 writes to different bank groups.
2. DDR5 DRAM supports an optional fixed BL32 mode and optional BL32 OTF (On the fly) mode for x4 devices only.

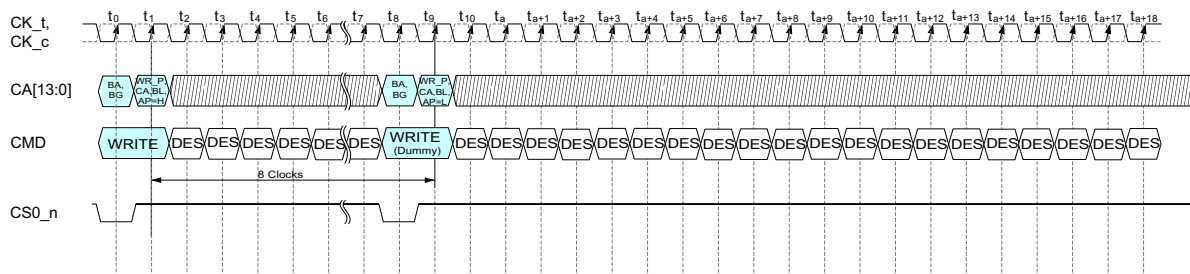
Figure 56 — Write to Write to Different Bank Group for BL16 in BL32 OTF



NOTES:

1. Figure shows back to back BL16 writes to same bank group using a timing of tCCD_L_WR.
2. Back to Back BL32 writes to same bank group shall have a minimum separation of 16 clocks.
3. DDR5 DRAM supports an optional fixed BL32 mode and optional BL32 OTF (On the fly) mode for x4 devices only.

Figure 57 — Write to Write to Same Bank Group for BL16 in BL32 OTF

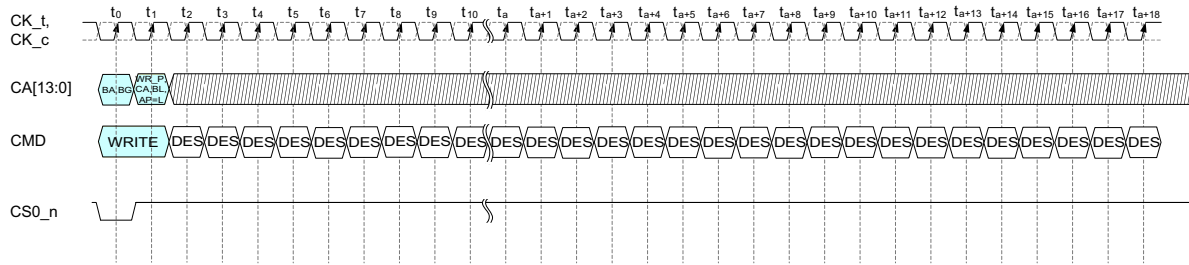


NOTES:

1. AP bit must be set HIGH for first CAS and LOW for dummy CAS command.
2. DDR5 DRAM supports an optional fixed BL32 mode and optional BL32 OTF (On the fly) mode for x4 devices only.
3. CA bits other than C10 and AP in dummy CAS command are the same as the first CAS command.

Figure 58 — Write with Auto-Precharge for Fixed BL32 and BL32 in BL32 OTF Mode

4.8.4 Write Burst Operation for Optional BL32 Mode (cont'd)



NOTES:

1. AP bit must be set to LOW with the CAS command.
2. DDR5 DRAM supports an optional fixed BL32 mode and optional BL32 OTF (On the fly) mode for x4 devices only.

Figure 59 — Write with Auto-Precharge for BL16 in BL32 OTF Mode

4.8.5 Same Bank Group Write to Write Timings

DDR5 devices will have separate same bank group write timings, based on whether the second write requires an RMW (Read-Modify-Write) access or JW (Just-Write) access. In JW access, DDR5 updates all 128 bits of data on the addressed codeword, while in RMW access, a part of 128 bits is updated.

Table 51 — JW (Just-Write) Access and RMW (Read-Modify-Write) Access Definition

Configuration	BL16		BC8		Optional BL32	Notes
	Normal	Data Mask	Normal	Data Mask		
x4	RMW	—	RMW	—	JW	1, 2, 3
x8 / x16	JW	RMW			—	1, 2, 3

NOTE 1 BC8 refers to BC8 OTF mode enabled by MR0 OP[1:0]=01B, where Write command is issued with BL=L in CA5.
 NOTE 2 Optional BL32 refers to BL32 OTF mode (enabled by MR0 OP[1:0]=11B), where Write command is issued with BL=L in CA5, or BL32 fixed mode (enabled by MR0 OP[1:0]=10B).
 NOTE 3 Data Mask refers to Data Mask mode enabled by MR5 OP[5]=1B, where Write command is issued with WP=L in CA11.

Table 52 — Same Bank-Group Write Access to RMW Access Timings

From	To		Notes
	BL16	BC8	
BL 16	tCCD_L_WR	tCCD_L_WR	1, 2, 3
BC8	tCCD_L_WR	tCCD_L_WR	1, 2, 3
Optional BL32	8nCK+tCCD_L_WR	—	1, 2, 3, 4, 5

NOTE 1 BC8 refers to BC8 OTF mode enabled by MR0 OP[1:0]=01B, where Write command is issued with BL=L in CA5.
 NOTE 2 Optional BL32 refers to BL32 OTF mode (enabled by MR0 OP[1:0]=11B), where Write command is issued with BL=L in CA5, or BL32 fixed mode (enabled by MR0 OP[1:0]=10B).
 NOTE 3 In Optional BL32 case, this timing table affects to the 1st Write command only, not the dummy Write command.
 NOTE 4 There is no BL32 to BC8 case.
 NOTE 5 When Write CRC is enabled, the timing increases by 1nCK, i.e., 9nCK+tCCD_L_WR

4.8.5 Same Bank Group Write to Write Timings (cont'd)

Table 53 — Same Bank-Group Write Access to JW Access Timings

From	To		Notes
	BL16	Optional BL32	
BL16	tCCD_L_WR2	tCCD_L_WR2	1, 2, 3
BC8	tCCD_L_WR2	—	1, 2, 3, 4
Optional BL32	—	8nCK+tCCD_L_WR2	1, 2, 3, 5
NOTE 1 BC8 refers to BC8 OTF mode enabled by MR0 OP[1:0]=01B, where Write command is issued with BL=L in CA5.			
NOTE 2 Optional BL32 refers to BL32 OTF mode (enabled by MR0 OP[1:0]=11B), where Write command is issued with BL=L in CA5, or BL32 fixed mode (enabled by MR0 OP[1:0]=10B).			
NOTE 3 In Optional BL32 case, this timing table affects to the 1st Write command only, not the dummy Write command.			
NOTE 4 There is no BC8 to BL32 case.			
NOTE 5 When Write CRC is enabled, the timing increases by 1nCK, i.e., 9nCK+tCCD_L_WR			

4.8.6 Different Bank-Group Write to Write Timings

Table 54 — Different Bank-Group Write to Write Timings

From	To			Notes
	BL16	BC8	Optional BL32	
BL16	8nCK	8nCK	8nCK	1, 2, 3
BC8	8nCK	8nCK	—	1, 2, 3, 4
Optional BL32	16nCK	—	16nCK	1, 2, 3, 4
NOTE 1 BC8 refers to BC8 OTF mode enabled by MR0 OP[1:0]=01B, where Write command is issued with BL=L in CA5.				
NOTE 2 Optional BL32 refers to BL32 OTF mode (enabled by MR0 OP[1:0]=11B), where Write command is issued with BL=L in CA5, or BL32 fixed mode (enabled by MR0 OP[1:] =11B).				
NOTE 3 In Optional BL32 case, this timing table affects to the 1st Write command only, not the dummy Write command.				
NOTE 4 There is no BC8 to BL32 case.				

4.8.7 Write Timing Violations

4.8.7.1 Motivation

Generally, if Write timing parameters are violated, a complete reset/initialization procedure has to be initiated to make sure that the DRAM works properly. However, it is desirable, for certain violations as specified below, the DRAM is guaranteed to not “hang up,” and that errors are limited to that particular operation.

For the following, it will be assumed that there are no timing violations with regards to the Write command itself (including ODT, etc.) and that it does satisfy all timing requirements not mentioned below.

4.8.7.2 Data to Strobe Eye Height or Width Violations

Should the required data to strobe timing or voltage parameters be violated (Such as: $t_{Rx_RDQ_tMargin}$, $t_{Rx_DQS2DQ_Skew}$, VRx_DQS , VRx_DQ , etc.), for any of the data/strobe timing edges or data/strobe voltage limits associated with a write burst data eye, then incorrect data might be written to the memory locations addressed with this WRITE command.

Subsequent reads from that location might results in unpredictable read data, however the DRAM will work properly otherwise.

4.8.7.3 Strobe and Strobe to Clock Timing Violations

Should the strobe timing requirements (t_{WPRE} , t_{WPST}) or the strobe to clock timing requirements (t_{DQSS} , $t_{DQSoffset}$) be violated for any of the strobe edges associated with a Write burst, then wrong data might be written to the memory location addressed with the offending WRITE command. Subsequent reads from that location might result in unpredictable read data, however the DRAM will work properly otherwise with the following constraints:

- (1) Both Write CRC and data burst OTF are disabled; timing specifications other than t_{WPRE} , t_{WPST} , t_{DQSS} , $t_{DQSoffset}$ are not violated.
- (2) The offending write strobe (and preamble) arrive no earlier or later than six DQS transition edges from the Write-Latency position.
- (3) A Read command following an offending Write command from any open bank is allowed.
- (4) One or more subsequent WR or a subsequent WRA {to same bank as offending WR} may be issued t_{CCD_L} later but incorrect data could be written; subsequent WR and WRA can be either offending or non-offending Writes. Reads from these Writes may provide incorrect data.
- (5) One or more subsequent WR or a subsequent WRA {to a different bank group} may be issued t_{CCD_S} later but incorrect data could be written; subsequent WR and WRA can be either offending or non-offending Writes. Reads from these Writes may provide incorrect data.
- (6) Once one or more precharge commands (PRE_{pb} , PRE_{sb} , or PRE_{ab}) are issued to DDR5 after offending WRITE command and all banks become precharged state (idle state), a subsequent, non-offending WR or WRA to any open bank shall be able to write correct data.
- (7) DQS strobes including preamble must align to each Write commands data burst length configuration. If the DRAM fails to capture or incorrectly de-serializes the incoming data stream because of misalignment or missing strobe edges, errors may occur. These errors will propagate indefinitely until the DRAM is put into an idle state, i.e., all banks are in the precharged state with TRP satisfied.

4.8.8 Write Enable Timings

4.8.8.1 Introduction

The following specifies the relationship between the write enable timing window $tWPRE_EN_ntck$ and the DRAM related DQS to CK drift window $tDQSD$ as well as the system related DQS to CK drift window $tDQSS$ around the final DQS to CK offset trained pass/fail point $tDQSoffset$ based on write leveling feedback in order to support n-tck pre-amble mode. Functional operation requires that the following condition is met:

- $tWPRE_EN_ntck \geq |tDQSSmin| + tDQSSmax + |tDQSDmin| + tDQSDmax$

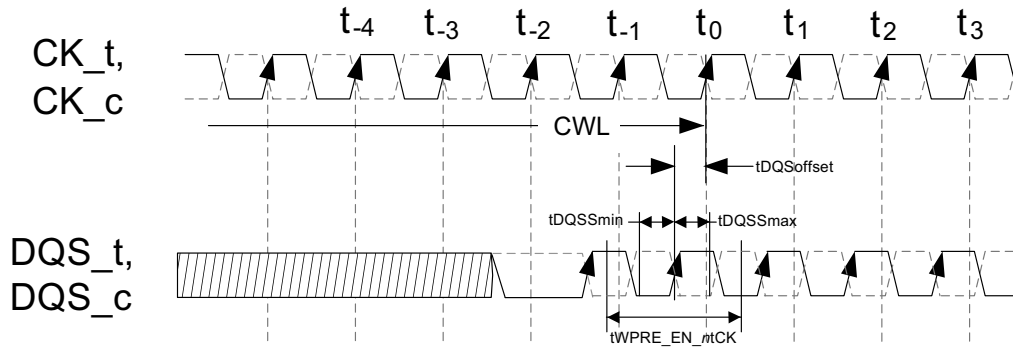


Figure 60 — $tDQSS$: DRAM External CLK-to-DQS Variation

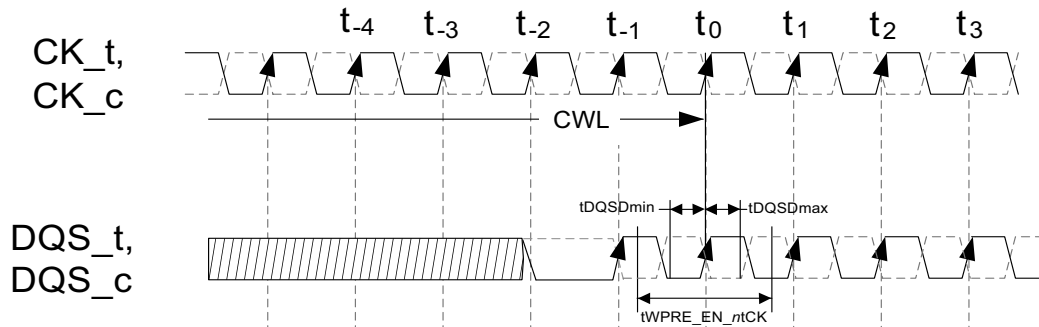


Figure 61 — $tDQSD$: DRAM Internal CLK-to-DQS Variation

4.8.8.1 Introduction (cont'd)

Table 55 — Write Enable Timing Parameters DDR5 3200 to 8400

Parameter	Symbol	Speed Bins DDR5 3200-8800		Unit	Notes
		Min	Max		
2-tck Write pre-amble enable window	tWPRE_EN_2tck	1.5	-	tCK	
3-tck Write pre-amble enable window	tWPRE_EN_3tck	2.5	-	tCK	
4-tck Write pre-amble enable window	tWPRE_EN_4tck	2.5	-	tCK	
Final trained value of host DQS_t-DQS_c timing relative to CWL CK_t-CK_c edge	tDQSoffset	-0.5	0.5	tCK	2
DRAM voltage/temperature drift window of first rising DQS_t pre-amble edge relative to CWL CK_t-CK_c edge	tDQSD	-0.25* tWPRE_EN_ntCK (min)	0.25* tWPRE_EN_ntCK (min)	tCK	1
Host and system voltage/temperature drift window of first rising DQS_t pre-amble edge relative to CWL CK_t-CK_c edge	tDQSS	-0.25* tWPRE_EN_ntCK (min)	0.25* tWPRE_EN_ntCK (min)	tCK	1, 3
NOTE 1 Measured relative to the write leveling feedback, after write leveling training has been completed.					
NOTE 2 When measuring the tDQSoffset, tWLS/H are reflected in the tDQSoffset result.					
NOTE 3 DDR5-3200 timings apply for data rates <2933 MT/s. For example, at 2000 MT/s, tDQSS(max) = (2000/2933)*0.25*tWPRE_EN_ntck(min) = 0.17*tWPRE_EN_ntck(min).					

Symbol	Description	Min	Max	Unit
tWLS/H	Write Leveling Setup/Hold Time	-80	+80	ps

4.9 Self Refresh Operation

The Self-Refresh command can be used to retain data in the DDR5 SDRAM, even if the rest of the system is powered down. When in the Self-Refresh mode, the DDR5 SDRAM retains data without external clocking. The DDR5 SDRAM device has a built-in timer to accommodate Self-Refresh operation. While in Self Refresh, the DDR5 SDRAM adjusts and updates its internal average periodic refresh interval, as needed, based on its own temperature sensor. The internal average periodic refresh interval adjustment (increasing, decreasing or staying constant) does not require any external control.

Self Refresh entry is command based (SRE), while the Self-Refresh Exit Command is defined by the transition of CS_n LOW to HIGH with a defined pulse width tCSH_SRExit, followed by three or more NOP commands (tCSL_SRExit) to ensure DRAM stability in recognizing the exit. This is described below in more detail.

Before issuing the Self-Refresh-Entry command, the DDR5 SDRAM must be idle with all bank precharge state with tRP satisfied. 'Idle state' is defined as all banks are closed (tRP, etc. satisfied), no data bursts are in progress, and all timings from previous operations are satisfied (tMRD, tRFC, etc.). A Deselect command must be registered on the last positive clock edge before issuing Self Refresh Entry command. Once the Self Refresh Entry command is registered, Deselect commands must also be registered at the next positive clock edges until tCPDED is satisfied. After tCPDED has been satisfied, CS_n must transition low. After CS_n transitions low at the end of tCPDED, the CS_n shall stay low until exit. The DDR5 SDRAM may switch to a CMOS based receiver to save more power and that transition should coincide with CS_n going low.

When the CS_n is held low, the DRAM automatically disables ODT termination and sets Hi-Z as termination state regardless RTT configuration for the duration of Self-Refresh mode. Upon exiting Self-Refresh, DRAM automatically enables ODT termination and set RTT_PARK (for DQs) asynchronously during tXSDLL when RTT_PARK is enabled. CA/CS/CK ODT shall revert to its strapped or its MR ODT Setting state if previously applied. During normal operation (DLL on) the DLL is automatically disabled upon entering Self-Refresh and is automatically enabled (including a DLL-Reset) upon exiting Self-Refresh.

When the DDR5 SDRAM has entered Self-Refresh mode, all of the external control signals, except CS_n and RESET_n, are "don't care." For proper Self-Refresh operation, all power supply and reference pins (VDD, VDDQ, VSS, VSS, and VPP) must be at valid levels. DRAM internal VrefDQ and/or VrefCA generator circuitry may remain ON or turned OFF depending on DRAM design. If DRAM internal VrefDQ and/or VrefCA circuitry is turned OFF in self refresh, when DRAM exits from self refresh state, it ensures that VrefDQ and/or VrefCA and generator circuitry is powered up and stable within tXS period. First Write operation or first Write Leveling Activity may not occur earlier than tXS after exit from Self Refresh. The DRAM initiates a minimum of one Refresh command internally once it enters Self-Refresh mode.

The clocks must stay on until tCKLCS but can be DON'T CARE after tCKLCS expires but it should be noted that shortly after tCPDED, the termination for the clocks will be off. The clock is internally disabled (in the DRAM) during Self-Refresh Operation to save power. The minimum time that the DDR5 SDRAM must remain in Self-Refresh mode is tCSL. The user may change the external clock frequency or halt the external clock tCKLCS after Self-Refresh entry is registered, however, the clock must be restarted and stable tCKSRX before the device can exit Self-Refresh operation.

The procedure for exiting Self-Refresh requires a sequence of events. Since the DRAM will switch to a CMOS based driver to save power, the DRAM will trigger Self-Refresh exit upon seeing the CS_n transition from low to high and stay high for tCSH_SRExit. tCASRX prior to CS_n transitioning high, the CA bus must be driven high. Once tCSH_SRExit is satisfied, three NOP commands must be issued, otherwise the DRAM could be put into an unknown state. The clocks must be valid for tCKSRX prior to issuing the NOP commands that completes the Self Refresh exit sequence. Once a Self-Refresh Exit is registered, the following timing delay must be satisfied:

1. Commands that do not require locked DLL:

tXS - ACT, MPC, MRW, PDE, PDX, PRE(ab,sb,pb), REF(ab,sb), RFM(ab,sb), SRE, VREFCA

2. Commands that require locked DLL:

tXS_DLL - RD, MRR, WR, WRP

There are some Host system requirements and restrictions to use the optional SRX/NOP Clock-Sync feature. The Clock-Sync feature is intended to address issue in a system where Host Clock has Duty Cycle Distortion that is different from cycle to cycle in even/odd clock in repeating manner where Clock mis-alignment between Host Clock and DRAM internal 4-phase clock can results in worse performance than initially trained. If Host Clock can guarantee the consistent duty cycle on every even/odd clock as far as meeting CK input spec, the Clock-Sync feature is not needed. If Host CLK has the consistent duty cycle on every even/odd clocks but changes direction (i.e., from 46% to 54%) after SRX, the Clock-Sync feature won't be able to address such issue.

4.9 Self Refresh Operation (cont'd)

This Clock-Sync feature can be used only with Host system tracking clock from the first SRX Clock-Sync prior to DCA training and guarantee to issue subsequent SRX with exact same phase. Host system applications that cannot guarantee the consistent system clock phase between the first SRX Clock-Sync prior to DCA training and subsequent SRX at the first NOP (for example, like a system where system clock is stopped to put host system in low power mode or to do frequency change and put DRAM into self-refresh) cannot use the Clock Sync feature and therefore should not enable this feature.

The SRX/NOP Clock-Sync scheme helps the DRAMs that have 4-phase internal clock to determine whether ICLK or IBCLK will be the first one that is synchronized with Host Clock after Self-Refresh Exit (SRX). It also helps host to identify the right DCA MR setting that was pre-trained during the initialization process to the clock edge. DRAMs are responsible to align the first NOP after SRX (t_{c+1} in Figure 58 and Figure 59) to the internal clock edge that existed during the DCA training. For instance on the DRAM that aligns the first NOP with internal Odd clock, while in SREF, the DRAM detects whether the first NOP arrives via the Even or the Odd command-decode pipeline. Then if the first NOP arrives via the Odd pipeline, the DRAM does do nothing. If the first NOP arrives via the Even Pipeline, the DRAM does initiate the Clock-Sync scheme to sync the first NOP after SRX to the internal Odd clock. For all subsequent SRE/SRX sequences, the DRAM will sync its internal clocks with what existed during the DCA training. In order for DRAM to do this, host is required to send SRX first NOP on the exact same clock phase that was used during the DCA training. In this way, the SRX/NOP Clock-Sync feature enforces repeatability of clock phases to an Odd phase upon SRX. It also helps on minimizing repeatability of the additive/subtractive effect of the DCA training on Tx jitter. If host sends SRX first NOP on different clock phase than the phase used during DCA training while the Clock-Sync feature is enabled, DCA adjustment may be in the wrong direction and could result in worse performance than without DCA adjustment. The SRX/NOP Clock-Sync scheme is an optional feature on DDR5. MR13:OP[5] indicates host whether the SRX/NOP Clock-Sync feature is supported or not.

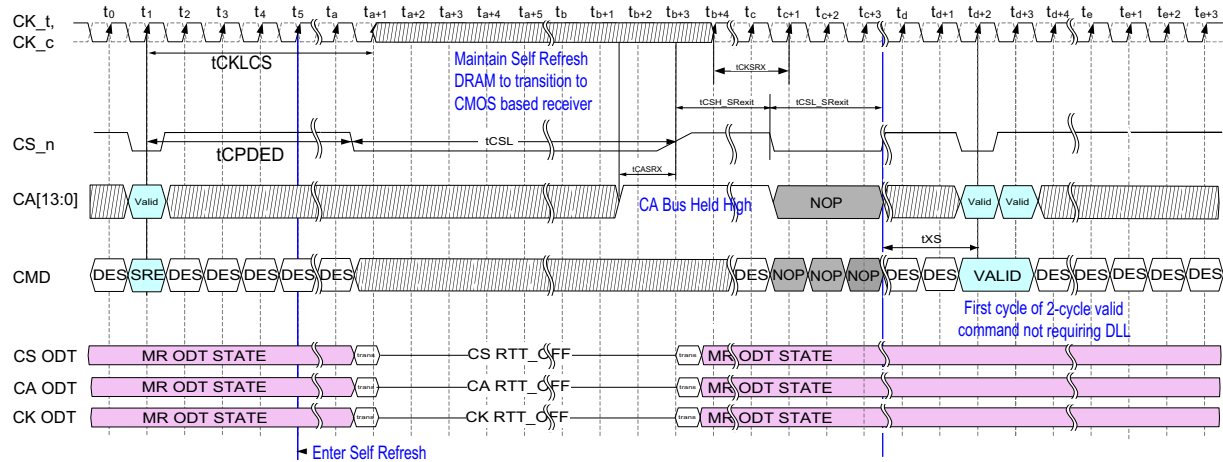
When SRX/NOP Clock-Sync feature is enabled by MRW to MR13:OP[5]=1, to make the Clock-Sync Scheme fully functional, (1) Host (including RCD, Clock Driver) is responsible to ensure the phases of the system clock before and after SRX. (2) Host shall issue an initial SRE/SRX pair after CSTM/CATM prior to DCA training to ensure that the Clock-Sync is in effect during DCA training, (3) the continuous system clock during Self-Refresh is not required to make the Clock-Sync feature fully functional, but every first NOP registered after each t_{CSH_SRExit} shall be the same phase of system clock (for example, Odd) that existed at the first NOP at t_{c+1} cycle in the initial SRX sequence prior to DCA training so that DRAM can get the consistent phases of system clock when DRAM does do the clock sync, and (4) for DCA training at multiple frequencies, the first NOP after SRX for a given clock frequency shall be the same phase of system clock (for example, Odd) that existed at exit Self Refresh w/Frequency Change mode prior to DCA training for that frequency.

Depending on the system environment and the amount of time spent in Self-Refresh, ZQ calibration commands may be required to compensate for the voltage and temperature drift as described in "ZQ Calibration Commands". To issue ZQ calibration commands, applicable timing requirements must be satisfied.

Upon exiting Self Refresh, t_{REFI} begins when CS_n rises with t_{CSH_SRExit} (t_{b+3} in Figure 62). One additional refresh shall be issued in addition to refreshes normally scheduled. This additional refresh is not counted toward the computation of the average refresh interval (t_{REFI}). If this refresh isn't issued by t_{REFI} after CS_n rises with t_{CSH_SRExit} , then it counts toward the maximum number of refreshes which may be postponed. This additional refresh does not replace the regularly periodic refresh that's also scheduled at t_{REFI} after CS_n rises with t_{CSH_SRExit} (both of these refresh commands count toward the maximum number of refreshes which may be postponed). The additional refresh consists of a single REFab command or $n * REFsb$, where n is the number of banks in a bank group. If Self Refresh is to be re-entered and no regularly scheduled periodic refresh commands have been issued, the additional refresh shall be issued prior to Self Refresh re-entry. Plus, FGR mode may require extra Refresh command(s), in addition the aforementioned additional Refresh, depending on the condition of the Self Refresh entry (refer to Section 4.13.7 for more information)

The exit timing from self-refresh exit to first valid command not requiring a locked DLL is t_{XS} . The value of t_{XS} is (t_{RFC}). This delay is to allow for any refreshes started by the DRAM to complete. t_{RFC} continues to grow with higher density devices so t_{XS} will grow as well.

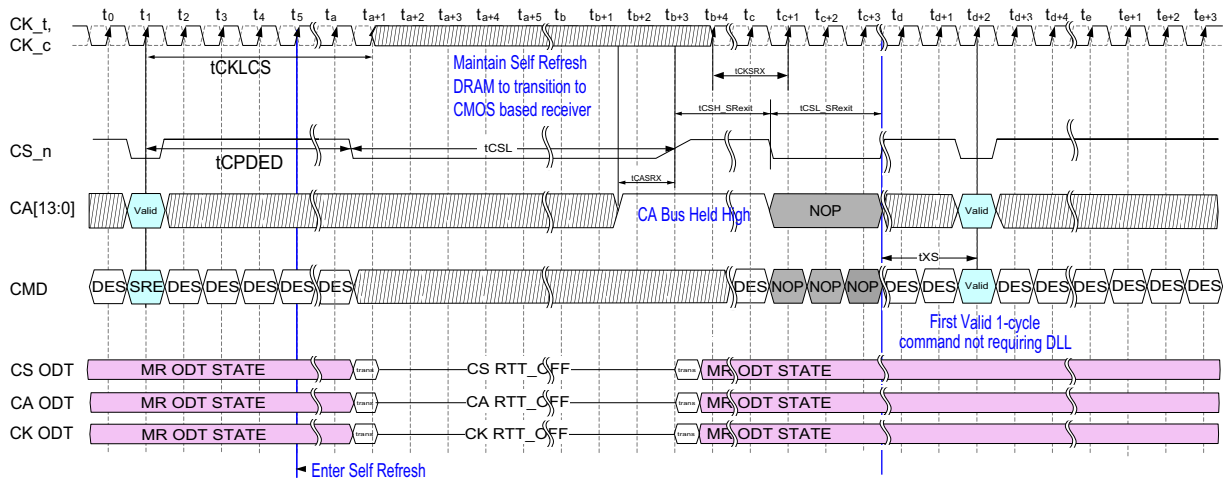
4.9 Self Refresh Operation (cont'd)



NOTES:

1. While in 2N mode, tCSL_SRExit will not be statically held low (as shown above), as it will pulse for each 2 cycle period. Refer to the 2N mode section for more details.
2. Both tCSH_SRExit and tCSL_SRExit timings must be satisfied to guarantee DRAM operation.
3. When tCSH_SRExit,min expires, the CA bus is allowed to transition from all bits High to any valid (V) level. Prior to CS_n being registered Low at tc+1, the CA bus must transition to NOP conforming to the CAI state of the DRAM and complying with applicable DRAM input timing parameters.

Figure 62 — Self-Refresh Entry/Exit Timing with 2-Cycle Exit Command



NOTES:

1. While in 2N mode, tCSL_SRExit will not be statically held low (as shown above), as it will pulse for each 2 cycle period. Refer to the 2N mode section for more details.
2. Both tCSH_SRExit and tCSL_SRExit timings must be satisfied to guarantee DRAM operation.
3. When tCSH_SRExit,min expires, the CA bus is allowed to transition from all bits High to any valid (V) level. Prior to CS_n being registered Low at tc+1, the CA bus must transition to NOP conforming to the CAI state of the DRAM and complying with applicable DRAM input timing parameters.

Figure 63 — Self-Refresh Entry/Exit Timing with 1-Cycle Exit Command

4.9 Self Refresh Operation (cont'd)

Table 56 — Self-Refresh Timing Parameters

Parameter	Symbol	Min	Max	Unit	Note
Command pass disable delay	tCPDED	max(5ns, 8nCK)	-	nCK, ns	
Self-Refresh CS_n low Pulse width	tCSL	10	-	ns	
Self-Refresh exit CS_n High Pulse width	tCSH_SRexit	13	200	ns	
Self-Refresh exit CS_n Low Pulse width	tCSL_SRexit	3nCK	30ns	nCK, ns	1
Valid Clock Requirement before SRX	tCKSRX	max(3.5ns, 8tCK)	-	nCK, ns	
Valid Clock Requirement after SRE	tCKLCS	tCPDED + 1nCK	-	nCK, ns	
Self-Refresh exit CS_n high	tCASRX	0		ns	
Exit Self-Refresh to next valid command NOT requiring a DLL	tXS	tRFC1	-	ns	
3DS exit Self-Refresh to next valid command NOT requiring a DLL	tXS_3DS	tRFC1_slr+10ns	-	ns	2, 3, 4
Exit Self-Refresh to next valid command requiring a DLL	tXS_DLL	tDLLK		ns	

NOTE 1 While in 2N mode, tCSL_SRexit will not be statically held low, as it will pulse for each 2-cycle period for a min of 6nCK. Refer to the 2N mode section for more details.

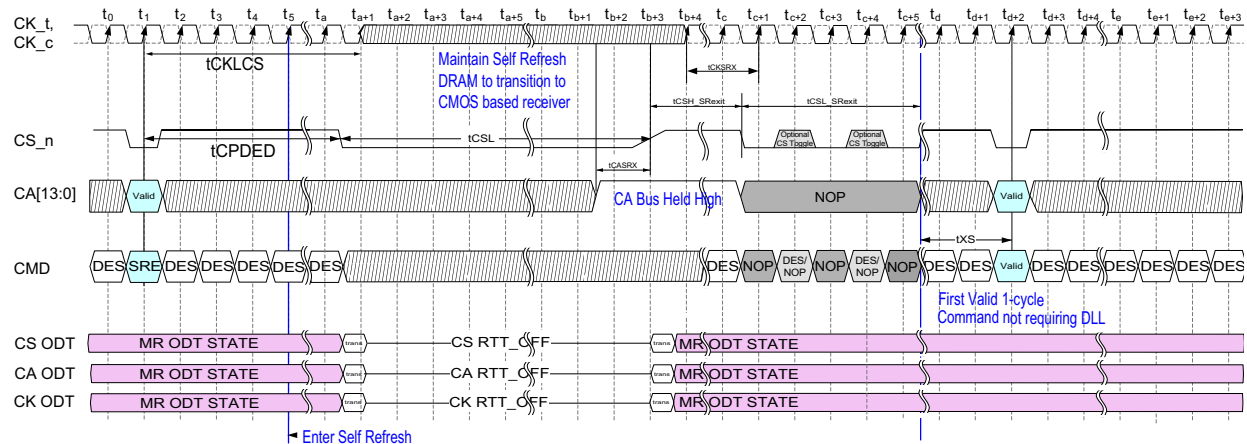
NOTE 2 Upon exit from Self-Refresh, the 3D Stacked DDR5 SDRAM requires a minimum of one extra refresh command to all logical ranks before it is put back into Self-Refresh Mode.

NOTE 3 This parameter utilizes a value that varies based on density. Refer to the 3DS Refresh section for more information.

NOTE 4 These timings are for x4 2H and 4H 3Ds devices.

4.9.1 Self Refresh in 2N Mode

The timing diagram in Figure 64 shows details for Self Refresh entry/exit in 2N Mode. Only SRX, with a pulsing CS_n (NOP-DES-NOP-DES-NOP) during tCSL_SRexit, to a 1-cycle command is shown, but behavior is similar for SRX to a 2-cycle command. Behavior is similar for Frequency Change during Self Refresh, with or without VREF and/or ODT changes. Pulsing CS_n during tCSL_SRexit is not required for Self Refresh exit (for example, CS_n may be held low for the full tCSL_SRexit duration).



NOTES:

- Both tCSH_SRexit and tCSL_SRexit timings must be satisfied to guarantee DRAM operation.
- When tCSH_SRexit,min expires, the CA bus is allowed to transition from all bits High to any valid (V) level. Prior to CS_n being registered Low at tc+1, the CA bus must transition to NOP conforming to the CAI state of the DRAM and complying with applicable DRAM input timing parameters.

Figure 64 — Self-Refresh Entry/Exit Timing in 2N Mode with 1-Cycle Exit Command

4.9.2 Partial Array Self Refresh (PASR)

PASR has been deprecated starting with spec working revision 1.90, JESD79-5C-v1.30. All MR60 bits will behave as RFU on devices that do not support PASR.

DDR5 DRAMs may contain an optional feature that disables refresh to selected segments in each bank when in self refresh. The feature allows for lower self refresh power if portions of the DRAMs are not required to retain data. Each bank is divided into six or eight segments based on the three highest row address bits supported by the DRAM device's density. Non-Binary density devices are divided into 6 segments as the 110 and 111 encodings of the PASR segment row bits are not used. Binary densities are divided into 8 segments.

MR60 provides the segment mask for all banks, with one bit per segment. A 0 (default) in the bit position provides normal refresh operation for the segment while a 1 masks that segment. Masked segments are NOT refreshed during self refresh. Note that this affects Self Refresh only. All segments are refreshed by refresh command when out of self refresh.

Segments which are masked are not guaranteed to retain their data if self refresh is entered. ECS Transparency will not produce accurate results if any mask bit is set, but ECS scrubbing will still occur if enabled.

MR19 bit 7 indicates whether the DRAM supports PASR. 0 = Not Supported, 1 = Supported

Table 57 — MR60 Definition

Segment (PASR Row Bits)	Type	Operand	Data	Notes
Segment 0 (000)	W	OP0	0=Normal, 1=Masked	
Segment 1 (001)	W	OP1	0=Normal, 1=Masked	
Segment 2 (010)	W	OP2	0=Normal, 1=Masked	
Segment 3 (011)	W	OP3	0=Normal, 1=Masked	
Segment 4 (100)	W	OP4	0=Normal, 1=Masked	
Segment 5 (101)	W	OP5	0=Normal, 1=Masked	
Segment 6 (110)	W	OP6	0=Normal, 1=Masked	Must be 0 for 24 Gbit and 48 Gbit devices.
Segment 7 (111)	W	OP7	0=Normal, 1=Masked	Must be 0 for 24 Gbit and 48 Gbit devices.

Table 58 — PASR Segment Row Address Bits

DRAM Density	PASR Row Bits	Segments
8 Gbit	R15:13	8
16 Gbit	R15:13	8
24 Gbit	R16:14	6
32 Gbit	R16:14	8
48 Gbit	R17:15	6
64 Gbit	R17:15	8

4.10 Power-Down Mode

DDR5's Power-Down (PD) mode is new to the DDR family, as it no longer has a CKE pin to control entry and exit. Instead, the Power-Down Entry (PDE)/Power-Down Exit (PDX) move to command based, triggered by the CS_n. Once in PD mode, the CS_n acts effectively like the historic CKE pin, waiting for it to transition from HIGH to LOW (with its command). In PD mode, CS_n should be sampled on every edge.

4.10.1 Power-Down Entry and Exit

Power-Down is entered when the PDE command is registered. Unlike Self Refresh Mode, CS_n will NOT be held low constantly while in Power-Down. Timing diagrams are shown in Figure 65 with details for entry and exit of Power-Down.

The DLL should be in a locked state when Power-Down is entered for fastest Power-Down exit timing. SDRAM design provides all AC and DC timing and voltage specification as well as proper DLL operation as long as DRAM controller complies with SDRAM specifications.

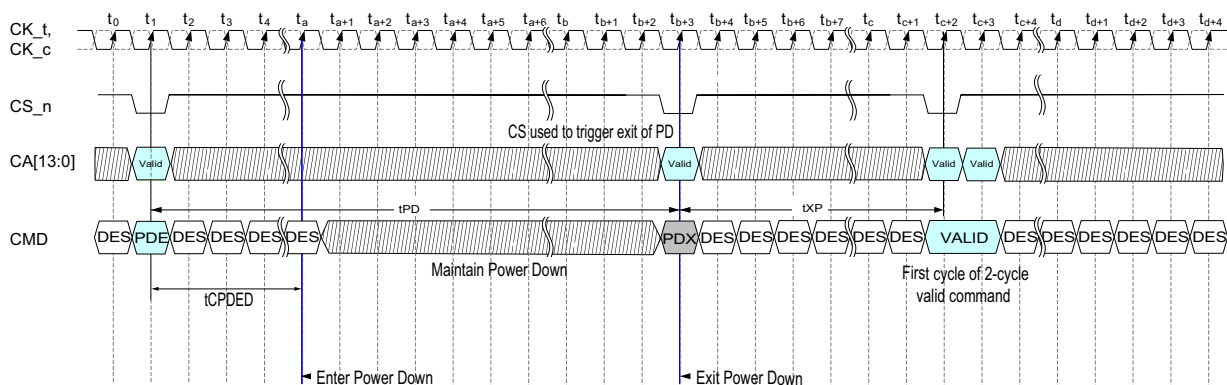
During Power-Down, if all banks are closed after any in-progress commands are completed, the device will be in precharge Power-Down mode; if any bank is open after in-progress commands are completed, the device will be in active Power-Down mode.

Entering Power-Down deactivates the input and output buffers, excluding CK_t, CK_c, CS_n, RESET_n which remain enabled throughout the duration of Power-Down. If CA11=L during the PDE command, command input buffers CA1 and CA4 remain enabled, allowing the appropriate Non-Target (NT) On-Die Termination (ODT) command to be passed through and decoded by the non-target DRAM in Power-Down (i.e. the DRAM will monitor commands that utilize NT ODT and will not exit Power-Down as a result of a valid NT ODT command being registered). Optionally, command input buffers CA0, CA2, CA3 and CA5 may remain enabled if needed by the DRAM to properly decode the NT ODT commands during PDE and PDX.

If CA11=H during the PDE command, only the PDX command, qualified by CS_n, is legal during Power-Down. If CA11=L during the PDE command, only NT ODT commands and PDX commands, qualified by CS_n, are legal during Power-Down. Refer to the command truth table for more information.

Although MRR is BL16 regardless of the MR0 Burst Length setting, the host shall not issue MRR NT ODT commands during Power-Down when MR0:OP[1:0]=01.

When Power-Down is entered with NT ODT control enabled (CA11=L), the DRAM will continue to accept NT commands throughout the Power-Down process, including entry and exit. Upon Power-Down entry during the tCPDED period, the DRAM will be switching from decoding all CA bus command bits to only decoding CA1 and CA4. During this time all CA command bits must be valid when CS_n is asserted with the full Read or Write command, as the DRAM may still be decoding the full command. Following tCPDED, only CA1 and CA4 need be valid as the DRAM will be ignoring the others. Following the PDX command, all CA command bits must be valid for NT termination commands also, as the DRAM will be transitioning to decoding all bits. In the optional case where command input buffers CA0, CA2, CA3 and CA5 remain enabled, all CA command bits must be valid when CS_n is asserted between the PDE command and the completion of tXP upon PDX. It is only the time between tCPDED completion and tXP start that non-CA command bits (CA13:6) and CA command bits not being used for the NT ODT command decode need not be valid.



NOTES:

1. There is no specific PDX command. In the case of systems with register using CA1, the encoding out of the register may be inverted from a NOP type command.
2. Diagram above is shown with a valid 2-cycle command after tXP for simplicity. 1-cycle valid commands are also legal.
3. CS_n shall be held HIGH, not toggled, during Power-Down, except Non-Target ODT command when PDE with CA11=L is asserted.

Figure 65 — Power-Down Entry and Exit Mode

4.10.1 Power-Down Entry and Exit (cont'd)**Table 59 — Power-Down Entry Definitions**

Status of DRAM	DLL	PD Exit	Relevant Parameters
Active (One or more banks open)	On	Fast	tXP
Precharge (All banks precharged)	On	Fast	tXP

The DLL is kept enabled during Precharge Power-Down or Active Power-Down. (If RESET_n goes low during Power-Down, the DRAM will be out of PD mode and into reset state). Power-down duration is limited by tPD(max) of the device.

Table 60 — Power Down Timing Parameters

Parameter	Symbol	Min	Max	Unit	Note
Command pass disable delay	tCPDED	max(5ns, 8nCK)	-	ns, nCK	
Power-Down duration	tPD	max(7.5ns, 8nCK)	5 * tREFI1 (Normal) 9 * tREFI2 (FGR)	ns, nCK	8
Power-Down Exit command to next valid command (excluding NT ODT commands when PDE with CA11=L)	tXP	max(7.5ns, 8nCK)		ns, nCK	
Activate command to Power Down Entry command	tACTPDEN	2		nCK	1
Precharge, Precharge All, or Precharge Same Bank command to Power-Down Entry command	tPRPDEN	2		nCK	1
Read or Read w/Auto Precharge command to Power-Down Entry command	tRDPDEN	CL+RBL/2+1		nCK	4, 6
Write command to Power-Down Entry command	tWRPDEN	CWL+WBL/2+WR+1		nCK	2, 4, 7
Write w/Auto Precharge command to Power-Down Entry command	tWRAPDEN	CWL+WBL/2+WR+1		nCK	2, 3, 4, 7
Refresh All or Refresh Same Bank command to Power-Down Entry command	tREFPDEN	2		nCK	
MRR command to Power-Down Entry command	tMRRPDEN	CL+8+1		nCK	4
MRW command to Power-Down Entry command	tMRWPDEN	tMRD(min)		nCK	
MPC command to Power-Down Entry command	tMPCPDEN	tMPC_delay		nCK	5

NOTE 1 Power-Down command can be sent while operations such as Activation, Precharge, Auto-Precharge or Refresh are in progress but IDD spec will not be applied until the operations are finished.

NOTE 2 Write Recovery (WR) is calculated from tWR by using the Rounding Algorithm. Refer to the Rounding Definitions and Algorithm section for more information.

NOTE 3 Write Recovery (WR) in clock cycles as programmed in MR6.

NOTE 4 Read/Write/MRR can refer to both Target command and Non-Target command when CA11=H during PDE command.

NOTE 5 tMPC_delay is a valid timing parameter for all MPC functions except:

a) Enter CS training Mode, Enter CA Training Mode, PDA Enumerate ID Program Mode because Power Down Entry is not supported for these MPC commands.

b) Apply VrefCA, VrefCS and RTT_CA/CS/CK because this MPC command requires waiting for VrefCA_time/VREFCS_time.

NOTE 6 RBL: Read burst length associated with Read command

a) RBL = 32 (36 with RCRC on) for fixed BL32 and BL32 in BL32 OTF mode

b) RBL = 16 (18 with RCRC on) for fixed BL16 and BL16 in BL32 OTF mode

c) RBL = 16 (18 with RCRC on) for BL16 in BC8 OTF mode and BC8 in BC8 OTF mode

NOTE 7 WBL: Write burst length associated with Write command

a) WBL = 32 (36 with WCRC on) for fixed BL32 and BL32 in BL32 OTF mode

b) WBL = 16 (18 with WCRC on) for fixed BL16 and BL16 in BL32 OTF mode

c) WBL = 16 (18 with WCRC on) for BL16 in BC8 OTF mode and BC8 in BC8 OTF mode

NOTE 8 tPD(max) shall always be less than or equal to 5*tREFI1(max) during Normal Refresh Mode and less than or equal to

NOTE 9 tREFI2(max) during Fine Granularity Refresh Mode, and when using the rounding algorithms, nPD(max) shall always be less than or equal to 5*nREFI1(max) during Normal Refresh Mode and less than or equal to 9*nREFI2(max) during Fine Granularity Refresh Mode.

4.10.1 Power-Down Entry and Exit (cont'd)

Table 61 — Valid Command During Power-Down with ODT Enabled

CA1	CA4	Command	Operation of DRAM in Power Down
L	L	Write	DRAM will enable ODT_WR_NOM
L	H	Read	DRAM will enable ODT_RD_NOM
H	L	Illegal	Illegal. CS_n will NOT be asserted to a powered down DRAM with this combination
H	H	PDX(NOP)	Exit Power Down
NOTE 1 MRR NT ODT commands during Power-Down are not allowed when MR0:OP[1:0]=01.			

4.11 Input Clock Frequency Change

Once the DDR5 SDRAM is initialized, the DDR5 SDRAM requires the clock to be “stable” during almost all states of normal operation. This means that, once the clock frequency has been set and is to be in the “stable state”, the clock period is not allowed to deviate except for what is allowed for by the clock jitter and SSC (spread spectrum clocking) specifications.

The input clock frequency can be changed from one stable clock rate to another stable clock rate under Self Refresh with Frequency Change mode. Outside Self-Refresh w/Frequency Change mode, it is illegal to change the clock frequency.

Prior to entering Self-Refresh with Frequency Change mode, the host must program tCCD_L/tCCD_L_WR/tCCD_L_WR2/tDLLK via MPC Command to update MR13 shadow register to the desired target frequency and configure VREFCA, VREFCS, RTT_CK, RTT_CS and RTT_CA (MR11, MR12, MR32, and MR33) shadow registers if needed.

Once the DDR5 SDRAM has been successfully placed into Self-Refresh w/Frequency Change mode and tCKLCS has been satisfied, the state of the clock becomes a don't care. Once a don't care, changing the clock frequency is permissible, provided the new clock frequency is stable prior to tCKSRX. During tCSL_FreqChg and prior to exiting Self-Refresh, the DRAM will automatically apply the changes to tCCD_L/tCCD_L_WR/tCCD_L_WR2/tDLLK, VREFCA, VREFCSRTT_CK, RTT_CS, and RTT_CA. When entering and exiting Self-Refresh mode for the sole purpose of changing the clock frequency, the Self-Refresh entry and exit specifications must still be met as outlined in Section 4.9 “Self-Refresh Operation”. For the new clock frequency, Mode Registers may need to be configured (to program the appropriate CL, Preambles, Write Leveling Internal Cycle Alignment, DFE, DCA, etc.) prior to normal operation.

The DDR5 SDRAM input clock frequency is allowed to change only within the minimum and maximum operating frequency specified for the particular speed grade.

4.11.1 Frequency Change Steps

The following steps must be taken:

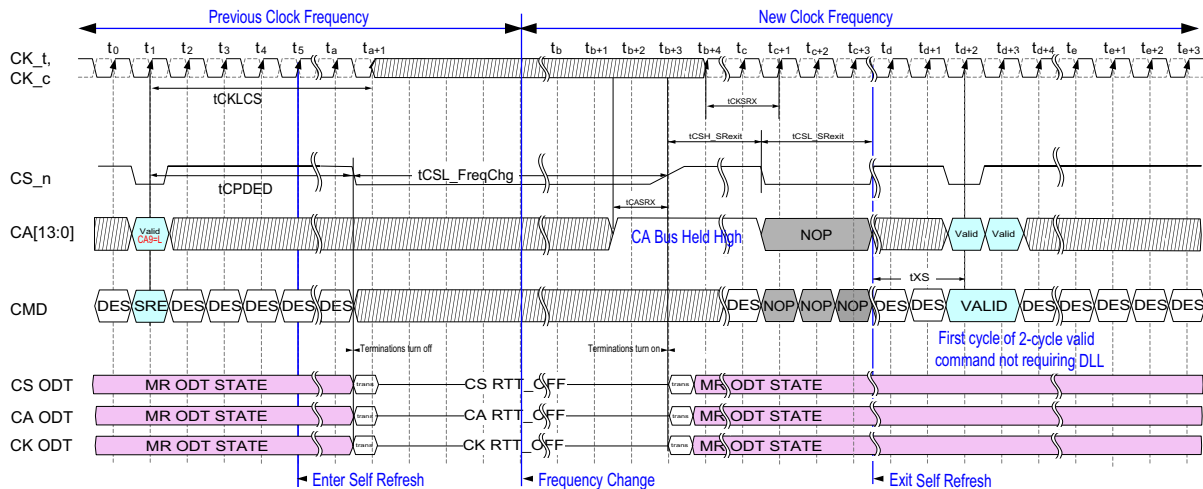
- Prior to executing the SREF command, the shadow register of Mode Registers MR11/MR12/MR13/MR32/MR33 can be pre-loaded anytime if DRAM is in an IDLE state:
 - The host MUST program tCCD_L/tCCD_L_WR/tCCD_L_WR2/tDLLK via MR13:OP[3:0] to the desired target frequency. During this stage, the values are stored in the MR13 shadow register and shall be applied to MR13 automatically when running SREF w/ CA9="L" on the DRAM. This change occurs during tCSL_FreqChg when exiting Self-Refresh with Frequency Change.
 - The host can configure the appropriate CS/CA/CK ODT settings via Mode Register (MR32 and MR33) if new values are needed for the new target frequency. During this stage, the values are stored in the MR32 and MR33 shadow registers and shall be applied to MR32 and MR33 automatically when running SREF with CA9="L" on the DRAM. This change occurs during tCSL_FreqChg when exiting Self-Refresh w/Frequency Change.
 - The host can configure the VREFCA and VREF CS via the VREFCA or VREFCS command(s). During this stage, the values are stored in the MR11 and MR12 shadow registers and shall be applied to MR11 and MR12 automatically when running SREF with CA9="L" on the DRAM. This change occurs during tCSL_FreqChg when exiting Self-Refresh with Frequency Change.
 - Entering and exiting Self-Refresh with the SREF command shall replace the values of MR11, MR12, MR13, MR32, and MR33 with their shadow register values.
- Enter Self Refresh with Frequency Change by sending the SREF command (SRE with CA9='L').

4.11.1 Frequency Change Steps (cont'd)

3. After $t_{CPDED}(\min)$, the host will transition CS_n low, indicating to the DRAM that the terminations are safe to turn off.
4. After t_{CKLCS} , the host can turn the clocks off.
5. Device enters Self Refresh.
6. At this time, changing the clock frequency is permissible, provided the new clock frequency is stable prior to t_{CKSRX}
7. Exiting Self-Refresh w/Frequency Change follows the same process as normal Self-Refresh exit.
8. After t_{XS} , any additional mode registers requiring the DLL that are needed for the new frequency can be configured or other commands not requiring a DLL may be issued. (ACT, MPC, MRW, PDE, PDX, PREab, PREsb, PREpb, REFab, REFsb, RFMab, RFMsb, SRE, VREFCA, VREFCS)
9. After t_{XS_DLL} , normal operations resume and all commands are legal.

Table 62 — Self Refresh with Frequency Change (for Reference)

Function	Abbreviation	CS	CA Pins														NOTES
			CA0	CA1	CA2	CA3	CA4	CA5	CA6	CA7	CA8	CA9	CA10	CA11	CA12	CA13	
Self Refresh Entry with Frequency Change	SREF	L	H	H	H	L	H	V	V	V	V	L	L	V	V	V	1
NOTE 1 See Command Truth Table for details																	



NOTES:

1. While in 2N mode, t_{CSL_SRexit} will not be statically held low (as shown above), as it will pulse for each 2 cycle period. Refer to the 2N mode section for more details.
2. Both t_{CSH_SRexit} and t_{CSL_SRexit} timings must be satisfied to guarantee DRAM operation.
3. Diagram above is shown with a valid 2-cycle command after t_{XS} for simplicity. 1-cycle valid commands are also legal.
4. When $t_{CSH_SRexit, \min}$ expires, the CA bus is allowed to transition from all bits High to any valid (V) level. Prior to CS_n being registered Low at t_{c+1} , the CA bus must transition to NOP conforming to the CAI state of the DRAM and complying with applicable DRAM input timing parameters.

Figure 66 — Frequency Change during Self Refresh

Table 63 — Self-Refresh Frequency Change Timing Parameters

Parameter	Symbol	Min	Max	Unit	Note
Self-Refresh CS_n low Pulse width with Freq Change	$t_{CSL_FreqChg}$	V_{refCA_time}	-	ns	1

NOTE 1 Since frequency can require VREFCA, VREFCS and CA/CK/CS ODT Changes, the min time is longer than the traditional t_{CSL} when the SRE command with CA9=L is used.

4.12 Maximum Power Saving Mode (MPSM)

When Maximum Power Saving Mode is enabled by setting the MPSM enable (MR2:OP[3]) bit to '1' using MRW command, the device enters Maximum Power Saving Mode Idle (MPSM Idle) state. When Maximum Power Saving Mode for Device 15 is enabled by setting the Device 15 MPSM enable bit (MR2:OP[5]) to '1' using MRW command, and the device's PDA Enumerate ID (MR1 bits OP[3:0]) are equal to 15, the device enters Maximum Power Saving Mode Idle (MPSM Idle) state. Setting the Device 15 MPSM enable bit to '1' must be done after PDA device enumeration is complete. Once the DRAM is placed into the MPSM Idle state, it can stay in that state indefinitely, or it can further enter either Maximum Power Saving Mode Power Down (MPSM Power Down) state or Maximum Power Saving Mode Self Refresh (MPSM Self Refresh) state.

Data retention is not guaranteed when DRAM is in any of MPSM states. Mode register status and Soft PPR information is preserved.

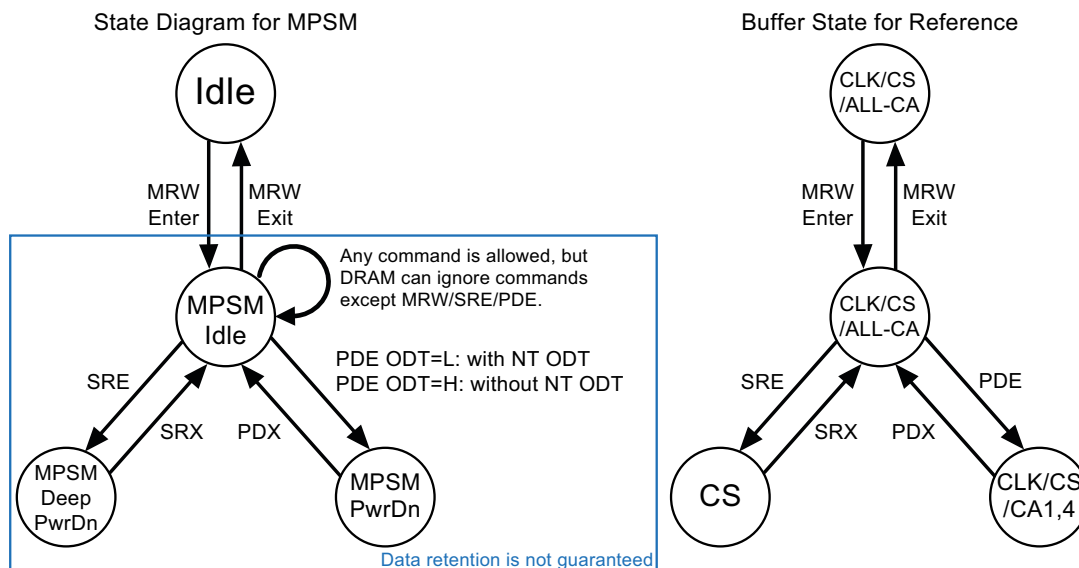


Figure 67 — State Diagram for Maximum Power Saving Mode

Table 64 — MPSM Configuration Options

MPSM MR2:OP[3]	Device 15 MPSM MR2:OP[5]	PDA Enumeration ID MR1:OP[3:0]	Action
1	X	X	Enter MPSM on MRW
X	1	1111	Enter MPSM on MRW
0	0	X	Exit MPSM on MRW
0	X	Not equal to 1111	Exit MPSM on MRW

4.12.1 MPSM Idle State

When DDR5 SDRAM is in this state, it ignores all types of commands except MRW, ODT, Power Down Entry (PDE) and Self Refresh Entry (SRE) commands. Only MRW commands to Exit MPST Idle, as described in Table 64, shall be issued. For any other MRW command the DRAM shall first be taken out of MPSM Idle, then the MRW command may be issued. ODT, PDE and SRE commands are executed normally. DRAM shall not respond to any other command except these four command types. DLL status is same as in normal idle state. DRAM continues to drive CA ODT as programmed.

Normal command timing parameters are applied in this state, except that tREFI doesn't need to be satisfied as Refresh command doesn't need to be issued in this state.